

STAT 145 Course Pack

UWL Department of Mathematics & Statistics

1.1 Introduction to Data

Scientists seek to answer questions using rigorous methods and careful observations. These observations — collected from the likes of field notes, surveys, and experiments — form the backbone of a statistical investigation and are called **data**. Statistics is the study of how best to collect, analyze, and draw conclusions from data. In this first chapter, we focus on both the properties of data and on the collection of data. We will introduce the building blocks of statistical thinking: observations, variables, data frames, and the types of questions that statistics can help us answer.

Key Concepts

- Recognize that a data set consists of cases and variables
- Identify variables as either categorical or quantitative
- Determine explanatory and response variables where appropriate
- Determine whether a study is observational or experimental

Why we collect data

All statistical analyses start with a question or an idea. This question or idea is posed about a *population*, in particular a population *parameter*. The main purpose of collecting data is to learn about this unknown population parameter. We often cannot measure data for every individual in a population because it would cost too much money or take too long, so a *sample*, or subset of the population, is collected instead. Using the information in the sample, a *statistic* is calculated and used to estimate the corresponding population parameter. This process of using a sample statistic to learn about a population parameter is called *statistical inference*. One of the main goals of this class is to introduce you to the procedures for making statistical inferences in a variety of settings.

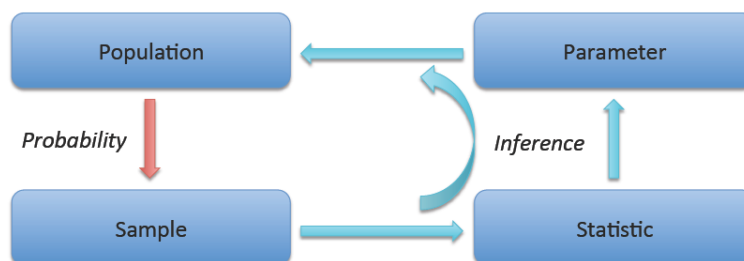
A *population* is a well-defined collection of *all* individuals or objects of interest.

A *parameter* is a number that describes a population and is almost always unknown.

A *sample* is a subset of the population on which we measure data.

A *statistic* is a number that describes a sample and is almost always known.

Inference is the process of drawing conclusions about a *population parameter* from a *sample statistic*.



Data Basics

After data is gathered, it needs a place to be stored. In statistics, this is called a *data set*, or *data frame*. Datasets can be as simple as an Excel spreadsheet or as complicated as a major database. It all depends on the complexity of the question.

Structure: Cases and variables

In general, when data is stored in a table, each row of the table represents an *observation*, or a *case*, in the dataset. Each column of the table represents a *variable* in the dataset.

Note: Many datasets use the first column as an identifier. This would not be considered a variable.

A *data set*, or *data frame* is used to store the set of measurements taken on individual units.

Observations, or *Cases*, are the subjects/objects that we obtain information about. These are most generally known as *experimental units*.

A *Variable* is a characteristic that is measured and recorded for each case.

Class Example 1.1.1 Distinguishing between cases and variables

Let's get to know your classmates. Find two people near you and ask them the following questions:

- What is your first name?
- What is your major?
- How tall are you?
- How many credit hours are you taking this semester?
- How often do you drink caffeine (never, once a week, few times a week, everyday)?

Name	Major	Height	Credit Hours	Caffeine

How many cases are there?

How many variables are there?

Types of Variables

Before we can know how to answer a question, we need to think about the kinds of variables we have recorded. Variables that split cases into groups, called *levels*, are called *categorical*, or *qualitative*, variables. Variables that are numerical quantities are called *quantitative* variables.

Categorical variables can further be subdivided:

- Nominal variables* have levels with no natural ordering. The variable *major* from Example 1.1 is nominal, there is no reason to rank one major over another.
- Ordinal variables* have levels with a meaningful order. The variable *caffeine* has a natural ordering.

Quantitative variables can further be subdivided:

- Continuous variables* can take any value within a range, including decimals. The variable *height* is continuous.
- Discrete variables* can only take counting values (typically whole numbers: 0, 1, 2, ...). The variable *credit hours* is discrete.

Note: In most of this course, we will treat categorical variables as nominal (un-

Categorical, or *Qualitative* variables divide the cases into groups, placing each case into exactly one of two or more non-numerical, non-overlapping categories.

Quantitative variables take on numerical values for each case, often by measuring or counting. Numerical operations like adding and averaging make sense for quantitative variables.

ordered) and quantitative variables as continuous. The distinction between nominal and ordinal, and between discrete and continuous becomes more important in advanced courses when special methods are used to take advantage of the ordering.

The ability to identify whether a variable is categorical or quantitative is important because different types of variables are summarized in different ways.

Class Example 1.1.2 *Computer ownership at UWL*

UWL would like to estimate the proportion of students who own a computer. The university asks 500 students.

- Identify the cases and variable(s) recorded.
- What is the sample?
- What is the population?
- Is the variable categorical or quantitative?
- Identify an unmeasured quantitative variable of interest.
- Is the result of this study a statistic or parameter?

Explanatory and Response Variables

In this class, we will discuss two ways in which data are used to answer questions. The first is looking only at a single variable, and the second is to determine if there is a relationship between two variables. When two variables show some connection or pattern with one another, they are called *associated* variables (or related variables). If two variables are not associated — there is no evident relationship between them — they are said to be *independent*.

In studies that look for an association, many times we want to know if one variable explains the behavior of another variable. In this kind of study, the *explanatory* variable would explain the behavior in the *response* variable.

Associated variables show some connection or pattern with one another.

Independent variables have no evident relationship between them.

An *explanatory* variable is one that helps us understand or predict the value of another variable.

The *response* variable is the variable that we are trying to understand.

Class Example 1.1.3 *Explanatory and response variables*

For the following research questions, identify the explanatory variable and the response variable.

- Does eating yogurt cause people to lose weight?

- Does meditation help reduce stress?

- Is hyperactivity in children affected by sugar consumption?

Class Example 1.1.4 *2023 Hollywood Movies*

Here is a small part of a dataset that includes information on all 136 movies to come out of Hollywood in 2023. “Rating” is audience rating on a 100-point scale, “Budget” is budget in millions of dollars, and “Opening” is opening weekend gross, in millions of dollars.

Title	Lead Studio	Rating	Genre	Budget	Opening
Oppenheimer	Atlas	93	Biography	\$100	\$82
Barbie	Warner Bros	88	Comedy	\$145	\$162
Spider-Man: Across the Spider-Verse	Sony	95	Action	\$100	\$120.5
Guardians of the Galaxy Vol. 3	Disney	82	Sci-fi	\$250	\$118
The Holdovers	Focus	97	Drama	\$13	\$2.7
Killers of the Flower Moon	Paramount	93	Drama	\$200	\$23

a) What are the cases in this dataset?

b) What are the variables in this dataset? For each variable, identify whether it is categorical or quantitative.

c) Name a question we might ask about this dataset that is about

- a single variable
- a relationship between two of the variables

Observational Studies and Experiments

There are two primary types of data collection: experiments and observational studies. Understanding the difference is critical, because the type of study determines what conclusions we can draw.

If the researcher is only observing what happened, the study is an *observational study*. In statistics, the word observational does not mean that the researcher is watching the participant do every part of the study. Instead, observational refers to the fact that the researcher takes a dataset and makes observations about the data in the set. If the researcher is actively involved with assigning different values of the explanatory variable to the participants, then the study is an *experiment*. In general, observational studies can provide evidence of a naturally occurring *association* between variables, but they cannot by themselves show a *causal* connection. This is because in observational studies, researchers cannot control for all the other variables that might be influencing the outcome.

Warning: Association does not imply causation.

In an *experiment* the researcher actively controls one or more of the explanatory variables.

In an *observational study* the researcher simply observes the values as they naturally exist.

Two variables are *associated* if values of one variable tend to be related to the values of the other variable.

Two variables are *causally associated* if changing the value of one variable influences the value of the other variable.

Class Example 1.1.5 Experimental or Observational Studies

Determine whether each of the following is an experiment or an observational study:

- a) A pharmaceutical company randomly assigns 200 patients to receive either a new drug or a placebo and measures their cholesterol levels after 6 months.

- b) A university surveys its graduates to determine whether those who participated in study-abroad programs earn higher salaries five years after graduation.
- c) A teacher assigns students in her morning class to use a new learning app and students in her afternoon class to study using traditional methods, then compares exam scores.

Summary

This chapter introduced you to the world of data. Here are the key ideas:

- *Data* are observations collected about the world around us. *Statistics* is the science of collecting, analyzing, and drawing conclusions from data.
- A *data set*, or *data frame*, organizes data so that each row represents an *observation* and each column represents a *variable*.
- Variables come in two fundamental types: *quantitative* (continuous or discrete) and *categorical* (nominal or ordinal).
- The *explanatory* variable is the variable we think might influence the *response* variable.
- *Experiments* randomly assign subjects to treatments and can establish causation. *Observational* studies merely observe what happens and can identify associations, but generally cannot establish causation.
- Association does not imply causation — this is one of the most important principles in statistics.

Many of the ideas from this chapter will be revisited as we move on to doing end-to-end data analyses. In the next chapter, you will learn about how we can design studies to collect the data we need to make conclusions with the desired scope of inference.

1.2 Data Collection and Study Design

*Before digging into the details of working with data, we pause to think about how data come to be. If data are to be used to draw broad conclusions, then it is important to understand who or what the data represent. One important aspect is **sampling** — knowing how the observational units were selected from a larger group allows us to generalize back to the population from which the data were drawn. Additionally, by understanding the structure of the study, we can separate causal relationships from mere associations. A good question to ask before analyzing any dataset is: How were these observations collected? You will learn a lot about the data by understanding its source.*

Key Concepts

- Distinguish between a sample and a population
- Recognize when it is appropriate to use sample data to make inferences about the population
- Critically examine the way a sample is selected, identifying possible sources of bias
- Recognize that random sampling is a powerful way to avoid sampling bias
- Identify other potential sources of bias that may arise in studies on humans
- Distinguish between an observational study and a randomized experiment
- Recognize that only randomized experiments can lead to claims of causation
- Distinguish between a randomized comparative experiment and a matched pairs experiment
- Identify potential confounding variables in a study

Motivation

A state in the upper Midwest is currently considering a bill that would provide amnesty to underage persons who call for help from the authorities with an overly intoxicated person. One of the state senators who represents a district that contains a university is hesitant to initially support the bill, so she has commissioned a study to learn about how the residents of her district feel about the bill. The senator will select from one of the three study options.

Study 1: “Would you support a bill that prevents authorities from giving a ticket to an underage person who has asked for help with an overly intoxicated person?” Participants would be recruited by purchasing ads on the local radio stations.

Study 2: “Would you support a bill that has the potential to allow underage people to drink with no fear of retribution?” Participants would be a random sample of all taxpayers in the district gathered by door-to-door interviews.

Study 3: “Would you support a bill that helps reduce the chances of undergraduate death due to alcohol poisoning by guaranteeing that underage people who request emergency medical assistance for someone who is very intoxicated will not receive disciplinary sanctions from the university?” Participants would be a random sample of all property owners in the district contacted to fill out a survey via email.

- a) Rank the three studies based on the question they ask. Briefly justify your ranking.

- b) Rank the three studies based on the sampling method. Briefly justify your ranking.

- c) If you could mix the questions and sampling methods, which combination do you think would be the best?

A *population* includes **all** individuals or objects of interest.

A *census* occurs when every case in the population is measured.

The *sampling frame* is a list of the names of all subjects or objects in the population.

Data are collected from a *sample*, which is a subset of the population.

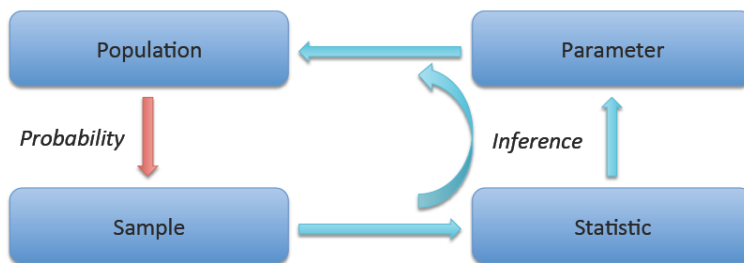
Statistical inference is the process of using data from a sample to gain information about the population.

Populations and Samples

The first step in conducting research is to identify the question to be investigated. A clearly stated research question helps identify what subjects or cases should be studied and what variables are important, or the *population*. It is also important to consider how data are collected so that they are reliable and help achieve the research goals.

One way we could find an answer is to look at every case by taking a *census*. However, most of the time this is not feasible. Taking a census could take months and is often very costly. Sometimes items are destroyed in the sampling process, such as measuring the tensile strength of a wire or the length of life of a light bulb or battery. Collecting a census in these cases would be foolish. So, most of the time, researchers take a *sample* from the population and use what they find from the sample to describe the population.

In order to make sure that all units in the population are included, a *sampling frame* is compiled and used to select the subjects for the sample. This process of using a sample to describe the population is called *statistical inference*, and the accuracy of the inference can be greatly impacted by the quality of the sample and how well the sample reflects the characteristics of the population.



Class Example 1.2.1 *Population Identification*

For the following research questions, identify the target population and what represents an individual case

- a) What is the average mercury content in swordfish in the Atlantic Ocean?
- b) Over the last five years, what is the average time to complete a degree for UW–La Crosse undergrads?

- c) Does a new drug reduce the number of deaths in patients with severe heart disease?

Parameters and Statistics

In most statistical analyses, the research question boils down to understanding a numerical summary — perhaps a quantity you already know (like the average) or one you will learn about in this course.

A numerical summary can be calculated on either the sample, called a *sample statistic*, or the entire population, called a *population parameter*. However, measuring every unit in the population is usually impossible. So we calculate the summary from a sample and use it to estimate the corresponding population value.

A *parameter* is a number that describes a population and is almost always unknown.

A *statistic* is a number that describes a sample and is almost always known.

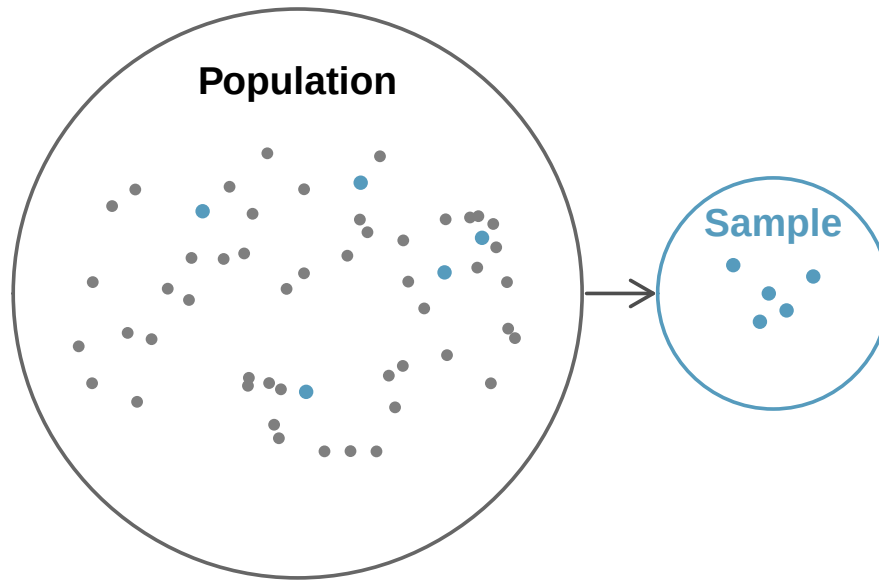
Class Example 1.2.2 Parameter and Statistic Identification

For the following research questions, identify the population parameter and the sample statistic.

- a) A sample of 60 swordfish have an average of 0.97 ppm
- b) An average of 4.37 years is reported from 25 UWL students as the time to complete their degree.
- c) After 1000 patients started taking a new drug for severe heart disease, 28% were reported to have died within 5 years.

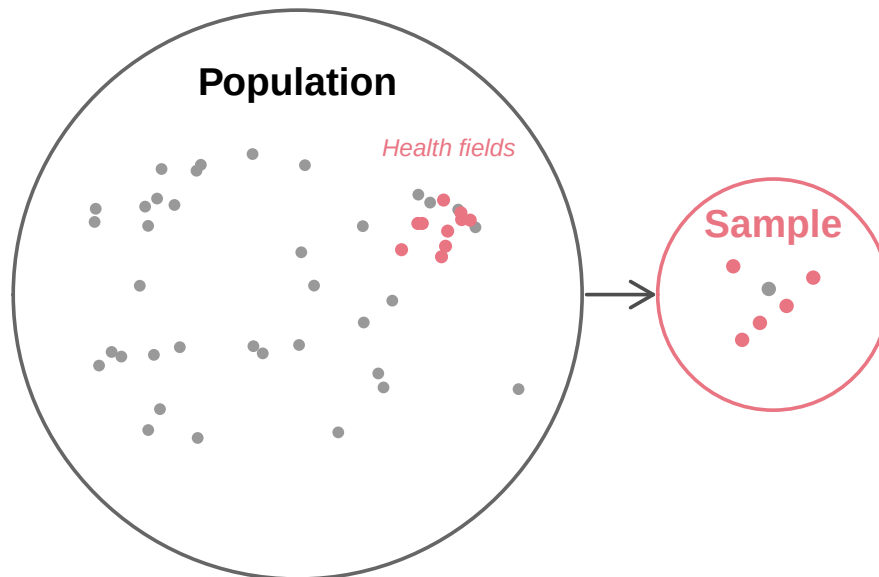
Sampling from a Population

We might try to estimate the time to graduation for UW–La Crosse undergraduates by collecting a sample of graduates. All graduates in the last five years represent the *population*, and graduates who are selected for review are collectively called the *sample*. In general, we always seek to *randomly* select a sample from a population. The most basic type of random selection is equivalent to how raffles are conducted. For example, we could write each graduate's name on a raffle ticket and draw 10 tickets. The selected names would represent a random sample of 10 graduates.



If we ask a student who happens to be majoring in nutrition to select several graduates for the study, they might pick a disproportionate number of graduates from health-related fields. When selecting samples by hand, we run the risk of picking a *biased* sample, even if our bias is unintentional.

Bias occurs when the results from the sample differ from the population.



If someone is permitted to pick and choose exactly which individuals are included in the sample, it is entirely possible that the sample will overrepresent that person's interests — even if the bias is unintentional. Sampling randomly helps address this problem.

The act of taking a simple random sample helps minimize bias. However, bias can crop up in other ways. For example, asking sensitive questions may result in *response bias* because participants do not feel comfortable answering the question truthfully. If a lot of people do not respond to a survey, it will suffer from *non-response bias*. Finally, if the population from which the sample is drawn excludes groups of people,

Response bias is the tendency for people to not answer questions honestly.

Non-response bias occurs when the response rate is very low.

Undercoverage occurs when part of the population is not considered for inclusion in the sample.

then the sample will suffer from *undercoverage*.

Class Example 1.2.3 Wording bias

A survey is to be conducted using a random sample of citizens in a town, asking them if they support raising taxes to increase funding for the public school.

a) Write the question in a way that is likely to bias the results toward more *yes* answers.

b) Write the question in a way that is likely to bias the results toward more *no* answers.

Sampling Methods

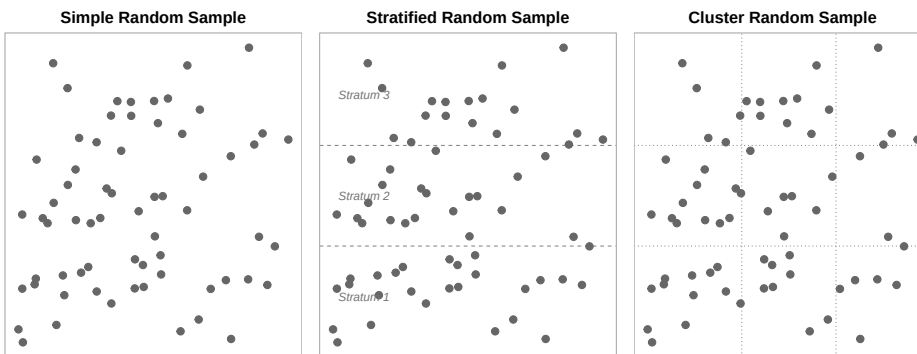
Good Ways to Obtain a Sample

There are a number of appropriate ways to randomly select your sample. The most basic type of sampling is the *simple random sample* (SRS). Taking a SRS avoids sampling bias and tends to produce a good representative sample of the population. However, there are many times when we can't conduct a simple random sample. *Stratified random sampling* occurs when the population is divided into homogeneous, non-overlapping groups (called *strata*) and a simple random sample is selected from each group. *Cluster random sampling* occurs when the population is divided into many non-overlapping groups (called *clusters*), a simple random sample of clusters is obtained, and data is collected from every case in the cluster.

A *simple random sample* is obtained when every sample of a given size is equally likely to be the one selected.

Stratified random sampling occurs when the population is divided into homogeneous, non-overlapping groups (called *strata*) and a simple random sample is selected from each group.

Cluster random sampling divides the population into many non-overlapping groups (called *clusters*), randomly selects a subset of clusters, and collects data from every case in the cluster.



Class Example 1.2.4 Picking a good sampling method

Which type of sampling method is most appropriate in the following scenarios?

a) Researchers want to test a new math curriculum for third graders. They only have the budget to try the curriculum in 10 elementary schools.

b) Processor chips are continuously manufactured at a production plant. To ensure that they meet size specifications so that they will fit inside an iPhone, manufacturers want to test 100 chips throughout the day.

- c) A player's agent wants to know the average salary for each position of major league baseball.

The table below summarizes the key features of each sampling method:

Method	How it works	Best used when...
SRS	Every case has an equal chance of selection	Population is relatively homogeneous and accessible
Stratified	Divide into similar groups (strata), then SRS within each	Known subgroups differ on the outcome of interest
Cluster	Randomly select entire groups (clusters)	Clusters are internally diverse but similar to each other
Multistage	Randomly select clusters, then SRS within each	Same as cluster, but clusters are large

Random sampling caution: Random is not the same as haphazard. We cannot obtain a random sample by haphazardly picking a sample on our own. We must use a formal random sampling method such as technology or drawing names out of a hat.

Class Activity: Sampling from Gettysburg Address Worksheet

Bad Ways to Take a Sample

Without a random sample, we may encounter *sampling bias* which occurs when the sample does not accurately represent the population of interest. One common sampling technique that will nearly always be biased is *voluntary response sampling*. With this kind of sampling, a researcher asks people to volunteer to be a part of a study. Another common type of sampling that produces biased results is the *convenience sample*. In a convenience sample, a researcher simply uses the most convenient group available as the sample.

Sampling bias occurs when the sample does not accurately represent the population of interest.

Voluntary response sampling occurs when a researcher asks people to volunteer to be participants instead of randomly selecting possible participants.

Convenience sampling happens when a researcher just uses the

Class Example 1.2.5 Driving with a pet on your lap

Over 30,000 people participated in an online poll on cnn.com conducted in April 2012 asking "Have you ever driven with a pet on your lap?" We see that 34% of the participants answered yes and 66% answered no.

- What is the sample?
- Is the variable in this study quantitative or categorical?
- Can we conclude that 34% of all drivers have driven with a pet on their lap?
- Explain why is not appropriate to generalize these results to all drivers, or even to all drivers who visit cnn.com. What is the problem with this method of data collection?

Experiments

Well-designed *experiments* incorporate four key principles:

1. **Controlling.** Researchers assign treatments to cases and do their best to *control* any other differences between the groups. For example, when patients take a drug in pill form, some patients might take the pill with a sip of water while others drink a full glass. To control for the effect of water consumption, a doctor might instruct every patient to drink a 12-ounce glass of water with the pill.
2. **Randomization.** Researchers randomly assign subjects to treatment groups to account for variables that cannot be controlled. For example, some patients may be more susceptible to a disease than others due to their dietary habits. Randomly assigning patients to treatment or control groups helps even out such differences across groups.
3. **Replication.** The more cases researchers observe, the more accurately they can estimate the effect of the explanatory variable on the response. In a single study, we **replicate** by collecting a sufficiently large sample. At a minimum, we want multiple subjects per treatment group. Replication also refers to repeating an entire study to verify earlier findings. (The *replication crisis* refers to the ongoing problem in several scientific disciplines where past findings have failed to be replicated in new studies.)
4. **Blocking.** Researchers sometimes know or suspect that variables other than the treatment influence the response. They may first group individuals based on this variable into *blocks* and then randomize cases within each block to the treatment groups. For instance, in studying the effect of a drug on heart attacks, researchers might first split patients into low-risk and high-risk blocks, then randomly assign half the patients from each block to the treatment and half to the control group.

To demonstrate the differences between some of the more commonly used experimental designs, consider the following study. A running shoe company wants to know if the type of shoe impacts the speed of runners. One aspect of the shoe that they are considering is the heel-toe drop, or the difference in height between the heel of the shoe to the toe of the shoe. Minimalist shoes reduce the heel-toe drop to 0 – 4 mm, whereas a more traditional shoe has an 8 – 12 mm heel-toe drop. This company wants to know if a lower heel-toe drop will cause runners to run more quickly, so they recruit 100 marathon runners of varying ability. For the study, the runner will run 5k, and their time will be used to establish their pace.

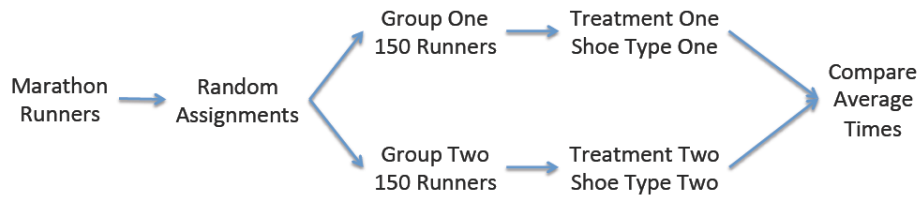
Randomized comparative design

In a *randomized comparative design*, we take all the participants in the study and randomize them to a different treatment group.

In an *experiment* the researcher actively controls one or more of the explanatory variables.

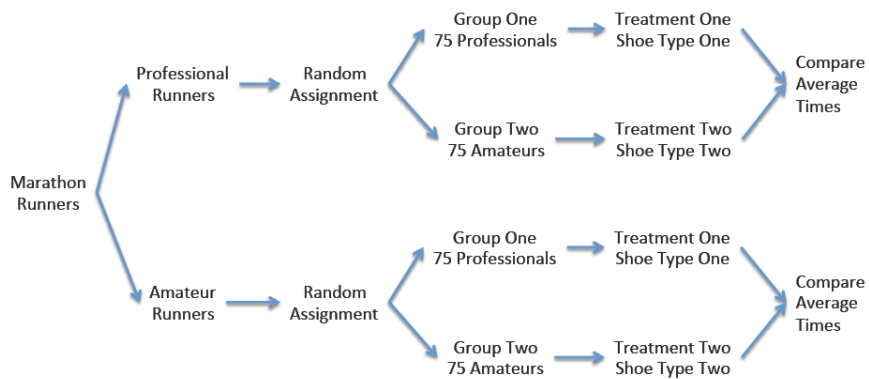
In a *randomized experiment* the value of the explanatory variable for each unit is determined randomly, before the response variable is measured.

In a *randomized comparative design*, we randomly assign cases to different treatment groups and then compare results on the response variable(s).



Block design

In a *block design*, we first group participants by some logical grouping that we think might have an impact on the response in addition to the variable we are studying.



In a *block design*, cases are first split into groups (called *blocks*), and within each group the value of the explanatory variable for each unit is determined randomly.

Class Example 1.2.6 *Picking the blocking factor*

Why might it make sense to group the runners into professionals vs amateurs?

Matched pairs design

In a *matched pairs design*, each participant is assigned to each of two treatment settings, responses are measured for each treatment, and the differences in responses are computed for each participant. Because two measurements are obtained for each individual, the measurements are *dependent* upon each other. The two treatments often come in the form of “Before vs. After” or “Method A vs. Method B.” In the “Method A vs. Method B” type of pairing, it is important to randomize the order of the treatments to avoid a confounding *order effect*.

In a *matched pairs* design, cases get both treatments (or cases are paired) and we examine differences in the response between the two treatments.

Class Example 1.2.7 *Describing the matched pairs running shoe experiment*

Thinking about the matched pairs design for the experiment about running shoes

a) What are the variables in the study? Are they categorical or quantitative?

b) What are the explanatory and response variables?

c) Explain how the treatments would be assigned.

d) Why might a matched pairs design be beneficial in this experiment?

Reducing Bias in Human Experiments

For an experiment, it is important to be able to determine if the treatment actually is making a difference, or if we are seeing results that would have happened regardless of the treatment. The purpose of a *control group* is to establish what we would expect to happen regardless of the treatment. Often times, in order to make sure that people adhere to the study guidelines, we use a *placebo* that is designed to look like the treatment.

The placebo makes it much easier to conduct *single-blinded* experiments or *double-blinded* experiments. Not all randomized experiments make use of control groups, placebos, or blinding; however, in studies involving humans, these techniques are often used to help make effective comparisons between groups.

The *control group* does not receive a treatment.

A *placebo* is a fake treatment that is designed to look like the treatment but has no therapeutic value.

In a *single-blind* study, participants do not know if they are in the treatment or control group.

In a *double-blind* study, neither the participants nor the people interacting with the participants knows if they are in the treatment or control group.

Class Example 1.2.8 *Depression and Prozac*

Researchers at a prominent Midwestern university would like to determine if Prozac is effective in reducing depression as determined by patient responses to a questionnaire. The researchers recruited 50 individuals

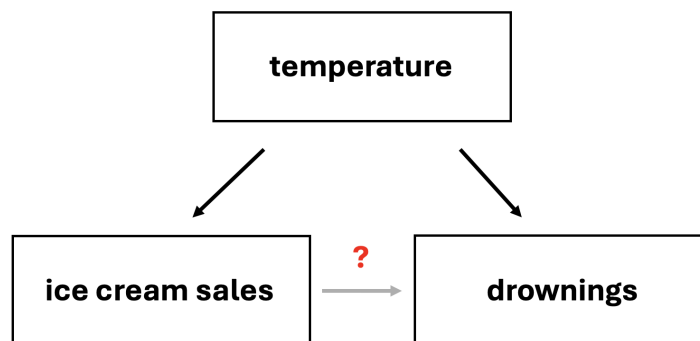
suffering from depression. Describe a completely randomized double-blind experiment using a placebo to test this association is a causal association.

Confounding Variables

A *confounding variable* is a variable that is associated with both the explanatory variable and the response variable. Because of this dual association, a confounding variable makes it impossible to determine whether the explanatory variable truly caused the observed response.

A *confounding variable* is a third variable that is associated with both the explanatory and response variable, but not necessarily of interest in the study.

Consider a classic example: ice cream sales and drowning rates both increase during summer months. If we looked only at the association between ice cream sales and drownings, we might (absurdly) conclude that ice cream causes drowning. The confounding variable is temperature (or season) — warm weather drives both increased ice cream consumption and increased swimming, which leads to more drowning incidents.



Confounding diagram.

Summary

In this chapter, we explored how data are collected and why the method of collection matters for the conclusions we can draw.

Key concepts:

- The *population* is the entire group of interest; a *sample* is a subset from which we collect data. A *census* collects data on every member of the population but is usually impractical.
- Good sampling methods include *simple random sampling (SRS)*, *stratified sampling*, and *cluster sampling*. Each has advantages depending on the population's structure and practical constraints.
- *Bias* can enter through non-random sampling (selection bias), non-response, or convenience sampling.
- In an *experiment*, researchers actively assign treatments to subjects. In an *observational study*, researchers simply observe without intervening.
- *Confounding variables* are associated with both the explanatory and response

variables and can create misleading associations.

- *Blinding* (single and double) and placebos help reduce bias in human experiments.

1.3 Exploring Categorical Data

This chapter focuses on **categorical data** — variables whose values are categories or labels rather than numbers. We begin with visualizations for a single categorical variable (bar charts, pie charts, waffle charts), then move to tools for exploring the relationship between two categorical variables (contingency tables and stacked bar charts). We close with principles for creating effective, honest data visualizations.

Key Concepts

- Display information from a categorical variable in a table or a graph
- Use information about a categorical variable to find and report a proportion, using correct notation
- Display information about a relationship between two categorical variables in a contingency table
- Use a contingency table to find proportions

Technology for Chapter

StatLens Use the [One Categorical Variable](#) explorer for bar charts and pie charts. Use the [Two Categorical Variables](#) explorer for contingency tables and stacked bar charts.

jamovi Use Analyses>Exploration>Descriptives for bar charts for one categorical variable. Use Analyses>Frequencies>Independent Samples for two categorical variables.

One Categorical Variable

When we want to summarize one categorical variable, we use a *proportion* which tells us what fraction of the sample is in each group. In this course, we will use p to represent the population proportion, and $\hat{p}$ (p-hat) to represent the sample proportion. The sample proportion is given by

$$\hat{p} = \frac{x}{n}$$

where x is the number of cases in a given category and n is the total number of cases.

Numerical displays

A single categorical variable is often summarized using a *frequency table* to display the number of observations that fall into each category. An alternative to the frequency table is the *relative frequency table* which displays the proportion of the sample that is in each category.

The *proportion* (or *relative frequency*) in a category is found by dividing the number of cases in that category by the total number of cases.

A *frequency table* is a table that displays the count for each category.

A *relative frequency table* is a table that displays the proportion of the sample in each category.

Class Example 1.3.1 Hair Color

The table below gives the frequency of hair color for a random sample of 55 UWL students. Find the proportions of students with each hair color and use them to fill in the relative frequency column.

	Frequency	Relative Frequency
Brown	30	
Blonde	14	
Red	2	
Black	5	
Colorful	3	
Total	54	

Frequency table for hair color.

Graphical summaries

The standard graphical summaries for a categorical variable are the *bar chart* and the *pie chart*. A bar chart is the most common way to display the distribution of a single categorical variable. Each category gets its own bar, and the height (or length) of the bar represents either the count or the proportion of observations in that category. A pie chart divides a circle into slices, where each slice represents a category and the size of the slice corresponds to the proportion of observations in that category.

A *bar chart* includes one bar for each category, and the height of the bar is the frequency the category appears.

In a *pie chart*, the proportions correspond to the areas of sectors of a circle.

Class Example 1.3.2 *Hair Color revisited*

Sketch a bar chart and a pie chart for the hair color of random sample of 55 UWL students.

Pie charts can give a quick high-level overview, especially when a variable has only a few levels or when each level represents a simple fraction (one-half, one-quarter, etc.). However, bar charts are almost always easier to read because our eyes are better at comparing lengths than angles or areas.

Two Categorical Variables

Sometimes our goal is to examine whether there is a relationship between two categorical variables. To investigate these relationships, we use a *contingency table*, or *two-way table*. In a two-way table, the rows represent the groups of the first category, and the columns represent the groups of the second category. Each cell of the table contains the number of cases that are in both the row and the column category.

A *contingency table* summarizes the data for two categorical variables.

The rows represent the levels of one variables, the columns represent the levels of the other.

Class Example 1.3.3 *Relationship status and class level*

169 college students were asked about their relationship status and class year (lower=freshman/sophomore, upper=junior/senior). The results are given in the table.

	Upper	Lower	Total
In a relationship	32	10	42
It's complicated	12	7	19
Single	63	45	108
Total	107	62	169

Contingency table for relationship status and class year.

- a) What proportion of students in this sample are in a relationship?

- b) What proportion of lower class students in this sample are in a relationship?

- c) What proportion of upper class students in this sample are in a relationship?

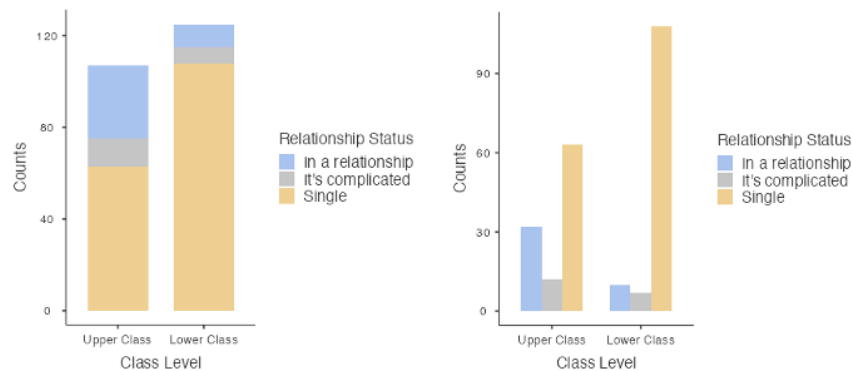
- d) Using $\hat{p}_U$ to represent the proportion of upper class students in a relationship and $\hat{p}_L$ to represent the proportion of lower class students in a relationship, find the difference in proportions $\hat{p}_U - \hat{p}_L$

- e) What proportion of the people who are in a relationship in this sample are upper class students?

- f) What proportion of the people in this sample are upper class students and in a relationship?

- g) Answers c), e) and f) are all ways of representing upper class students in a relationship. Why are they different numbers?

Contingency tables can be displayed graphically using an extension of the bar chart. There are two common variants: a stacked bar chart and a side-by-side bar chart.



Class Example 1.3.4 *Handedness and occupation*

In a study of handedness in occupations, 10 of 118 psychiatrists were left-handed, 26 of 148 architects were left-handed, 5 of 132 orthopedic surgeons were left-handed, and 16 of 105 lawyers were left-handed.

a) Make a two-way table of this relationship.

b) What proportion of the people in the sample are left-handed?

c) What proportion of left-handed people are architects?

d) Is your answer in part c) the same or different than the proportion of architects who are left-handed?

Effective Data Visualization

Graphs can powerfully communicate ideas directly and quickly. However, there are times when a visualization conveys a message that is inaccurate or misleading. The following are guiding principles for creating clear, honest, and effective visualizations.

- *Keep it simple:* Colors should be used purposefully and can be used to draw attention
 - Colors used only for decoration can be distracting and can confuse readers
- *Tell a story:* Include annotations (text, lines or labels that provide context beyond the raw data)
- *Order matters:* The arrangement of categories in a bar chart can help or hinder understanding
 - Alphabetical order is rarely the most informative choice
- *Make labels readable*
- *Select meaningful colors:* Default or rainbow color schemes are not always the best choice

Summary

This chapter introduced tools for exploring categorical data.

-For a single categorical variable, *bar charts* are the primary visualization, with *pie charts* reserved for simple cases.

-For two categorical variables, *contingency tables* organize counts, and stacked and side-by-side bar charts provide visual comparisons.

-Effective visualizations follow principles of simplicity, purposeful use of color, meaningful ordering, and accessibility.

1.4 Exploring Quantitative Data

*In the previous chapter we explored categorical data. Now we turn to **quantitative data** — variables that take numerical values where arithmetic makes sense (heights, incomes, interest rates). This chapter teaches you to both see and measure distributions. We introduce dot plots and histograms alongside the mean and median, box plots alongside the IQR, and the standard deviation alongside histogram shapes. By the end, you will be able to describe any distribution in terms of its shape, center, spread, and unusual features.*

Key Concepts

- Understand the difference between displays of categorical and quantitative data
- Use a dotplot or histogram to describe the shape of a distribution
- Calculate the mean and the median for a set of data values, with appropriate notation
- Identify the approximate locations of the mean and the median on a dotplot or histogram
- Explain how outliers and skewness affect the values for the mean and median
- Recognize the uses and meaning of the standard deviation
- Interpret a five number summary
- Use the range, the interquartile range, and the standard deviation as measures of spread
- Describe the advantages and disadvantages of the different measures of spread
- Identify outliers in a dataset based on the IQR method
- Use a boxplot to describe data for a single quantitative variable
- Use the empirical rule to estimate the middle 68%, 95%, or 99.7% for bell-shaped distributions
- Use technology to compute summary statistics for a quantitative variable

Technology for Chapter

StatLens Use the [One Quantitative Variable](#) to visualize distributions as histograms, dot plots, or box plots, and compute means, medians, standard deviations, and quartiles for any dataset.

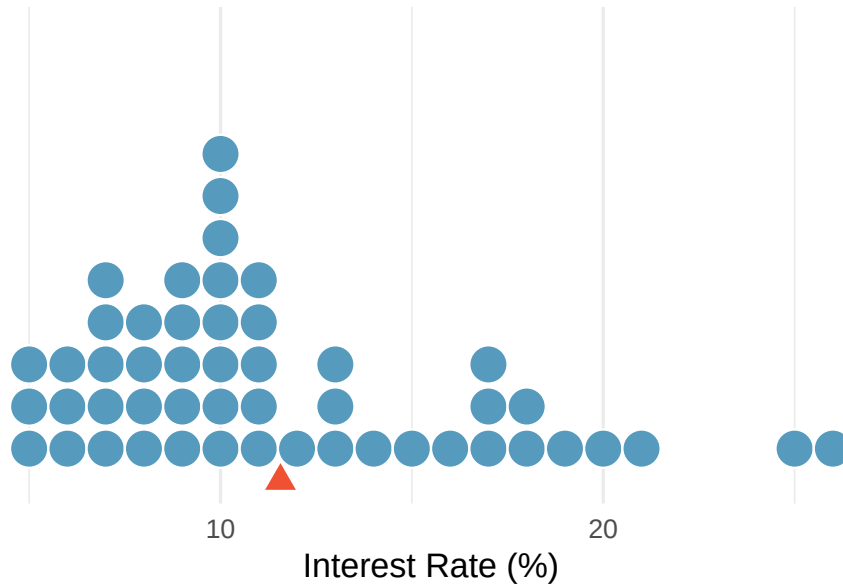
jamovi Use Analyses>Exploration>Descriptives to visualize distributions as histograms or box plots, and compute means, medians, standard deviations, and quartiles for any dataset.

Dot Plots and Histograms

Dot Plots

A *dot plot* is the simplest display for a single quantitative variable. Each observation is represented by a dot placed above its value on a number line. When observations share the same value (or are rounded to the same value), the dots stack vertically.

Consider the interest rates on 50 loans from a lending platform. In a dot plot, each of the 50 loans is a single dot. The red triangle marks the mean.



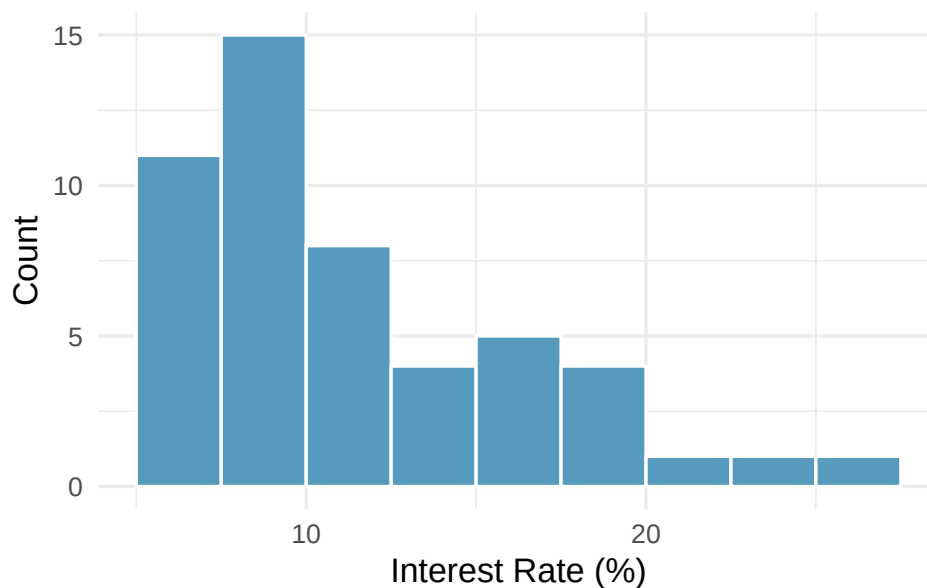
Dot plots are most useful for small datasets (up to about 50–100 observations) where you want to see every individual value. For larger datasets, histograms provide a better summary.

A *histogram* groups quantitative data into bins (intervals) and displays the count or proportion of observations in each bin as a bar. Unlike bar charts for categorical data, histogram bars touch each other because the bins represent a continuous range of values.

Histograms

To build a histogram, we:

1. Choose a bin width (e.g., 2.5 percentage points for interest rates)
2. Count how many observations fall in each bin
3. Draw bars whose heights equal the counts



A distribution is considered *symmetric* if we can fold the plot over a vertical center line and both sides match closely.

A distribution is considered *skewed left* if the longer tail of the distribution is to the left.

A distribution is considered *skewed right* if the longer tail of the distribution is to the right.

Histograms are especially useful for identifying the *shape* of a distribution. One key characteristic is symmetry. When we say a distribution is *symmetric* it means that the left-side of a distribution is very similar to the right-side of the distribution. We say a distribution is *skewed left* when the tail of the distribution is longer on the left. Similarly, we say that a distribution is *skewed right* when the tail of the distribution is longer on the right.

Class Example 1.4.1 *Shape of the interest rates*

Describe the shape of the histogram for the interest rate plot.

Measures of Center: Mean and Median

One type of numerical summaries you will see for a dataset is called measures of *center*. Measures of center, provide us with a single number that best describes a quantitative variable's distribution. Essentially, they give us a guess of what a typical value would look like. There are two measures of center we will use in this class: the *mean* and *median*.

The mean

The *mean* of a sample tells us the average. For a sample of observations $x_1, x_2, \dots, x_n$, the mean is given by

$$\text{Mean} = \frac{x_1 + x_2 + \dots + x_n}{n} = \frac{\sum_{i=1}^n x_i}{n}$$

where n is the number of observations.

We will use μ (pronounced “mew”) to represent the population mean and $\bar{x}$ (read “x-bar”) to represent the sample mean. We often cannot measure μ directly because we rarely have access to every member of the population. Instead, we estimate μ using the sample mean $\bar{x}$.

The median

The *median*, denoted m , divides the sample into two groups of equal size. We find the median by following two steps:

1. Sort the sample values.
2. Find the middle value:
 - If n is odd, the median is the middle value in the sorted list.
 - If n is even, the median is the average (midpoint) of the middle two values in the ordered list.

The dot plot below shows both the mean and the median marked on the interest rate data. Notice that the median (9.93%) is lower than the mean (11.57%) — the few loans with very high interest rates pull the mean to the right, but the median stays anchored in the middle of the data.

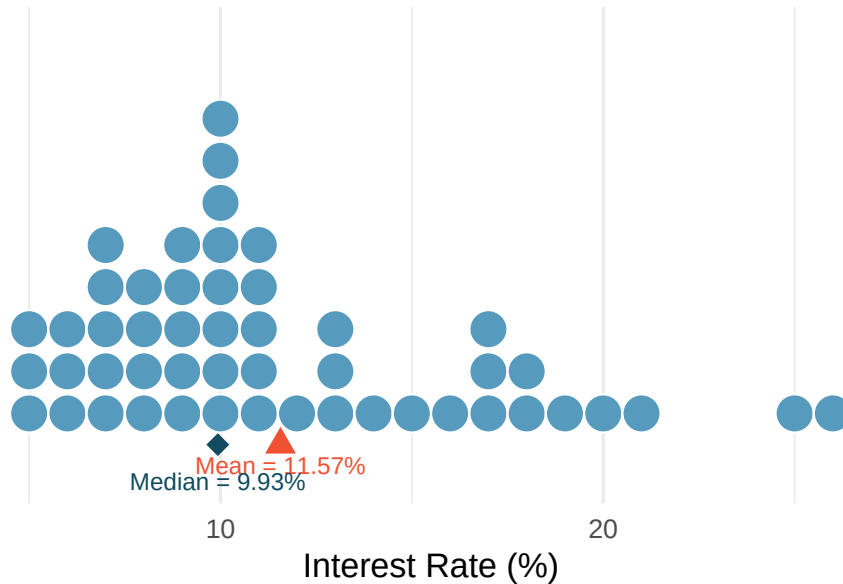
The *mean* of a data set is the sum of all its values divided by the number of observations.

The *median* of a dataset is the middle entry of an ordered list of data. If there are an even number of entries, then we take the average of the middle two values.

Notation:

μ : population mean

$\bar{x}$: sample mean


Class Example 1.4.2 *Ants on a sandwich*

The number of ants climbing on a piece of a peanut butter sandwich left on the ground near an anthill for a few minutes was measured 7 different times, and the results are: 43, 59, 22, 25, 36, 47, 19

- Calculate the mean number of ants.
- Calculate the median number of ants.
- Suppose one of the sandwich bits was extremely appealing to the ants, and instead of 59, the number of ants for that bit was actually 159, giving the 7 values as: 43, 159, 22, 25, 36, 47, 19. Compute the mean and median for this new dataset.
- Which statistic (mean or median) is impacted most by the outlier?

How skewness affects the mean and median

In a symmetric distribution, the mean and the median are approximately equal ($\text{mean} \approx \text{median}$). In a right-skew distribution, the mean is pulled towards the right tail and is larger than the median ($\text{mean} > \text{median}$). In a left-skewed distribution, the mean is pulled towards the left tail and is smaller than the median ($\text{mean} < \text{median}$).

When to use the mean vs. the median

As we saw in the last example, the median is not impacted by extreme data values (outliers) like the mean is. We say that the median is resistant. Although the mean

is a more commonly used measure of center, the median may be preferred in cases with extreme values or a high degree of skew.

If the concern is the total or aggregate, the mean is the best. If the concern is a typical observation, the median is preferred, especially when the distribution is skewed or contains extreme values.

Measures of Spread

Knowing the center of a distribution is important, but it tells only part of the story. Two distributions can have the same center but look very different if one is tightly clustered and the other is widely dispersed. We need measures of *spread* (also called variability or dispersion).

A measure of *spread* tells us how much variability exists in the data.

The range

The simplest measure of spread is the *range*: the difference between the maximum and minimum values.

The *range* is the difference between the maximum and minimum of the data.

$$\text{Range} = \text{Maximum} - \text{Minimum}$$

While easy to compute, the range has a major weakness: it depends entirely on the two most extreme observations. A single extreme value can dramatically increase the range, giving a misleading impression of the overall variability. For this reason, we usually prefer the IQR or standard deviation.

The interquartile range (IQR)

The *interquartile range* measures the spread of the middle 50% of the data. To find the IQR, we need to first estimate Q_1 and Q_3 , the first and third quartiles. The first quartile, Q_1 , is the value which 25% of the data falls below; and the third quartile, Q_3 , is the value which 75% of the data falls below. To find Q_1 , we need to take the median of all the values below the median. To find Q_3 , we need to take the median of all the values above the median. The IQR is given by

The *IQR* (Interquartile range) is the difference between the 1st and 3rd quartiles of the data.

$$\text{IQR} = Q_3 - Q_1.$$

The five-number summary

A way to combine the information from the range, the IQR, and a measure of center is through the *five-number summary*. The five-number summary consists of the minimum, Q_1 , the median, Q_3 , and the maximum of the data set.

Class Example 1.4.3 *Ants revisited*

In the previous example, we found that for the 7 different times, the results were 19, 22, 25, 36, 43, 47, 59 (sorted in ascending order)

a) Find the five number summary for this data.

a) Find the IQR for this data

b) Find the range for this data

c) Find the range and IQR if the value 59 was actually 159?

d) Which statistic (range or IQR) is impacted most by the extreme value?

Variance and standard deviation

The **standard deviation** is the most commonly used measure of spread. The standard deviation gives an estimate for the average distance each observation is from the mean.

We start with the concept of a **deviation**: the distance of an observation from the mean.

$$\text{deviation of } x_i = x_i - \bar{x}$$

Some deviations are positive (observations above the mean) and some are negative (observations below the mean). If we simply averaged the deviations, the positives and negatives would cancel out, giving us zero. To avoid this, we *square* each deviation before averaging. The average of the squared deviations, using $n - 1$ in the denominator is the *variance* and is given by

$$\text{Variance} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1} = \frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1}$$

The *standard deviation* is the square root of the variance:

$$\text{Standard Deviation} = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

In this course, we will use σ^2 (pronounced “sigma squared”) to represent the population variance and s^2 to represent the sample variance. We will use σ (pronounced “sigma”) to represent the population standard deviation and s to represent the sample standard deviation. When the standard deviation is large, it means there is a lot of variability amongst the observations. When the standard deviation is small, it means there is a lot of consistency amongst the observations.

Looking back at the interest rate data with shaded bands marking one and two standard deviations from the mean. The standard deviation of 5.05 percentage points gives us a sense of how spread out the data are around the mean of 11.57%.

The *standard deviation* of a variable is a rough estimate of the typical distance of a data value from the mean.

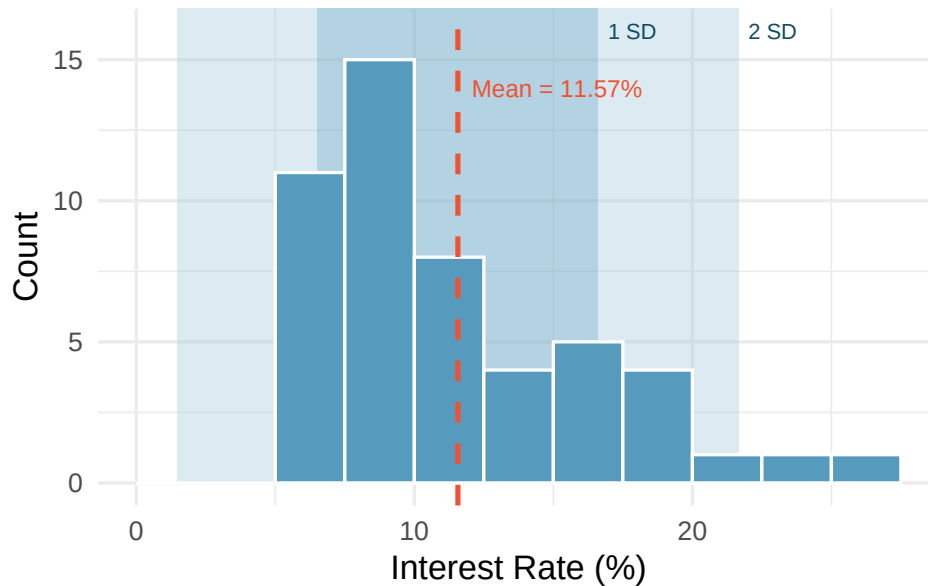
A *deviation* is the difference between an observation and the mean

The *variance* is the average of the squared deviations.

Notation:

σ : population standard deviation

s : sample standard deviation



The standard deviation tells us that interest rates typically deviate about 5 percentage points from the mean (11.57%).

Class Example 1.4.4 *Ants revisited*

- Find the standard deviation of the ants on a sandwich data.
- Find the standard deviation of the data if the value 59 was actually 159.

Outliers and Robust Statistics

Identifying outliers

An *outlier* is an observation that appears extreme relative to the rest of the data. A common rule for identifying outliers is the **1.5 IQR rule**: A data value, x_0 is considered to be an outlier if

$$x_0 < Q_1 - 1.5(IQR) \quad \text{or} \quad x_0 > Q_3 + 1.5(IQR).$$

An *outlier* is a data point that is unusual given the rest of the data.

Examining data for outliers serves several purposes:

- Identifying strong skew in the distribution
- Identifying possible data collection or data entry errors
- Providing insight into interesting or unusual properties of the data

Robust statistics

We call a statistic **robust** (or **resistant**) if extreme observations have little effect on its value.

We say that a statistic is *robust*, or *resistant*, if its is relatively unaffected by extreme values.

Class Example 1.4.5 *Resistant statistics*

Look back at your answers to the questions about ants on a sandwich.

a) Which measures of center (mean, median) are resistant?

b) Which measures of spread (IQR, range, standard deviation) are resistant?

Robust vs. non-robust statistics.

Robust (resistant to outliers)	Not robust (sensitive to outliers)
Median	Mean
IQR	Standard deviation
	Range

When a distribution is skewed or contains outliers, robust statistics (median and IQR) often give a more representative summary of the “typical” observation. When a distribution is roughly symmetric with no extreme outliers, the mean and standard deviation work well and carry additional mathematical advantages.

Box plots

So far in this class, we have used histograms and dot plots to represent data graphically. There is another very commonly used graphical summary called the *boxplot*. The boxplot gives us a graphical display of the *five-number summary* which consists of the minimum, Q_1 , median, Q_3 , and the maximum of a dataset.

The *boxplot* is a graphical display of the *five-number summary* which consists of the minimum, Q_1 , median, Q_3 , and the maximum of a dataset.

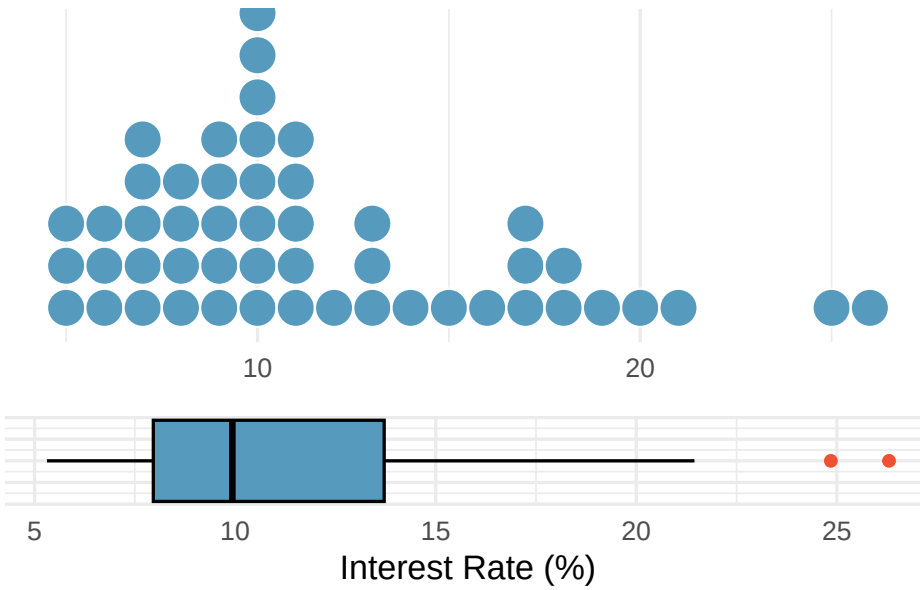
Constructing a box plot

The components of a box plot are:

1. *The box*: Spans from Q_1 (the 25th percentile) to Q_3 (the 75th percentile). This box contains the middle 50% of the data. The length of the box is the *interquartile range* ($IQR = Q_3 - Q_1$).
2. *The median line*: A line inside the box marks the median (the 50th percentile), which divides the data in half.
3. *The whiskers*: Lines extending from the box toward the smallest and largest observations that are *not* outliers. Specifically, the whiskers extend to the most extreme data points within $1.5 \times IQR$ of the box.
4. *Outlier points*: Any observations beyond $1.5 \times IQR$ from the edges of the box are plotted individually as dots. These are potential outliers.

The following gives a general sketch of a boxplot:

A box plot of interest rates shows: the left edge of the box at about 8%, the median line at about 10%, and the right edge of the box at about 14%. Two dots appear beyond the right whisker at about 25% and 26%.


Class Example 1.4.6 *Population of U.S. states*

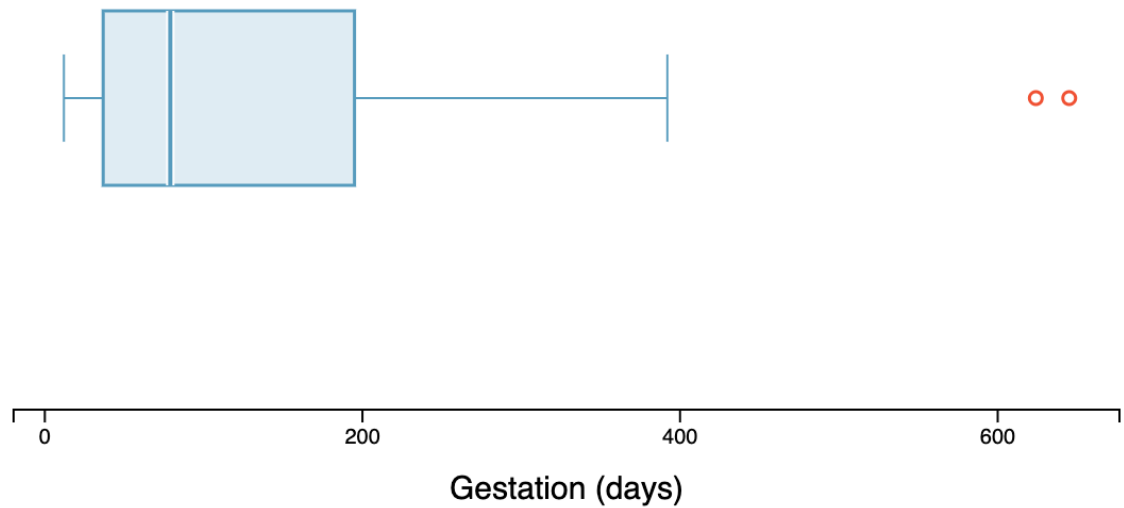
The following data represent the populations of 12 of the 50 U.S. states, in millions of people, in ascending order.

0.506	0.830	1.315	1.813	2.333	2.953
3.524	4.198	5.504	6.207	8.685	22.472

- Identify the five-number summary for the state populations.
-
- Draw a boxplot for the data. If there are any outliers, provide a justification as to why they are outliers and identify them on your boxplot.
-

Class Example 1.4.7 *Gross state product*

The figure below gives a boxplot of the gestation (in days) for 54 mammal species.



- Are there any outliers? If so, how many and estimate the values of the outliers.
- Estimate the median, IQR and range of the data.
- How would you describe the shape of this distribution?
- Do you expect the mean to be less than, approximately equal to, or greater than the median? Explain.

Describing a distribution

A complete description of a distribution should address three things: *shape*, *center*, and *spread*.

A good description of a distribution should always:

1. Name the *shape*: symmetric or skewed (and direction)
2. Give the *center*: report the mean and/or median with units
3. Give the *spread*: report the standard deviation and/or IQR with units
4. Note any *unusual features*: outliers, gaps, clusters

5. Use *context*: refer to the actual variable and its units, not just abstract numbers

A way to describe a distribution is as follows: “The distribution of commute times is right skewed, with a median of 22 minutes and an IQR of 15 minutes. A few commuters have commute times exceeding 90 minutes.”

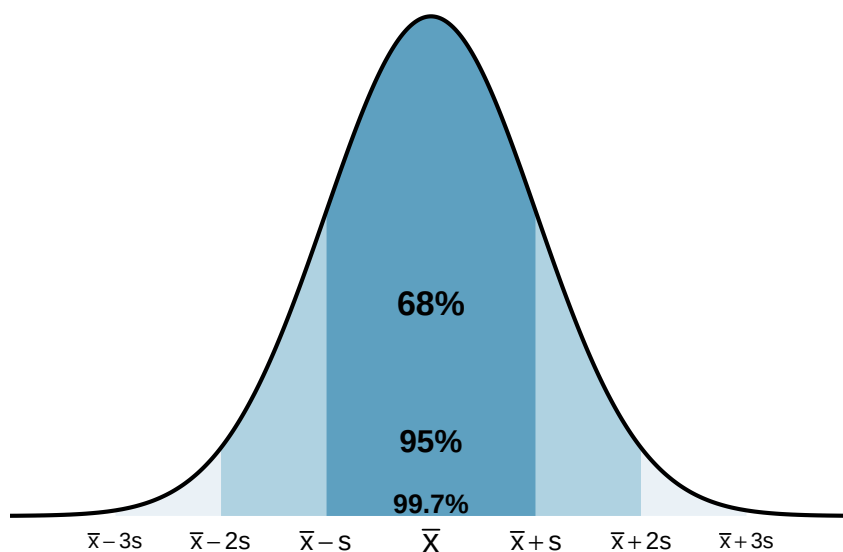
The Empirical Rule (68-95-99.7 rule)

For distributions that are approximately *bell-shaped*, the standard deviation provides a useful ruler. Statisticians call this particular bell shape the *normal distribution* — it comes up so often in statistics that it has its own name. We’ll study it formally later, but for now, just know that “bell-shaped” and “normal” mean the same thing.

The empirical rule.

For a roughly bell-shaped distribution:

- About **68%** of the data fall within 1 standard deviation of the mean: $(\bar{x} - s, \bar{x} + s)$
- About **95%** of the data fall within 2 standard deviations of the mean: $(\bar{x} - 2s, \bar{x} + 2s)$
- About **99.7%** of the data fall within 3 standard deviations of the mean: $(\bar{x} - 3s, \bar{x} + 3s)$



Class Example 1.4.8 *SAT scores*

The SAT scores (standardized test scores) are approximately bell-shaped with a mean of 1150 inches and a standard deviation of 150 inches. Use the empirical rule to estimate the range of scores for the middle 95% of students.

Summary

This chapter introduced the tools for exploring *quantitative* data — both visual and numerical.

- *Dot plots* show individual values for small datasets, *histograms* reveal the shape and density of a distribution, density plots provide smoothed alternatives, and *box plots* give a compact *five-number summary*.
- Distribution shape is described using symmetry (*left skewed*, *symmetric*, *right skewed*).
- The *mean* and *median* measure center, with the mean being the balancing point and the median being the middle value.
- The *standard deviation* and *IQR* measure spread, with the standard deviation roughly representing the typical deviation from the mean and the *IQR* representing the range of the middle 50%.
- The *median* and *IQR* are *robust* to outliers, while the mean and standard deviation are sensitive to extreme values.
- The *empirical rule* connects the standard deviation to the distribution shape for bell-shaped data: about 68%, 95%, and 99.7% of observations fall within 1, 2, and 3 standard deviations of the mean.

1.5 Relationships Between Variables

The preceding chapters focused on exploring one variable at a time. But the most interesting statistical questions involve **relationships between variables**. Does higher education lead to higher income? Is a drug more effective for one group than another? This chapter introduces tools for exploring relationships: side-by-side box plots and faceted histograms for comparing a quantitative variable across groups, and scatterplots for examining the association between two quantitative variables. We close with the critical distinction between association and causation.

Key Concepts

- Use a side-by-side graph to visualize a relationship between quantitative and categorical variables
- Examine a relationship between quantitative and categorical variables using comparative summary statistics

Technology for Chapter

StatLens Use the [Quantitative by Group](#) to compare distributions across groups with side-by-side box plots, histograms, and density overlays. Use the [Two Quantitative Variables](#) for scatterplots of two quantitative variables.

jamovi Use Analyses>Exploration>Descriptives, using the “Split by” input, to visualize distributions as histograms or box plots, and compute means, medians, standard deviations, and quartiles for any dataset. Use Analyses>Exploration>Scatterplot for scatterplots of two quantitative variables.

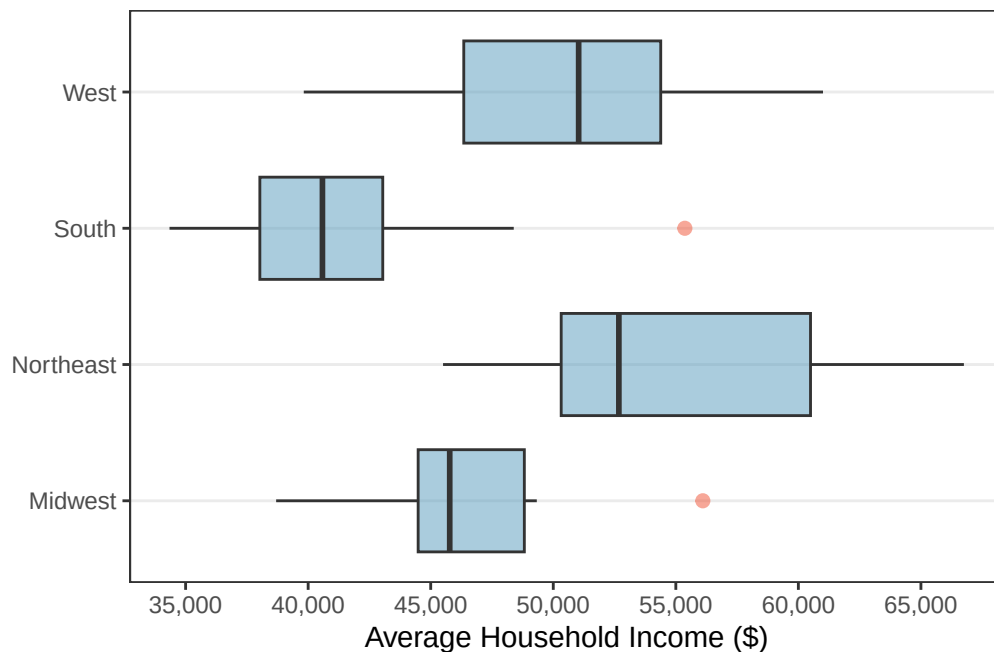
Comparing Groups

Some of the most interesting investigations involve comparing a quantitative variable across groups defined by a categorical variable. Sometimes when investigating the relationship between categorical and quantitative variables, it is helpful to plot them *side-by-side*.

We use *side-by-side* plots to visualize the relationship between a quantitative variable and a categorical variable.

Class Example 1.5.1 *Average household incomes broken down by U.S. geographical region*

The figure below shows a side-by-side boxplot of average U.S. household income broken down by geographical region of the U.S.



a) Use these side-by-side boxplots to discuss how the average U.S. household income differs by region.

b) Are there any outliers? If so, how many and estimate the values of the outliers.

c) For which region(s) would it be more appropriate to use the median instead of the mean to describe the average U.S. household income? Justify your answer.

Scatterplots

A *scatterplot* displays the relationship between two quantitative variables. Each observation is represented as a point, with one variable on the horizontal axis and the other on the vertical axis. When looking at a scatterplot, we should consider four different aspects of the plot:

A *scatterplot* is a graph of the relationship between two quantitative variables. If there are explanatory and response variables, we put the explanatory variable on the horizontal (x) axis and the response on the vertical (y) axis.

1. *Direction*: Is the association *positive* (both variables tend to increase together) or *negative* (one increases as the other decreases)?
2. *Form*: Is the relationship approximately *linear* (following a straight-line pattern) or *nonlinear* (curved)?
3. *Strength*: Is the association *strong* (points closely follow a pattern), *moderate*, or *weak* (points are loosely scattered)?
4. *Unusual observations*: Are there any *outliers* — points that deviate markedly from the overall pattern?

Example 5.2 Describing Associations

Consider a dataset of 50 loans, where we plot total income on the horizontal axis and loan amount on the vertical axis:



Describe the association seen in the scatterplot in Figure 2.

Associations vs. causation

We have explored the association between variables, but an association is not the same as causation. When two variables are associated, it means they tend to vary together. That does not necessarily mean one variable *causes* the other. There may be a *confounding variable* that is related to both, creating the illusion of a direct relationship.

A *confounding variable* is a third variable that is associated with both the explanatory and response variable, but not necessarily of interest in the study.

Example 5.3 GPAs and Breakfast

A study finds that students who eat breakfast tend to have higher GPAs than students who skip breakfast. Can we conclude that eating breakfast causes higher GPAs?

Summary

This chapter introduced tools for exploring relationships between variables.

- For comparing a *quantitative variable across groups* defined by a categorical variable, side-by-side box plots provide a compact comparison of centers and spreads, faceted histograms reveal full distributional shapes, and ridge plots overlay density curves for compact multi-group comparison.
- For *two quantitative variables*, scatterplots reveal the direction, form, strength, and unusual features of an association.
- We closed with the fundamental principle that *association does not imply causation* — confounding variables can create misleading associations, and establishing causation generally requires a randomized experiment.

2.1 Sampling Variability

So far in this course, you have learned how to collect data, visualize it, and compute summary statistics. But statistics is about more than describing the data you have — it is about using data to learn about a larger population. The central challenge is this: **every sample is different**. If you took another sample from the same population, you would get different results. This chapter introduces the foundational ideas that make statistical inference possible: the distinction between parameters and statistics, the concept of a sampling distribution, and the standard error as a measure of how much sample statistics vary. These ideas are the bridge from describing data to drawing conclusions about the world.

Key Concepts

- Distinguish between a population parameter and a sample statistic, recognizing that a parameter is fixed while a statistic varies from sample to sample
- Compute a point estimate for a parameter using an appropriate statistic from a sample
- Recognize that a sampling distribution shows how sample statistics tend to vary
- Recognize that statistics from random samples tend to be centered at the population parameter
- Estimate the standard error of a statistic from its sampling distribution
- Explain how sample size affects a sampling distribution

Parameters vs. Statistics

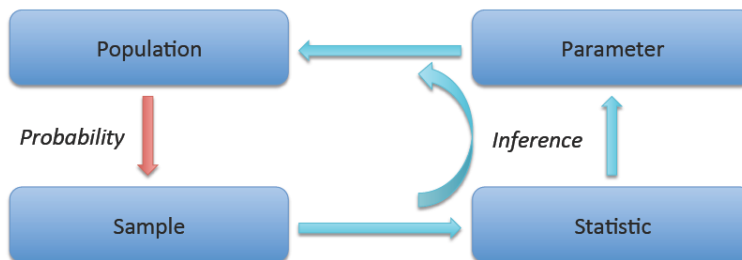
In Chapter 2, *statistical inference* was defined as the process of using a *statistic* calculated from a sample to estimate a *parameter*, or unknown value, from the population. The reason this cycle works is that when we gather data properly (i.e. use a random sampling technique), the statistics behave in a predictable way. Sample statistics are rarely, if ever, the exact population parameter, but they are usually close. If we only have one sample and we don't know the value of the population parameter, the sample statistic is our best estimate of the true population parameter (called a *point estimate* since it's a single number.)

Statistical inference is using the information in a sample to draw conclusions about the entire population.

A *parameter* is a number that describes some aspect of the population.

A *statistic* is a number that is computed from the data in a sample.

We use the sample statistic as a *point estimate* for the population parameter.



We use specific notation to distinguish parameters from statistics:

Quantity	Parameter (population)	Statistic (sample)
Proportion	p	$\hat{p}$
Mean	μ	$\bar{x}$
Standard deviation	σ	s
Difference in proportions	$p_1 - p_2$	$\hat{p}_1 - \hat{p}_2$
Difference in means	$\mu_1 - \mu_2$	$\bar{x}_1 - \bar{x}_2$

Class Example 2.1.1 *Parameter vs. Statistic*

For each of the following, determine whether the quantity described is a parameter or a statistic and give the proper notation.

- Average height of players on the 2024 Brazil World Cup team, using data from all 23 players on the roster.
- Average daily high temperature in La Crosse in June, based off of randomly selecting 100 June days over the last 20 years.
- Proportion of households in the U.S. who are renting their living accommodations using data from the 2020 Census.
- Proportion of UWL students who live on campus, using data from 30 students.

Sampling Distributions

To understand how far a statistic might be from its parameter, we need to understand what happens when we take many samples from the same population. A sampling distribution is the distribution of sample statistics computed for different samples of the same size drawn from the same population. The sampling distribution shows us how the sample statistic varies from sample to sample and gives us a way to see how close we can expect statistics to be to the true parameter value.

The *sampling distribution* shows how a statistic will vary from sample to sample.

Class Example 2.1.2 *Illustrating sampling distributions*

Go to StatLens at [?var:tool-url/](https://var-tool-url/) and click on Sampling Distributions under the Simulate Tab. We will use the Right-Skew population shape. We do not have access to the population, but we wish to estimate the (then unknown) population mean.

- What is the population mean (μ)?

- b) Use StatLens to select a sample size of 10. What was the sample mean ($\bar{x}$) of this data set? Is it close to the population mean?
-
- c) Keep taking samples of size 10, one by one. Visualize how the sample means of these samples change. Even though the sample means change from sample to sample, where do they appear to center?
-
- d) Now generate 1000 samples at once to get a fuller picture of the sampling distribution of the sample mean. Where is the distribution centered? What range do the sample means lie within?
-
- e) Reset the plot and now consider drawing samples of size 100 and exploring the sample mean. Begin with drawing samples one by one. Where do the sample means tend to center? Do they deviate much from this center?
-
- f) Now generate 1000 samples at once. Where is the sampling distribution centered? What range do the sample means lie within? How does this range compare to the range from part d)?
-
- g) What accounts for the differences in variability from part d) to part f)?
-

Facts about sampling distributions

For many common statistics, if the sample size is large enough and representative of the population (i.e. a random sample), the sampling distribution has three important features:

1. *Center*: The sampling distribution is centered at the population parameter. On average, the sample statistic equals the parameter.
2. *Spread*: The sampling distribution has a measureable spread that decreases as the sample size increases. Larger samples give more precise estimates.
3. *Shape*: For many statistics (e.g. $\bar{x}$ and $\hat{p}$), the sampling distribution is approximately bell-shaped (normal) when the sample size is large enough.

Sampling distributions tell us useful information about which sample statistics are likely to occur. Sample statistics that fall near the center of the sampling distribution are more likely to occur than those that fall far from the center. Understanding which values are likely is one of the primary goals in Statistics.

The *sampling distribution* is fundamentally different from the *data distribution*:

- The data distribution shows how individual observations vary.
- The sampling distribution shows how a statistic (like the mean or proportion)

varies from sample to sample.

Variability of sample statistics

When gathering data, there is a tradeoff between how large a sample is and the accuracy of the statistic calculated from the sample. This happens because with a larger sample size, we get a more complete picture of what the population looks like. So, as the sample size increases, we are more likely to see sample statistics that are closer to the true value of the population parameter (i.e. the variability of the sample statistics tends to decrease). We saw this in the last example where we explored the sampling distribution for the mean in StatLens.

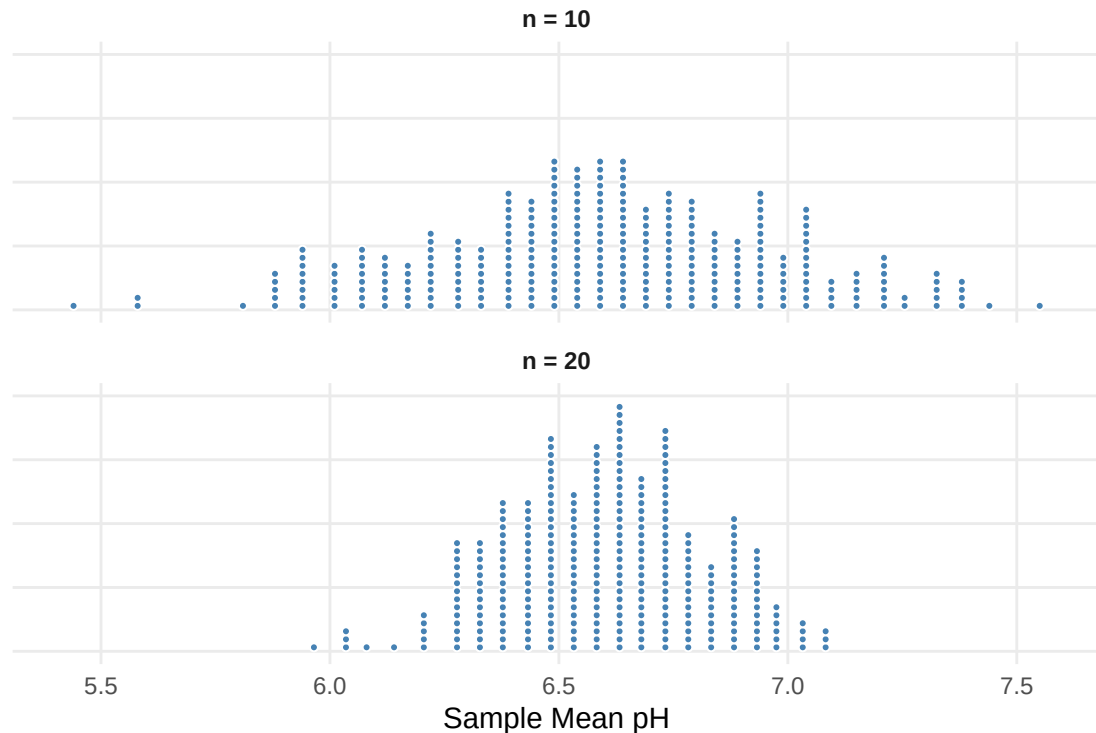
As the sample size *increases* the variability of a statistic will *decrease*.

Two factors control how much a sample statistic bounces around from sample to sample:

1. **Sample size (n):** Larger samples produce statistics that are closer to the parameter. This is intuitive — a sample of 1,000 tells you more about the population than a sample of 10.
2. **Population variability:** If the population is very homogeneous (everyone is similar), then any sample will look like any other sample, and the statistic won't vary much. If the population is very heterogeneous (lots of individual differences), then different samples can look quite different, and the statistic will vary more.

Class Example 2.1.3 Average pH levels in Florida lakes

The figure below shows two sampling distributions for the average pH level in Florida lakes for two different sample sizes.



- a) The difference between these two dotplots is that one comes from samples where $n = 10$ and the other comes from samples where $n = 20$. Identify which dotplot corresponds to each sample size.

- $n = 10$ Top Bottom
 - $n = 20$ Top Bottom
- b) What does each dot in the dotplots represent?

c) For which sample size are we more likely to find an average pH greater than 7?
 Circle one: $n = 10$ $n = 20$

Standard Error

We need a single number that quantifies how much a statistic varies from sample to sample. That number is the *standard error*. It measures the typical distance between a sample statistic and the population parameter. The standard error of a statistic is very important because it gives us a sense of how certain we are that our statistic is accurate, so it is important that we can easily estimate the standard deviation of the sampling distribution. When we use a computer to generate a sampling distribution, it is very easy to have the computer calculate the standard deviation of the sampling distribution. However, there is also a graphical technique we can use to estimate the standard deviation of a distribution.

In a sampling distribution the *standard error* is the standard deviation of the statistic.

Using some ideas from calculus, we can say the first standard deviation on either side of the mean occurs at the *inflection points*. Essentially, when tracing along the distribution from left to right, when your pen stops making an up-cup (a cup that can hold water) and starts making a down-cup (cannot hold water), that occurs approximately one standard deviation from the mean. The same occurs on the other side of the mean.

An *inflection point* is a point on a distribution where the up-cup becomes the down-cup.

Class Example 2.1.4 Estimating standard errors from a dot plot

Refer back to Figure 2 from Example 6.3. Use the method above to estimate the standard error for each data set.

Do not confuse the standard error with the standard deviation of the data.

- The *standard deviation* (s or σ) measures how much individual observations vary around the mean.
- The *standard error* (SE) measures how much a sample statistic varies around the population parameter from sample to sample.

Why the standard error matters

The standard error is the engine of statistical inference. Here is why:

- *Confidence intervals* are built by going a certain number of standard errors above and below the point estimate. A rough 95% confidence interval is:

$$\text{statistic} \pm 2 \times SE$$

- *Hypothesis tests* compare an observed statistic to what we would expect under the null hypothesis. The comparison is made in units of standard errors: "Is the observed result more than 2 standard errors from the null value?"
- *Sample size planning* uses the standard error to determine how many observations are needed to achieve a desired level of precision.

In the coming chapters, we will use simulation to *estimate* the standard error. In later chapters (Unit 3), we will use mathematical formulas to *calculate* it directly.

Words of caution

The statistics calculated from a sample are only reliable if we collect our data *correctly* and have a *representative* sample. When we do not use proper techniques to gather the data, then the statistics we calculate will be biased.

Summary

- A *population parameter* is a fixed but unknown numerical summary of the population. A *sample statistic* (point estimate) is computed from data and serves as our best guess of the parameter.
- Different samples from the same population yield different statistics. This is *sampling variability* — it is not a mistake, but a fundamental feature of working with samples.
- The *sampling distribution* of a statistic describes how the statistic varies across all possible samples of a given size. It is typically centered at the parameter, and its spread decreases as sample size increases.
- The *standard error* is the standard deviation of the sampling distribution. It measures the precision of the statistic as an estimate of the parameter.
- The standard error is not the same as the standard deviation of the data. The standard deviation measures variability of individual observations; the standard error measures variability of a statistic.
- Larger samples produce smaller standard errors, meaning more precise estimates.

2.2 Randomization Tests

Statistical inference is primarily concerned with understanding and quantifying the uncertainty of parameter estimates. While the equations and details change depending on the setting, the foundations for inference are the same throughout all of statistics. In this chapter, we introduce the **hypothesis testing framework** — a formal method for evaluating competing claims about a population using data. We learn how to build **randomization distributions** by shuffling data under the assumption that there is no effect, and we use these distributions to compute ***p*-values** that quantify the strength of the evidence.

Key Concepts

- Recognize when and why statistical tests are needed
- Specify null and alternative hypotheses based on a question of interest, defining relevant parameters
- Interpret a *p*-value as the probability of results as extreme as the observed results happening by random chance, if the null hypothesis is true
- Estimate a *p*-value from a randomization distribution
- Comparatively quantify strength of evidence using *p*-values
- Distinguish between one-tailed and two-tailed tests in estimating *p*-values
- Make a formal decision in a hypothesis test by comparing a *p*-value to a given significance level
- State the conclusion to a hypothesis test in context
- Interpret Type I and Type II errors in hypothesis tests
- Recognize a significance level as measuring the tolerable chance of making a Type I error
- Recognize that statistical discernibility is not always the same as practical significance

Motivation: Cereal example

Consider the following scenario: A sample of 80 four- to six-year olds participated in a study in which they were asked to taste two cereals, both called “Healthy Bits” (in



Boxes for the cartoon marketing cereal study.

random order). The cereal in each box was exactly the same, but the boxes differed in the fact that one of the boxes had pictures of cartoon penguins and the other did not.

- a) If the cartoons make no difference, because after all the cereal is the same in each box, how many of the 80 children do you expect to pick the cartoon box? What sample proportion is this?

- b) Suppose that in this study, 52 of the children chose the box with cartoons. What sample proportion is this? Does this lead you to believe that there was a preference toward the cartoon box, or would you expect this sample proportion to occur by chance even if there was no preference? (We will use StatKey to explore the sampling distribution for the proportion to determine what values are likely to occur.)

- c) Using the simulation from part b), what sample proportions do we expect to see if there is no preference toward the cartoons?

The hypothesis testing framework

There is almost always variability in data — one dataset will not be identical to a second dataset even if they are both collected from the same population using the same methods. However, quantifying the variability in the data is neither obvious nor easy. Answering the question “*how* different is one dataset from another?” is not trivial.

The question at the heart of hypothesis testing is: *Could the pattern we observe in the data have arisen from chance alone, or is there something real going on?*

Hypothesis testing

A *hypothesis test* is a statistical technique used to evaluate competing claims using data. The process involves:

1. *State the hypotheses*: Define a null hypothesis (H_0) and an alternative hypothesis (H_A).
2. *Collect data*: Gather evidence that bears on the hypotheses.
3. *Assess the evidence*: Determine how likely the observed data would be if the null hypothesis were true.
4. *Draw a conclusion*: Decide whether the evidence is strong enough to reject the null hypothesis.

If the null hypothesis and the data notably disagree, then we reject the null hypothesis in favor of the alternative hypothesis. There are many nuances to hypothesis testing, and we will discuss these ideas throughout this chapter and those that follow.

A *hypothesis test*, or *statistical test* uses data from a sample to assess a claim about a population.

Null and alternative hypotheses

In the first step of a hypothesis test, we must state the hypotheses. The hypotheses consist of the *null hypothesis* and the *alternative hypothesis*. The null hypothesis, denoted H_0 , often represents a skeptical perspective or a claim of “no difference” or “no effect.” It assumes that any observed pattern in the data is due to chance alone. The alternative hypothesis, denoted H_A , represents the claim under investigation, typically that there is a real difference, effect, or relationship. The alternative hypothesis is usually the reason the research was conducted in the first place.

Statistical tests, and in particular hypothesis tests, behave in much of the same way as the US justice system. The US court system considers two possible claims about a defendant: they are either innocent or guilty. We can set up these claims in a hypothesis framework. The skeptical perspective (null hypothesis) is that the person is innocent until evidence is presented that convinces the jury the person is guilty (alternative hypothesis).

Usually, a *null hypothesis*, notated H_0 , is a claim that there really is “no effect” or “no difference”, or represents the status quo.

An *alternative hypothesis*, notated H_A , is the claim for which we seek significant evidence, and is established by observing evidence that contradicts the null hypothesis and supports the alternative hypothesis.

H_0 : The defendant is innocent.

H_A : The defendant is guilty.

Caution

Notice that if a jury finds a defendant *not guilty*, this does not necessarily mean the jury is confident in the person’s innocence. They are simply not convinced of the alternative. This is also the case with hypothesis testing: *even if we fail to reject the null hypothesis, we do not accept the null hypothesis as truth*. Failing to find evidence in favor of the alternative is not the same as finding evidence that the null hypothesis is true.

Sets of common hypotheses

The first set of statistical hypotheses that we will investigate in this class are for population parameters and differences in population parameters. The hypotheses have the following forms:

1. Single Proportion (p_0 is the proportion claimed or status quo value):

- Null hypothesis:

$$H_0 : p = p_0$$

- Alternative hypothesis (pick one):

$$H_a : p \neq p_0$$

$$H_a : p > p_0$$

$$H_a : p < p_0$$

2. Single Mean (μ_0 is the average claimed or status quo value):

- Null hypothesis:

$$H_0 : \mu = \mu_0$$

- Alternative hypothesis (pick one):

$$H_a : \mu \neq \mu_0$$

$$H_a : \mu > \mu_0$$

$$H_a : \mu < \mu_0$$

3. Difference in Proportions (p_1 and p_2 are the population proportions for each group):

- Null hypothesis:

$$H_0 : p_1 - p_2 = 0$$

- Alternative hypothesis (pick one):

$$H_a : p_1 - p_2 \neq 0$$

$$H_a : p_1 - p_2 > 0$$

$$H_a : p_1 - p_2 < 0$$

4. Difference in Means (μ_1 and μ_2 are the population averages for each group):

- Null hypothesis:

$$H_0 : \mu_1 - \mu_2 = 0$$

- Alternative hypothesis (pick one):

$$H_a : \mu_1 - \mu_2 \neq 0$$

$$H_a : \mu_1 - \mu_2 > 0$$

$$H_a : \mu_1 - \mu_2 < 0$$

Alternative hypotheses containing “>” or “<” are called *one-sided* alternatives and alternative hypotheses containing “≠” are referred to as *two-sided* alternatives. One-sided alternatives specify a particular direction for the alternative, while two-sided alternatives do not.

Class Example 2.2.1 *Defining statistical hypotheses*

For each of the following scenarios, state the null and alternative hypotheses with the relevant parameter(s).

- a) We are testing to see if the average income is different in eastern states than western states.

- b) We are testing to see if the average income is greater in eastern states than western states.

- c) We are testing to see if a particular coin is weighted (i.e., not fair).

- d) We are testing to see if the average rent in La Crosse is less than \$850.

- e) We are testing to see if children have a preference for the cereal box with cartoons (above).

Case study: Opportunity cost

How rational and consistent is the behavior of the typical American college student? Here, we explore whether reminding students that money not spent now can be spent later causes them to be thriftier.

The participants were 150 students, all given a scenario about purchasing a video on sale for \$14.99. Half (the control group) were given these options:

- A. Buy this entertaining video.
- B. Not buy this entertaining video.

The other half (the treatment group) saw a slightly modified option (B):

- A. Buy this entertaining video.
- B. Not buy this entertaining video. Keep the \$14.99 for other purchases.

In this study, we define a “success” as a student who chooses *not* to buy the video and want to compare the proportion of students who do not buy the video in the control group compared to the treatment group.

Class Example 2.2.2 Opportunity cost

Using the opportunity cost case study, answer the following questions:

- a) Is this an observational study or an experiment? How does the type of study impact what can be inferred from the results?
-
- b) State the null and alternative hypotheses for this study.
-

The results are summarized below:

	Buy video	Not buy video	Total
Control	56	19	75
Treatment	41	34	75
Total	97	53	150

Class Example 2.2.3 Opportunity cost, revisited

Compute the difference in the proportion of choosing not to buy the video.

Building a randomization distribution

What would it mean if the null hypothesis were true? It would mean that each student would make the decision on whether to buy the video without regard to the change in the second option, (B). The 97 purchases and 53 non-purchases were determined by factors other than the verbiage of the options, and the difference we observed was just bad luck in how the files happened to be assigned.

Suppose the students' decision were independent of the phrasing of the choices. Then, if we conducted the experiment again with a different random assignment of choices to the students, differences in the proportions of the choices would be based only on random fluctuation in decisions. We can perform this *randomization*, which simulates what would have happened if the students' decisions had been independent of the phrasing but we had distributed the phrases differently.

The card-shuffling metaphor

Think of the data as 150 cards:

- 97 red cards (buy video)
- 53 white cards (not buy video)

In the actual study, 75 cards were dealt to the 'control' pile and 75 to the 'treatment' pile. Under the null hypothesis, the color of each card (whether to buy the video) has nothing to do with which pile it lands in.

To simulate what the null hypothesis looks like, we:

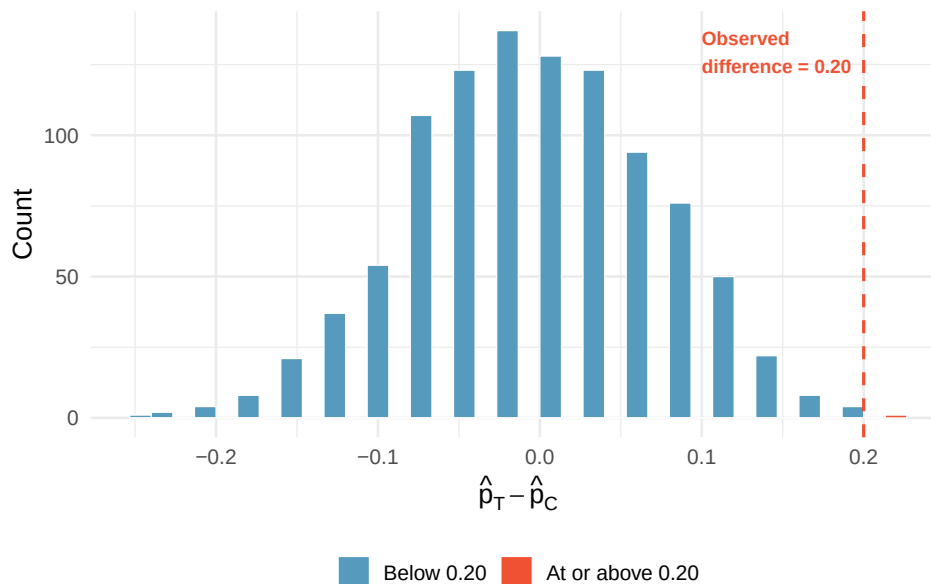
1. *Thoroughly shuffle* all 150 cards together.
2. *Deal 75* into a 'control' pile and 75 into a 'treatment' pile
3. *Count* the number of white (not buy video) in each pile and compute the difference in proportions.

This gives us one simulated difference *under the null hypothesis*, a difference that arose purely from the random shuffle, not from the phrasing.

Many simulations

One simulation is not enough. We need to repeat the shuffle-and-deal process many times to build up a picture of what chance alone looks like. Using a computer, we can do this hundreds or thousands of times creating a *randomization distribution* showing how likely different statistics are to occur if H_0 is true.

A randomization distribution approximates a sampling distribution from a population where the null hypothesis is true.



Under the null hypothesis, we would observe a difference of at least +20% about 0.6% of the time. That is very rare! We conclude the data provide strong evidence there is a treatment effect: reminding students that they could spend money later lowers their chance of making a purchase. Because this is an experiment (with

random assignment), we can make a causal claim.

P-Values

When we claim that we have evidence of our alternative hypothesis, the underlying question we are asking is

How unusual is it to see a sample statistics as extreme as the sample statistic we observed, if H_0 is true?

In statistics, we use the p - *value* to answer this question. The p -value is the proportion of the randomization distribution that is as extreme (or more extreme than) over observed statistic. Since the randomization distribution is built assuming the null hypothesis is true, the p -value is also defined as the probability of observing data at least as favorable to the alternative hypothesis as our current dataset, if the null hypothesis were true. We typically use a summary statistic of the data, such as a difference in the proportion, to compute the p -value. This summary statistic is called the *test statistic*.

If the test statistic lies in the tails of the distribution, we have strong evidence against H_0 in support of H_A . The smaller the p -value, the more evidence we have to reject the null hypothesis.

Randomization Distributions and p-values

To find the p -value using a randomization distribution, we:

- *For a one-sided alternative hypothesis:* Find the proportion of randomization samples that equal or exceed the original statistic in the direction (tail) indicated by the alternative hypothesis.
- *For a two-sided alternative hypothesis:* Find the proportion of randomization samples in the smaller tail at or beyond the original statistic and then double the proportion to account for the other tail.

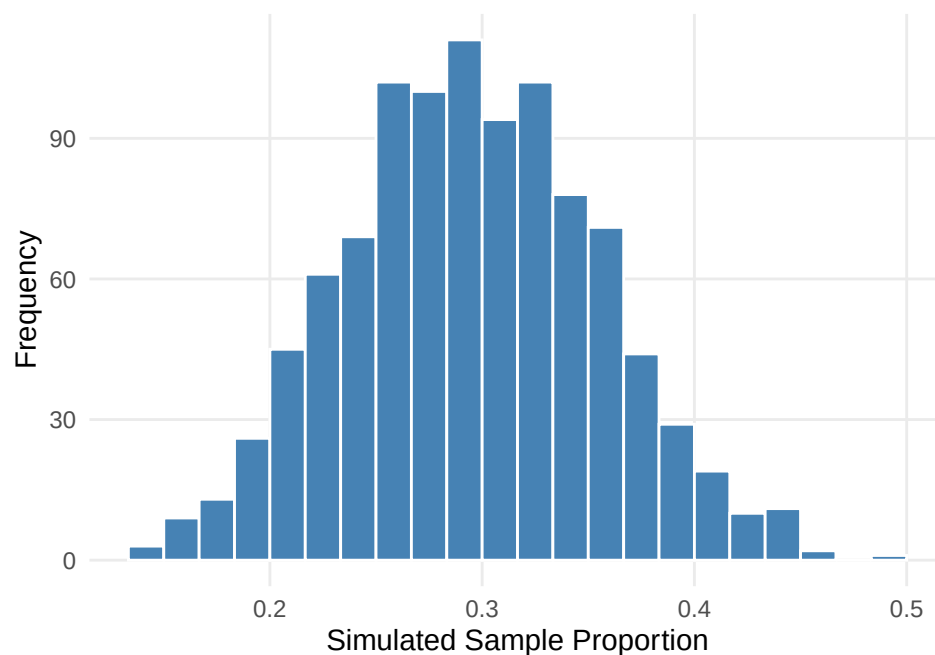
The p -value of the sample data in a statistical test is the probability of obtaining a sample statistic as extreme as (or more extreme than) the observed sample when the null hypothesis is *true*.

The *test statistic* is a summary statistic from the sample used in hypothesis testing.

In order to correctly estimate the p -value from a randomization distribution, we need to keep our alternative hypothesis in mind!

Class Example 2.2.4 *Estimating a p-value from a randomization distribution*

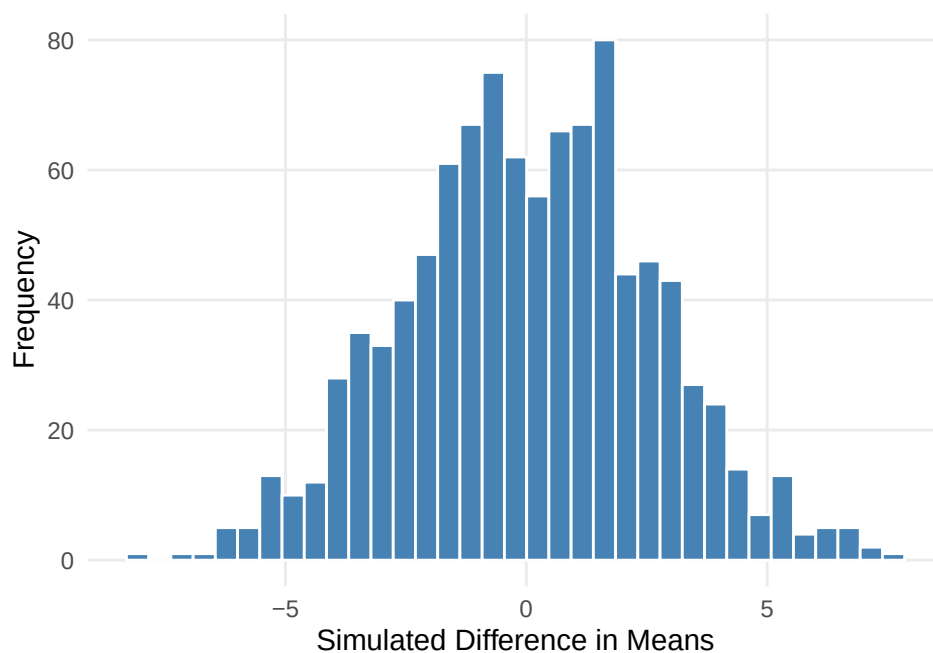
The figure below shows a randomization distribution for testing $H_0 : p = 0.3$ vs $H_a : p < 0.3$. In each case, use the distribution to decide which value is closer to the p -value for the observed sample proportion. There are 1000 randomizations given in the figure below.



- a) Why is the distribution centered at 0.3?
-
- b) The p -value for $\hat{p} = 0.15$ is closest to: 0.04 or 0.40?
-
- c) Which proportion, $\hat{p} = 0.20$ or $\hat{p} = 0.10$, provides the strongest evidence to suggest that H_0 is not true? Explain.
-

Class Example 2.2.5 *Estimating a p -value from a randomization distribution...continued*

The figure below shows a randomization distribution for testing $H_0 : \mu_1 - \mu_2 = 0$ vs $H_a : \mu_1 - \mu_2 \neq 0$. The statistic used for each sample is $D = \bar{x}_1 - \bar{x}_2$. In each case, use the distribution to decide which value is closer to the p -value for the observed sample difference in averages. There are 1000 randomizations given in the figure below.



- a) The p -value for $D = \bar{x}_1 - \bar{x}_2 = -2.9$ is closest to: 0.01 or 0.25?
-
- b) The p -value for $D = \bar{x}_1 - \bar{x}_2 = 1.2$ is closest to: 0.30 or 0.60?
-
- c) Of the two values of D above, which provides strongest evidence to suggest that H_0 is not true? Explain.
-

Class Example 2.2.6 *Migraine Acupuncture Trial*

Can acupuncture relieve migraine pain? Allais et al. (2011) conducted a randomized controlled study where 89 individuals who suffered from migraines were randomly assigned to receive either acupuncture (treatment, $n = 46$) or placebo acupuncture (control, $n = 46$). The placebo acupuncture involved placing needle at non-acupuncture-point locations. Researchers recorded whether each patient was pain-free 24 hours after treatment. The full data are in the data set *Migraine Acupuncture Trial* on StatLens.

- a) What are the null and alternative hypotheses for this study?
-
- b) What was the sample statistic for these data?
-
- c) What is the probability that the sample results (or those more extreme) occurred by random chance,

assuming that smiling had no effect?

d) How would your answer to c) change if the alternative hypothesis was two-sided?

Class Example 2.2.7 *Support for the death penalty*

In 1980 and again in 2021, a Gallup poll asked a random sample of 1000 US citizens people “Are you in favor of the death penalty for a person convicted of murder?”. In 1980, the proportion saying yes was 0.66. In 2021, it was 0.54. Does this data provide evidence that the proportion of US citizens favoring the death penalty was higher in 1980 than it was in 2021? Using p_1 for the proportion in 1980 and p_2 for the proportion in 2021:

a) State the null and alternative hypotheses

b) What is the sample statistic?

c) To create the randomization distribution, what do we have to assume?

d) What is the p -value for this study?

Class Example 2.2.8 *Cereal study revisited*

Recall the motivating cereal study. We were trying to determine if children had a preference for the cereal in the box with the cartoon. In the study 52 out of 80 children chose the box with cartoons. Use StatLens to find the p -value for this scenario.

Interpreting the p -value

- A *small p -value* (e.g., below 0.05) means that the observed result would be rare if the null hypothesis were true. This is evidence *against* the null hypothesis.
- A *large p -value* (e.g., above 0.10) means that the observed result is not particularly unusual under the null hypothesis. The data do not provide strong evidence against the null hypothesis.
- The p -value is *not* the probability that the null hypothesis is true. It is the probability of the observed data (or more extreme data) *given that* the null hypothesis is true.

Caution

A common mistake is to interpret the p -value as the probability that the null hypothesis is true (e.g., “there is a 2% chance that sex has no effect on promotions”). This is incorrect. The p -value tells us how *surprising* the data would be *if* the null hypothesis were true, not how likely the null hypothesis is to be true.

Statistical discernibility

Statistical tests begin under the assumption that the null hypothesis, H_0 , is true. In order to reject H_0 , sufficient evidence to its contrary must be collected – just like in order to convict a defendant, sufficient evidence must be presented to the judge and/or jury. What exactly is sufficient evidence? In our courtroom example, evidence must be provided so that decisions can be made “beyond a reasonable doubt.” In statistical tests, the data results must be unrealistic assuming H_0 is true. This means that we have found evidence to support H_A .

Decision framework

For hypothesis tests, there are two conclusions:

- We *reject* H_0 when the sample statistic is extreme assuming H_0 is true.
- We *do not reject* H_0 when the sample statistic is not too extreme assuming H_0 is true.

If we reject the null hypothesis, we say our finding is *statistically discernable*. Not rejecting the null hypothesis is *not* the same thing as accepting the null hypothesis. The null hypothesis is *never accepted* because the hypothesis test procedure is not designed to *rule in* H_0 , but only to rule it out. That is, evidence is never collected with the idea of favoring or proving the null hypothesis. Sample data are collected for the purpose of looking for evidence *against* H_0 . To arrive at the correct conclusion, the p -value must be small enough to be considered unusual.

What “statistically discernible” does and does not mean.

“Statistically discernible” means the p -value fell below the chosen discernibility level. It does *not* necessarily mean the effect is large or practically important. A very large sample can detect a tiny difference as “statistically discernible” even if the difference is too small to matter in practice.

Conversely, “not statistically discernible” does not mean there is no effect. It may simply mean the sample was too small to detect the effect.

How small is small enough?

We discussed how small p -values mean that a statistic is unlikely if H_0 is true. Now we need to think about some rules to determine when our p -value is small enough to say something is unusual assuming the null hypothesis is true.

The *discernibility level*, notated α , is the point at which a p -value is considered unusual. Some commonly used significance levels are $\alpha = 0.05$, $\alpha = 0.01$, or $\alpha = 0.10$. Any p -value that is smaller than the significance level is considered unusual. So, to use a p -value to determine whether or not the results are significant, you should use the following guidelines:

- When the p -value $< \alpha$, reject H_0 .
- When the p -value $\geq \alpha$, do not reject H_0 .

When we make a formal decision, we have two options:

- Reject H_0
- Do not reject H_0

If we reject the null hypothesis, we say our finding is *statistically discernable*.

We use the *discernibility level*, α , to decide the cutoff for our p -value to be significant.

Common values for α : 0.05, 0.01, 0.10

We reject H_0 when p -value $< \alpha$

We do not reject H_0 when p -value $\geq \alpha$

Class Example 2.2.9 Red wine and weight loss

Resveratrol, an ingredient in red wine and grapes, has been shown to promote weight loss in rodents. A recent study (*Science Daily*, 2010) investigates whether the same phenomenon holds true in primates. A sample of six

lemurs had their resting metabolic rate, body mass gain, food intake, and locomotor activity measured for one week prior to resveratrol supplementation and then the four indicates were measured again after treatment. For each of the following, use the p -value to make the appropriate conclusion (reject or do not reject) for the hypothesis tests. Use $\alpha = 0.05$ for each of your decisions.

a) In a test to see if mean resting metabolic rate is higher after treatment (p -value of 0.013).

b) In a test to see if mean body mass gain is lower after treatment (p -value of 0.007).

c) In a test to see if mean locomotor activity is affected by treatment (p -value of 0.980).

Interpreting your results

After you have made a decision to reject H_0 or not to reject H_0 , you need to interpret your conclusion in the context of the study. We always write our interpretation in the context of H_A .

A framework for interpreting your decision is as follows (fill in the blanks with the context of the study):

At $\alpha =$ (discernability level), we (do/do not) have evidence that (alternative hypothesis in words).

Class Example 2.2.10 Red wine and weight loss revisited

For each of the following, provide an interpretation of the conclusion in the context of the original study.

a) In a test to see if mean resting metabolic rate is higher after treatment (p -value of 0.013).

b) In a test to see if mean body mass gain is lower after treatment (p -value of 0.007).

c) In a test to see if mean locomotor activity is affected by treatment (p -value of 0.980).

Type I and Type II errors

Every hypothesis test ends with one of two decisions: reject H_0 or fail to reject H_0 . And the truth is one of two realities: H_0 is true or H_A is true. Combining these gives four scenarios:

Two of the four outcomes are correct decisions:

- Rejecting H_0 when H_A is true — we correctly detected the real effect.
- Failing to reject H_0 when H_0 is true — we correctly avoided a false alarm.

A *Type I error* occurs when we reject H_0 , but it is true (e.g., convict an innocent person).

A *Type II error* occurs when we do not reject H_0 , but it is false (e.g., let a guilty person go free).

Table 1: Four possible outcomes of a hypothesis test.

	H_0 is true	H_A is true
Reject H_0	Type I error	Correct decision
Fail to reject H_0	Correct decision	Type II error

The other two outcomes are errors:

- *Type I error* (upper-left): We rejected H_0 but it was actually true. False alarm.
- *Type II error* (lower-right): We failed to reject H_0 but H_A was actually true. Missed detection.

Class Example 2.2.11 *Red wine and weight loss revisited*

For each of the following, identify which kind of error could have occurred (Type I or Type II) using $\alpha = 0.05$.

a) In a test to see if mean resting metabolic rate is higher after treatment (p -value of 0.013).

b) In a test to see if mean body mass gain is lower after treatment (p -value of 0.007).

c) In a test to see if mean locomotor activity is affected by treatment (p -value of 0.980).

What can cause these errors?

The two types of errors that can be made both involve the hypothesis test leading to the wrong conclusion about the null hypothesis. This can happen because the process of random sampling does not give complete or perfect information about the population, and it is always possible to draw a rare sample just by chance. Recall that we have two options when making statistical inferences: *reject H_0* or *do not reject H_0* . It is important to carefully consider what these decisions could mean.

- Decision: Reject H_0 . The following outcomes are possible:
 1. You are making the correct decision; H_0 is not true.
 2. You are making a Type I error; the sample statistic was so far from the hypothesized value just by chance.
 3. Your sampling process or experiment was poor; the sample statistic is biased.
- Decision: Do not reject H_0
 1. You are making the correct decision; H_0 is true.
 2. You are making a Type II error; the sample statistic was so close from the hypothesized value just by chance.
 3. Your sampling process or experiment was poor; the sample statistic is biased.

Balancing out the errors

The discernability level, α , is the probability we make a Type I error. If we can choose the discernability level and it gives the chance of an error being made, why

The discernability level, α , is the probability we make a Type I error.

not always make the discernability level really, really low? This is because there are trade-offs between controlling for Type I and Type II errors, and balancing the risk of these errors is important.

1. Setting α to be very small makes it difficult to make a Type I error, but it also makes it easier to make a Type II error. Conversely, setting α to be large increases the making a Type I error, but it also reduces the risk of making a Type II error.
2. Since a Type I error is usually thought to be more serious than a Type II error, we typically use a low α = Type I error probabilities like 0.05, 0.1, or 0.01
3. We can reduce the chance of a Type II error without impacting the risk of a Type I error by increasing the sample size.
4. Statistical *power* tells us how likely we are to reject H_0 when we should (so we want *higher* power!), and it is related to the risk of a Type II error by $\text{power} = 1 - \text{chance of Type II error}$.
5. Calculating the probability of Type II error and power are fairly complicated for most cases, so we won't discuss that in this class. But understanding the relationships between these errors is important.

The *power* of a test tells us how likely we are to reject H_0 when we should.

The *power* of a test tells us how likely

Class Example 2.2.12 *Errors in a drug study*

A pharmaceutical company is researching a new anti-anxiety drug. Marketing a new drug is costly, and the company only makes a profit when the drug is effective for over 80% of patients. A random sample of 60 participants are placed on the drug and the number of participants that respond favorably to the drug will be recorded.

- a) Give the null and alternative hypothesis that you would use for determining if the drug should be marketed.
- b) Describe in words what a Type I error would mean in the context of this study.
- c) Describe in words what a Type II error would mean in the context of this study.
- d) Suppose the significance level is set at $\alpha = .05$, but the researchers find that this produces a Type II error rate of 0.4 and this is too high for their liking. List two ways they could reduce the Type II error rate.
- e) Given a Type II error rate of 0.4, what is the power of the test?

The randomization test procedure: a summary

We can summarize the randomization test procedure as follows:

1. *Frame the research question in terms of hypotheses.* The null hypothesis (H_0) usually represents no difference or no effect. The alternative hypothesis (H_A)

represents the claim of interest.

2. *Collect data.* If the research question focuses on associations but not causation, use an observational study. If you want to establish a causal connection, use an experiment with random assignment.
3. *Model what would happen under the null hypothesis.* Shuffle (randomize) the data to break any association between the explanatory and response variables. Compute the test statistic for each shuffle. Repeat many times to build the *randomization distribution* (also called the null distribution).
4. *Compute the p-value.* Determine what proportion of the randomization distribution is at least as extreme as the observed test statistic.
5. *Draw a conclusion.* If the p-value is small (less than α), reject H_0 and conclude there is evidence for H_A . If the p-value is not small, fail to reject H_0 . Write the conclusion in context, in plain language.

Summary

- A *hypothesis test* evaluates two competing claims: the null hypothesis (H_0), which represents no effect or no difference, and the alternative hypothesis (H_A), which represents the claim under investigation.
- A *randomization test* (or permutation test) assesses the null hypothesis by shuffling the data to simulate what would happen if there were truly no effect. The collection of simulated test statistics forms the *randomization distribution*.
- The *test statistic* is the summary value computed from the data (e.g., a difference in proportions) that we use to evaluate the hypotheses.
- The *p-value* is the probability of observing a test statistic at least as extreme as the one observed, assuming the null hypothesis is true. It quantifies the strength of the evidence against H_0 .
- We reject H_0 when the p-value is less than the *discernibility level* α (commonly 0.05). This is called a *statistically discernible* result.
- A *one-sided test* looks for effects in a specified direction; a *two-sided test* looks for effects in either direction.
- Failing to reject H_0 is not the same as proving H_0 is true.
- Every hypothesis test ends in one of four outcomes, organized by the *decision table*: two correct decisions and two errors.
- A *Type I error* rejects a true H_0 (a false alarm); its probability is the discernibility level α , which we choose.
- A *Type II error* fails to reject H_0 when H_A is true (a missed detection); its probability is β , which we do not directly control.
- Statistical discernibility does not imply practical importance.

2.3 Bootstrap Confidence Intervals

*In the sampling variability chapter, we saw that sample statistics vary from sample to sample. This variability is not a nuisance, it's the key to quantifying uncertainty. In this chapter, we learn **bootstrapping**: a simulation-based method for constructing confidence intervals that requires no formulas and no assumptions about the shape of the population.*

Key Concepts

- Construct a confidence interval for a parameter based on a point estimate and a margin of error
- Use a confidence interval to recognize plausible values of the population parameter
- Construct a confidence interval for a parameter given a point estimate and an estimate of the standard error
- Interpret (in context) what a confidence interval says about a population parameter
- Recognize that a bootstrap distribution tends to be centered at the value of the original statistic
- Use technology to create a bootstrap distribution
- Estimate the standard error of a statistic from a bootstrap distribution
- Construct a 95% confidence interval for a parameter based on a sample statistic and the standard error from a bootstrap distribution
- Construct a confidence interval based on the percentiles of a bootstrap distribution
- Explain how the width of an interval is affected by the desired level of confidence and the sample size
- Recognize when it is appropriate to construct a bootstrap confidence interval using percentiles or the standard error

The confidence interval

From our discussion about sampling distributions and bootstrap distributions, recall that sample statistics vary from sample to sample. Any time we give a point estimate for a population parameter, we want to describe that variability. This is commonly done using an *interval estimate* or a range of values that are plausible for the population parameter. One common form of an interval estimate is

$$\text{point estimate} \pm \text{margin of error}$$

The *margin of error* reflects the precision of our statistic as a point estimate, and it indicates how far out we should go to obtain a range of plausible values for the population parameter.

An *interval estimate* gives a range of plausible values for a population parameter, commonly formed using *point estimate* $\pm$ *margin of error*.

Margin of error is a number that reflects the precision of the sample statistic as a point estimate for the parameter.

Class Example 2.3.1 *Adopting a child in the US*

A survey of 1000 American adults conducted in January 2013 stated that “44% say it’s too hard to adopt a child in the US.” The survey goes on to say that “The margin of error is ± 3 percentage points with a 95% level of confidence.”

a) What is the relevant sample statistic? Give appropriate notation and the value of the statistic.

b) What population parameter are we estimating with this sample statistic?

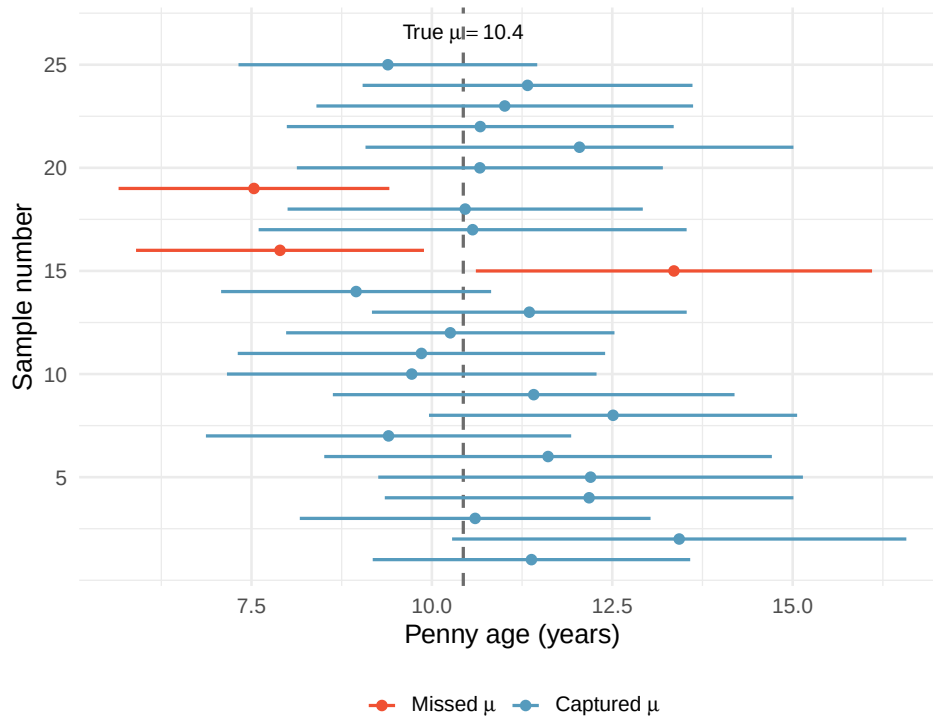
c) Use the margin of error to give a confidence interval for the estimate. Is 0.42 a plausible value of the population proportion? Is 0.50 a plausible value?

d) If we took a sample of 100 instead, do you expect the margin of error to increase or decrease?

using $\text{Statistic} \pm 2 \cdot SE$.

Typically in statistics, we refer to our interval estimates as *confidence intervals*. A confidence interval is an interval estimate along with a *confidence level*, denoted α , that tells us how successful our intervals would be at capturing the true (but unknown) parameter if we repeated our sampling thousands of times.

To illustrate the coverage concept, imagine we could treat our 648 pennies as if they were the entire population (with true mean $\mu \approx 10.4$ years). We then draw 25 smaller random samples of 50 pennies each and build a bootstrap CI from each one. This lets us check whether each interval captures the “true” mean — something we can never do with real data, because we don’t know the true parameter.



Each horizontal line represents a 95% CI from a different sample. Most intervals (blue) capture the true mean, but a few (red) miss it entirely. Over many samples, roughly 95% of the intervals succeed. This is the meaning of “95% confidence.”

What “95% confident” means: If we repeated the entire study many times — collecting a new sample each time and computing a new 95% confidence interval — approximately 95% of those intervals would contain the true population parameter. Any single interval either contains the parameter or it doesn’t; the 95% refers to the long-run success rate of the *procedure*.

Class Example 2.3.2 *Interpreting confidence level*

For each of the following indicate the number of confidence intervals that should capture the true population parameter.

a) We repeat our sampling 30 times and calculate 95% confidence intervals each time.

b) We repeat our sampling 100 times and calculate 99% confidence intervals each time.

One of the more common confidence intervals is the 95% confidence interval, and we can construct a 95% confidence interval using

$$\text{statistic} \pm 2 \cdot \text{standard error}.$$

Class Example 2.3.3 *Constructing confidence intervals*

For each of the following, use the information to construct a 95% confidence interval and give notation for the quantity being estimated.

a) $\bar{x} = 27$ with standard error 3.2.

Interval: Notation (circle one): $p, \mu, p_1 - p_2, \mu_1 - \mu_2$

b) $\hat{p}_1 - \hat{p}_2 = 0.05$ with margin of error for 95% confidence of 0.02.

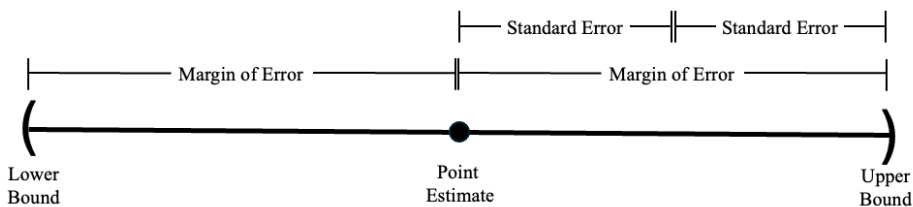
Interval: Notation (circle one): $p, \mu, p_1 - p_2, \mu_1 - \mu_2$

Recap of precision

Before learning about the interpretation of confidence intervals, it is important to highlight some differences between the measures of precision we have discussed thus far.

For a 95% confidence interval: $MOE = 2 \cdot SE$

- The *margin of error* is the amount we add and subtract in a confidence interval.
- The *standard error* is the standard deviation of the sampling distribution of the statistic.
- The *sample standard deviation* represents the average amount a single observation in our sample is from the sample average.



Understanding confidence intervals

Confidence intervals appear all over the place. These intervals appear in research articles and academic publications, but they also appear in mainstream news sources as well. For example, during almost any election cycle, it is almost impossible to find any article about the upcoming vote that does not cite some kind of poll with an associated interval estimate.

Class Example 2.3.4 *Have you ever been arrested?*

According to a recent study of 7335 young people in the US, 30% had been arrested for a crime other than a traffic violation by the age of 23. Crimes included such things as vandalism, underage drinking, drunken driving, shoplifting, and drug possession.

a) Is the 30%, that the study reports, a parameter or statistic?

b) The margin of error is 0.01 for a 95% confidence interval. Use this information to give a range of plausible values for the parameter.

- c) A correctional facility network claims that the true proportion of young people who had been arrested for a crime other than a traffic violation is less than 25%. Given the margin of error in part b), if we asked *all* young people in the US if they have ever been arrested, is it likely that the actual proportion is less than 25%?

Interpreting a confidence interval

We can interpret a 95% confidence interval by saying that we are 95% confident (or sure) that the parameter of interest is in the interval. It is important to note that the confidence interval only refers to the *population parameter*, it has nothing to do with individuals in the population or for the statistics calculated from other samples.

A framework for interpreting a confidence interval is as follows (fill in the blanks with the context of the study):

We are (confidence level%) confident, that the (population parameter in words) is between (lower bound) and (upper bound).

The following are common misinterpretations of confidence intervals that you should *avoid*.

1. ~~A 95% confidence interval contains 95% of the data in the population:~~ This statement is not what we mean by the 95%. The 95% indicates how confident that we are that we captured the population parameter (mean, proportion, etc.) and doesn't reflect how many individual cases are captured.
2. ~~We are 95% sure that the sample mean falls in the 95% confidence interval for the mean:~~ We know the sample mean and it will always (100% of the time) fall in the confidence interval.
3. ~~The probability that the population parameter is in this particular 95% confidence interval is 0.95:~~ Once an interval is constructed it either does or does not contain the population value.

Class Example 2.3.5 BPA in soup cans

To investigate BPA exposure from cans, 75 participants ate either canned soup or fresh soup for lunch for five days. On the fifth day, urinary BPA levels were measured. After a two-day break, the participants switched groups and repeated the process (matched pairs experiment). The difference in the BPA levels between the canned and fresh soup were measured for each participant, yielding a 95% confidence interval for the difference in means (canned minus fresh) of 19.6 to 25.5 micrograms/L.

- a) Provide an interpretation of the confidence interval in the context of the study.

- b) What is the margin of error for this confidence interval.

- c) Does this interval suggest that BPA levels differ for canned and fresh soup? Explain.

- d) If the study had included 500 participants instead of 75, would you expect the confidence interval to be wider or narrower? Explain.
-

Class Example 2.3.6 *Biomass in tropical forests*

Using a sample of 4079 inventory plots, scientists give a 95% confidence interval of 9600 to 13600 tons for the mean amount of carbon per square kilometer in tropical forests.

- a) What is wrong with this interpretation? “We believe that 95% of tropical forests will have an average of carbon per square kilometer that is between 9600 to 13600 tons.”
-

- b) Provide a correct interpretation of the confidence interval in the context of the study.
-

Class Example 2.3.7 *Tipping for pizza delivery*

Using a sample of 24 deliveries described in “Diary of a Pizza Girl”, we find a 95% confidence interval for the mean tip given for a pizza delivery to be \$4.18 to \$5.90. Which of the following is a correct interpretation of this interval? Circle the correct interpretation(s).

- a) I am 95% confident that all pizza delivery tips will be between \$4.18 and \$5.90.
- b) 95% of all pizza delivery tips will be between \$4.18 and \$5.90.
- c) I am 95% confident that the mean pizza delivery tip for this sample will be between \$4.18 and \$5.90.
- d) I am 95% confident that the mean tip for all pizza deliveries in this area will be between \$4.18 and \$5.90.
- e) I am 95% confident that the confidence interval for the mean pizza delivery tip will be between \$4.18 and \$5.90.

The big idea behind bootstrap sampling: resampling

In 2011, researchers collected a sample of 648 pennies in circulation and recorded how old each penny was (current year minus year minted). The question: what is the average age of all pennies currently in circulation in the United States?

The sample mean is $\bar{x} = 10.4$ years. This is our *point estimate* of the population mean μ . But how precise is this estimate? If we collected a different sample of 648 pennies, we’d get a different $\bar{x}$. We can’t keep collecting new samples — we only have this one.

The bootstrap idea: Treat your sample as a stand-in for the population and resample from it. Each *bootstrap resample* is drawn with replacement from your original sample, at the same sample size n . The *bootstrap statistic* is computed from the bootstrap resample and the collection of many bootstrap statistics from the *bootstrap distribution*.

A *point estimate* is a single numerical value used as a “best guess” or approximation to estimate an unknown population parameter.

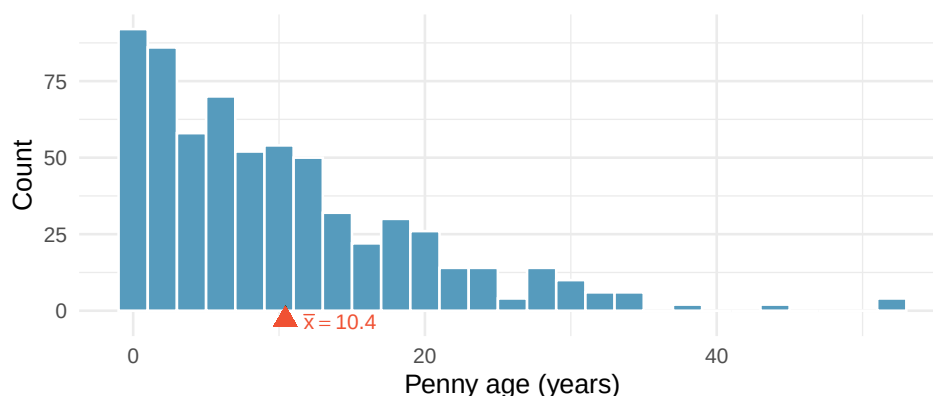
The *bootstrap distribution* is a collection of *bootstrap statistics*, each computed from a *bootstrap resample*. The bootstrap resample is a sample of size n drawn with replacement from the original sample.

Bootstrapping a sample mean

Let’s build a bootstrap confidence interval step by step.

Step 1: Collect data

Our sample of 648 pennies has a right-skewed distribution — many young pennies and a long tail of older ones:



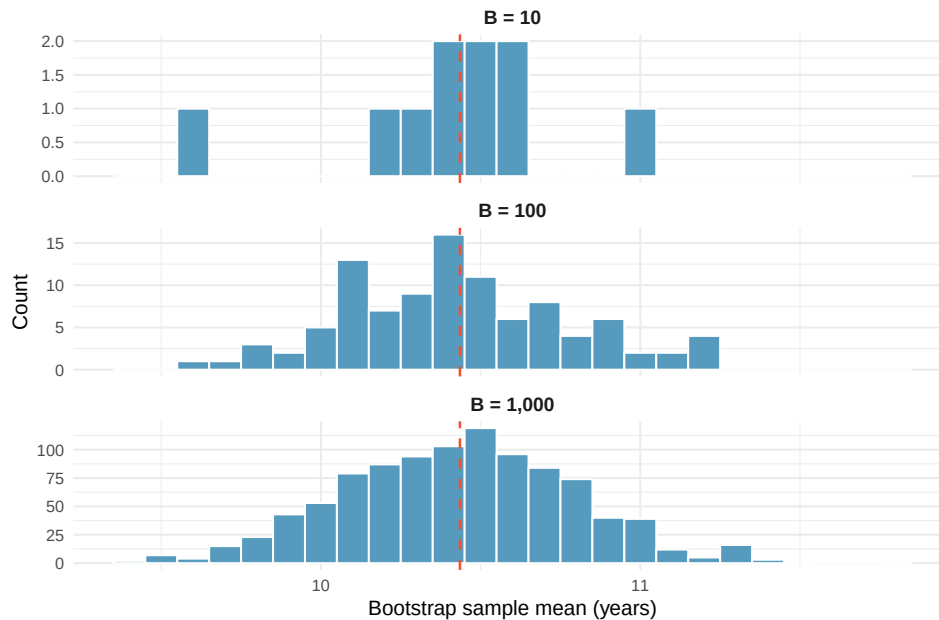
The sample mean is $\bar{x} = 10.4$ years. Notice the distribution is clearly not bell-shaped, it’s skewed right. Bootstrapping doesn’t care. It works regardless of the population shape.

Step 2: Resample with replacement

We draw a new sample of 648 values from the original 648, *with replacement*. Some pennies will appear more than once; others will be left out. Each such draw is one *bootstrap resample*, and the mean of that resample is a *bootstrap statistic* $\bar{x}^*$.

Step 3: Repeat many times

We repeat Step 2 a large number of times — typically $B = 1,000$ or more — and record the bootstrap mean $\bar{x}^*$ each time. This gives us a *bootstrap distribution*.

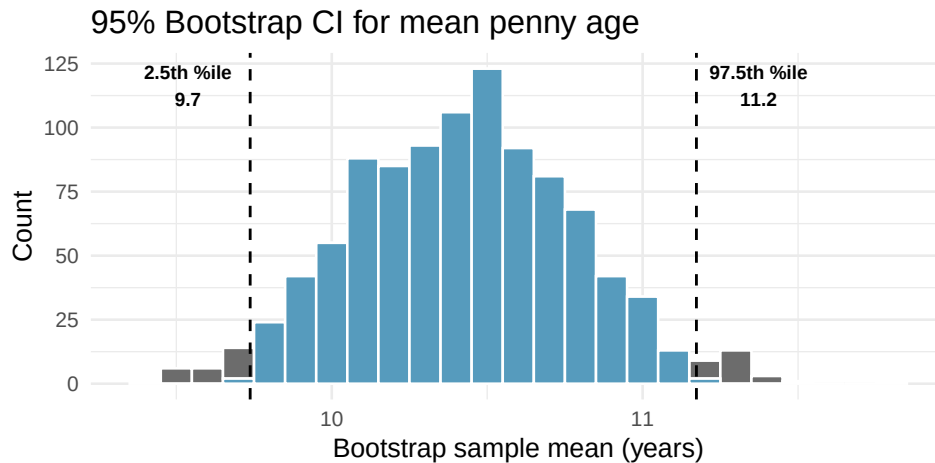


Notice two things: (1) the bootstrap distribution is centered near the original sample mean (dashed red line), and (2) even though the penny ages are right-skewed, the bootstrap distribution of the *mean* is approximately bell-shaped (what statisticians call a *normal* shape). This is the Central Limit Theorem at work, we'll explore it more in the next chapter.

Step 4: Find the confidence interval

For a 95% confidence interval, we find the middle 95% of the bootstrap distribution by taking the 2.5th and 97.5th percentiles.

The P th percentile is the value of a quantitative variable which is greater than P percent of the data



The 95% bootstrap confidence interval for the mean penny age is approximately (9.7, 11.2) years.

Class Example 2.3.8 *Can these be a bootstrap sample?*

We have a random sample of 5 exam grades: 85, 72, 79, 97, 88. For the following state whether or not it is a possible bootstrap sample. If it is not possible, provide a correction to the sample.

a) 79, 79, 97, 85, 82

b) 85, 88, 97, 72

c) 85, 85, 85, 85, 85

The percentile method

The approach used in the pennies example is called the *percentile method* for bootstrap confidence intervals.

For a $(1 - \alpha) \times 100\%$ confidence interval using the percentile method:

- **Lower bound** = the $(\alpha/2) \times 100$ th percentile of the bootstrap distribution
- **Upper bound** = the $(1 - \alpha/2) \times 100$ th percentile of the bootstrap distribution

For a 95% CI ($\alpha = 0.05$): use the 2.5th and 97.5th percentiles. For a 90% CI ($\alpha = 0.10$): use the 5th and 95th percentiles. For a 99% CI ($\alpha = 0.01$): use the 0.5th and 99.5th percentiles.

Class Example 2.3.9 *Increasing confidence level*

IF you increase the confidence level from 95% to 99%, what happens to the width of the confidence interval? Why?

Class Example 2.3.10 *Average penalty minutes in the NHL*

We have a random sample of the penalty minutes for $n = 26$ players from the 2018–2019 season for the NHL team the Ottawa Senators. Some percentiles from a bootstrap distribution of 1000 sample means are shown in the table.

	0.5%	1.0%	2.0%	2.5%	5.0%	95.0%	97.5%	98.0%	99.0%	99.5%
Percentile	13.8	14.6	15.4	15.8	17.2	33.1	35.1	35.3	36.4	38.0

Selected percentiles from bootstrap distribution {*.striped*}

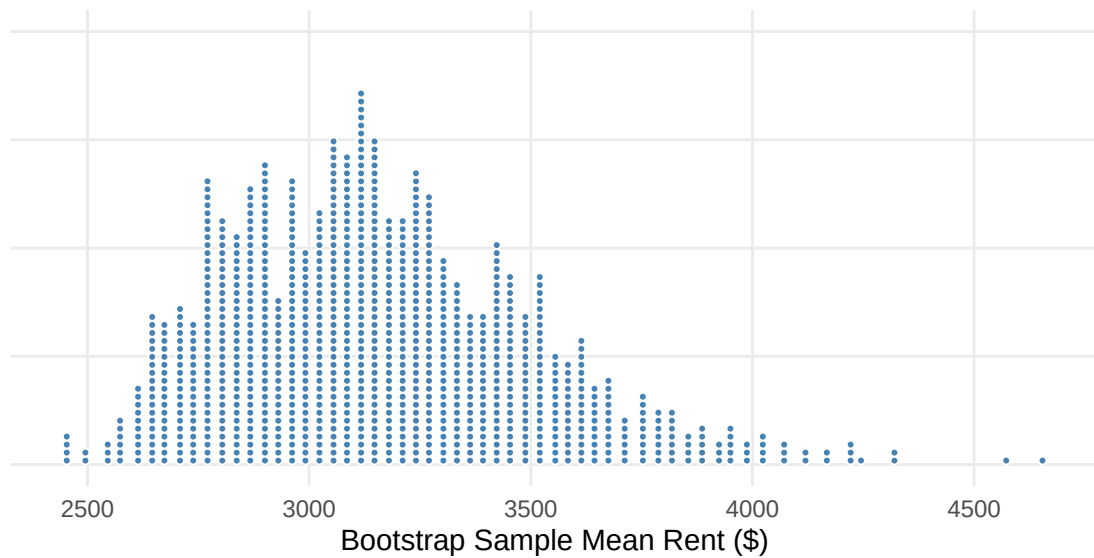
- What percentiles of the bootstrap distribution would be needed to create a 90% confidence interval?
- Create a 90% confidence interval for the average penalty minutes for a player on the Ottawa Senators in the 2018–2019 season using the percentile method.
- What percentiles of the bootstrap distribution would be needed to create a 98% confidence interval?
- Create a 98% confidence interval for the average penalty minutes for a player on the Ottawa Senators in the 2018–2019 season using the percentile method.
- Give an interpretation for one of the intervals (90% or 98%).
- Using the 90% interval, is it plausible to say that the Ottawa Senators players were given more than 34 penalty minutes on average per player? Explain.

Class Example 2.3.11 *Manhattan apartment rent*

The New York City housing authority wants to know about the average rent for a one-bedroom apartment in Manhattan. One of their employees goes to Craigslist and records the monthly rent for a randomly selected sample of 8 listed one-bedroom apartments in Manhattan. Her sample is given below

3150 2275 3100 2650 2925 3305 2325 5495

Use the dotplot and the table below to answer the following questions.



	1.0%	2.5%	5.0%	10.0%	90.0%	95.0%	97.5%	99.0%
Percentile:	2595.25	2651.85	2720.40	2788.53	3546.80	3679.48	3796.40	3913.68

Selected percentiles from bootstrap distribution {.striped}

- a) Write down one possible bootstrap sample from the original sample.
-
- b) Is it appropriate to use the bootstrap distribution to create a confidence interval for the average rent price in Manhattan? Justify your answer.
-
- c) Regardless of how you answered b), create and interpret a 95% confidence interval for the average rent in Manhattan using the percentile method.
-

The standard error method

The percentile method reads the confidence interval directly off the two tails of the bootstrap distribution. A second approach, the *standard error method*, instead summarizes the bootstrap distribution by its spread. It is the bridge to the formula-based confidence intervals we develop with the normal and *t*-distributions in Unit 3.

The *bootstrap standard error* (SE) is the standard deviation of the bootstrap distribution. When the bootstrap distribution is roughly symmetric and bell-shaped, an approximate 95% confidence interval is

The *standard deviation* of the bootstrap distribution provides an estimate of the *standard error* for our statistic.

$$\text{point estimate} \pm 2 \times SE.$$

The multiplier 2 comes from the normal model: about 95% of a bell-shaped distribu-

tion falls within two standard deviations of its center. (The more precise multiplier is 1.96, which we begin using in Unit 3.)

For the penny ages, the observed sample mean is 10.4 years and the bootstrap standard error, the standard deviation of the 1,000 bootstrap means, is 0.36 years. The standard error 95% confidence interval is

$$10.4 \pm 2(0.36) = (9.7, 11.2) \text{ years.}$$

This is essentially the same interval we found with the percentile method, as we should expect, because the bootstrap distribution of the mean penny age is nearly symmetric.

Two methods, usually one answer. For a symmetric, bell-shaped bootstrap distribution, the percentile method and the standard error method give almost identical intervals. They can disagree when the bootstrap distribution is *skewed*, there the percentile method, which follows the actual shape of the distribution, is the safer choice. The standard error method matters because it connects directly to the formula-based confidence intervals of Unit 3, where the standard error comes from a formula instead of from resampling.

Class Example 2.3.12 *Skateboard Prices*

In a random sample of 20 skateboard prices on eBay, we find the average price is \$67.59. A bootstrap distribution gives a standard error of 10.9.

- a) Find and interpret a 95% confidence interval.

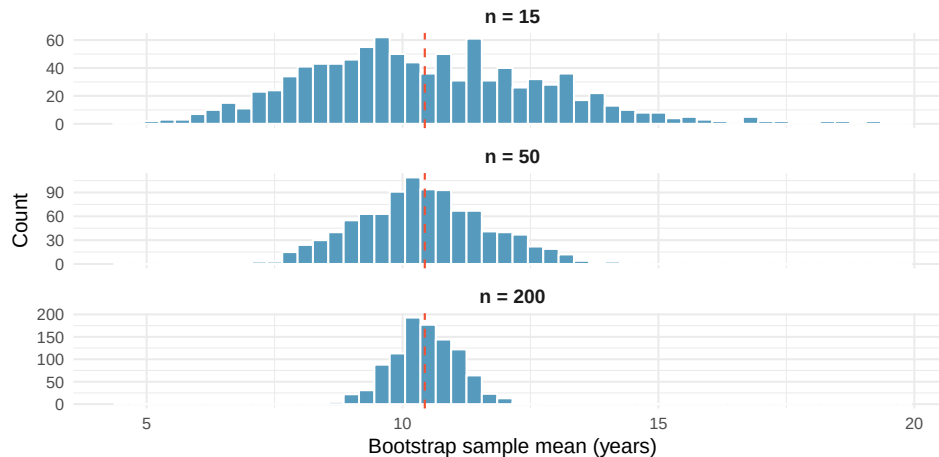
- b) Should a 90% confidence interval be wider or narrower than this interval?

- c) If we took a sample of 50 prices instead of 20, would the 95% confidence interval be wider or narrower than this interval? Explain.

What affects the width of the CI

Three factors control how wide a confidence interval is:

1. *Sample size (n):* Larger samples $\rightarrow$ narrower intervals (more information about the population)
2. *Variability in the data:* More spread $\rightarrow$ wider intervals (less precision)
3. *Confidence level:* Higher confidence $\rightarrow$ wider intervals (more certainty requires a wider net)



As sample size increases, the bootstrap distribution becomes tighter around the sample mean. This is why large studies produce more precise estimates than small ones.

When does bootstrapping work?

Bootstrapping is remarkably flexible, but it does require some conditions to produce reliable intervals:

1. *Independence*: The observations in the sample should be independent of one another. This is usually satisfied when data come from a random sample or a randomized experiment.
2. *Representative sample*: The sample should be reasonably representative of the population. If the sample is biased (e.g., only volunteers), the bootstrap CI will reflect that bias.
3. *Sufficient sample size*: Very small samples (e.g., $n < 10$) may not contain enough information about the population's shape. With small n , the bootstrap distribution can be unreliable.

Bootstrapping does *not* require the population to follow a bell-shaped (normal) distribution. This is one of its greatest strengths compared to traditional formula-based methods, which often do require normality. The penny ages data are clearly right-skewed, yet the bootstrap procedure works perfectly well. The bootstrap lets the data “speak for themselves.”

Summary

- The *bootstrap* constructs a confidence interval by resampling with replacement from the original sample.
- The *bootstrap distribution* approximates the sampling distribution of the statistic.
- The *percentile method* uses the middle $(1 - \alpha) \times 100\%$ of the bootstrap distribution as the CI.
- The *standard error method* uses point estimate $\pm 2 \times SE$, where the SE is the standard deviation of the bootstrap distribution; it agrees with the percentile method for symmetric distributions and bridges to the formula-based intervals of Unit 3.
- A *95% confidence interval* means that if we repeated the study many times, about 95% of the resulting intervals would capture the true parameter.
- CI width depends on sample size, data variability, and confidence level.
- Bootstrap CIs work for any statistic — means, medians, proportions, differ-

ences, and more.

- Bootstrapping requires independent observations from a representative sample with sufficient sample size, but does *not* require the population to be bell-shaped (normal).

2.4 From Simulation to Theory

*In the randomization tests and bootstrap confidence intervals chapters, we used simulation to answer two fundamental questions of inference: Is there an effect? (randomization tests) and How big is the parameter? (bootstrap confidence intervals). Both methods relied on the computer to generate distributions — randomization distributions and bootstrap distributions — that helped us quantify uncertainty. In this chapter, we step back to see the bigger picture. We examine the shape, center, and spread of these simulated distributions and discover a remarkable pattern: under certain conditions, they all look approximately **normal** (bell-shaped). This observation, formalized as the **Central Limit Theorem**, is the bridge from the simulation-based methods of Unit 2 to the formula-based methods of Unit 3.*

Key Concepts - WHAT TO PUT HERE?

- Explain the conclusion of the Central Limit Theorem (CLT)
- Recognize when the CLT can be applied to a study
- Identify when each of the three inference procedures should be used

The Central Limit Theorem

The **Central Limit Theorem (CLT)** tells us that for any population distribution (no matter how skewed or strange the population is), if we repeatedly take new random samples from this distribution and calculate the mean or proportion each time, then:

- As sample size increases, the sample average should get closer to the population average.
- As sample size increases, the averages will be less spread out.
- As sample size increases, the distribution of the sample averages will look more like a bell-shaped (normal) curve.

The **Central Limit Theorem (CLT)** states that when we take sufficiently large random samples from a population, the sampling distribution of many common statistics (such as $\hat{p}$ or $\bar{x}$) is approximately *normal* (bell-shaped), regardless of the shape of the underlying population distribution

Conditions for the CLT

The bell-shaped approximation does not apply in all situations. Two conditions must hold:

- **Independent observations.** The observations in the sample must be independent of each other. This is typically guaranteed by random sampling from a population, or by random assignment in an experiment.
- **Large enough sample.** The sample size cannot be too small. What counts as “large enough” depends on the context:
 - For proportions: we generally need at least 10 expected successes and 10 expected failures.
 - For means: the rule of thumb depends on how skewed the populations is. We will generally use $n > 30$ for STAT 145.

Class Example 2.4.1 Visualizing the CLT

Go to StatLens at [?var:tool-url/](https://statlens.org/) and click on Sampling Distributions under the Simulate Tab. We will use the Right-Skew population shape. We wish to visualize the distribution of the sample mean.

- Set the sample size to $n = 5$ and draw 1000 samples. What is the shape of the sampling distribution?

- b) Set the sample size to $n = 30$ and draw 1000 samples. How does the shape of the sampling distribution change?
-
- c) Set the sample size to $n = 100$ and draw 1000 samples. How does the shape of the sampling distribution change?
-
- d) Which sample size would you use if you wanted the shape of the sampling distribution to be bell-shaped?
-
- e) Now, selected a Normal population shape. Set the sample size to $n = 5$ and draw 1000 samples. How does the shape of this sampling distribution compare to the sampling distribution shape from part (a)?
-
- f) Set the sample size to $n = 30$ and draw 1000 samples. How does the shape of this sampling distribution change from part (e). How does it compare to the sampling distribution shape from part (b)?
-

Comparing randomization, bootstrap, and mathematical approaches

We now have three tools for statistical inference, each grounded in the same core ideas but using different mechanisms:

	Randomization	Bootstrap
Question answered	Is there an effect? (hypothesis test)	How big is the parameter? (confidence interval)
How variability is generated	Shuffle labels to simulate H_0	Resample with replacement from sample
What it models	Variability due to random assignment	Variability due to random sampling
Center of distribution	Null hypothesis value	Sample statistic
Conditions required	Minimal	Minimal
Computational cost	Moderate (need many simulations)	Moderate (need many resamples)

All three methods answer the same fundamental question: *How much does the statistic vary?* They just measure that variability in different ways.

- *Randomization* asks: “How much would the statistic vary if the null hypothesis were true?”
- *Bootstrap* asks: “How much would the statistic vary across different samples from this population?”
- *Mathematical models* ask: “How much does theory predict the statistic would vary?”

When conditions are met, all three give similar answers. The mathematical ap-

proach is faster and requires no simulation, which is why it is widely used. But the simulation approaches are more flexible and work even when conditions for the mathematical model are not met.

When is the normal approximation good enough?

The mathematical approach to inference (which we will develop in Unit 3) replaces simulation with the normal distribution. Instead of generating thousands of shuffles or resamples, we use a formula to compute the standard error and then rely on the bell curve to find p-values or construct confidence intervals.

But when can we trust this shortcut?

The success-failure condition for proportions

For a sample proportion $\hat{p}$, the sampling distribution is approximately normal when:

$$\text{Expected Successes: } np \geq 10 \quad \text{and} \quad \text{Expected Failures: } n(1 - p) \geq 10$$

where n is the sample size and p is the proportion of interest. In practice, since p is unknown, we use $\hat{p}$ as a stand-in:

$$n\hat{p} \geq 10 \quad \text{and} \quad n(1 - \hat{p}) \geq 10$$

This is called the **success-failure condition**.

The sample size condition for means

For a sample mean $\bar{x}$, the CLT tells us the sampling distribution is approximately normal when:

- The sample size is “large enough,” and
- The observations are independent.

A common guideline is $n \geq 30$, but this is only a rough rule of thumb. If the population distribution is nearly symmetric, the normal approximation can work well even for smaller samples. If the population is highly skewed or has extreme outliers, a larger sample may be needed.

Population shape	Minimum n for normal approximation
Symmetric, no outliers	$n \geq 15$ or even smaller
Slightly skewed	$n \geq 30$
Highly skewed or heavy-tailed	$n \geq 60$ or more

When in doubt, use simulation (bootstrap) to check whether the normal approximation is reasonable.

Class Example 2.4.2 *Applying the CLT*

For each of the following indicate whether the CLT can be applied. Why or why not?

- We are interested in approximating the sampling distribution of the sample mean with samples of size 10.
- We are interested in approximating the sampling distribution of the sample mean with samples of size 100.

-
- c) We are interested in approximating the sampling distribution of the sample proportion when $n = 62$ and $\hat{p} = 0.05$.
-
- d) We are interested in approximating the sampling distribution of the sample proportion when $n = 1000$ and $\hat{p} = 0.5$.
-
- e) We are interested in approximating the sampling distribution of the sample minimum with samples of size 100.
-

Summary

- Randomization distributions, bootstrap distributions, and theoretical sampling distributions all tend to be approximately *bell-shaped* (normal) under common conditions. This is the **Central Limit Theorem** (CLT).
- The CLT states that for sufficiently large samples with independent observations, the sampling distribution of many common statistics is approximately normal, regardless of the shape of the population.
- The *success-failure condition* ($n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$) is the practical check for whether the normal approximation is appropriate for proportions.
- The three inference approaches — randomization, bootstrap, and mathematical models — answer the same fundamental questions. When conditions are met, they produce similar results. Mathematical models are faster; simulation methods are more flexible.
- This chapter bridges Unit 2 (simulation-based inference) and Unit 3 (formula-based inference). The CLT is the theoretical justification for the formulas we will use in Unit 3.

3.1 Normal Distributions

Sometimes it's easier to use a theoretical model to answer a statistical question of interest. In order to do this, we must first talk about a few properties of density functions for random variables and the normal probability distribution in particular.

Key Concepts

- Estimate probabilities as areas under a density curve
- Recognize how the mean and standard deviation represent to the center and spread of a normal distribution
- Use technology to compute probabilities of intervals for normal distributions
- Use technology to find endpoint(s) of intervals with a specified probability for normal distributions
- Convert in either direction between a general normal distribution, denoted $N(\mu, \sigma)$, and a standard normal distribution, denoted $N(0, 1)$

Density curves

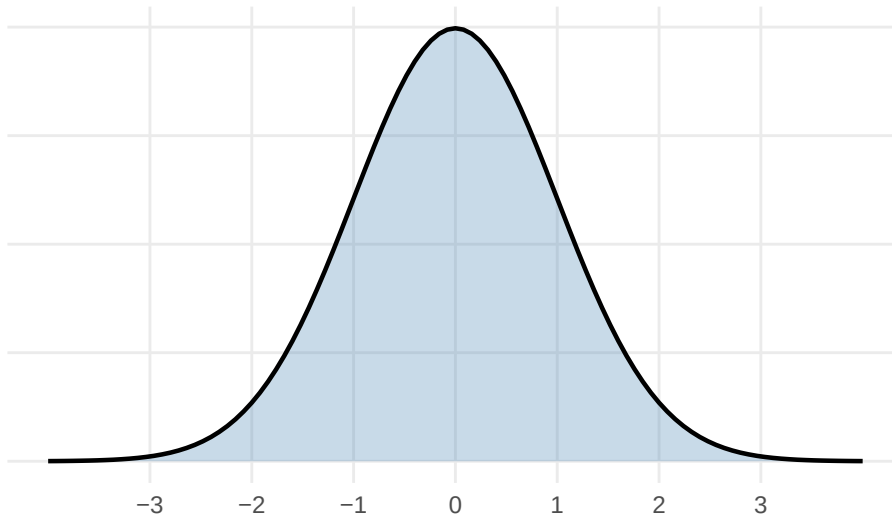
Although density curves come in many different shapes, there are two key characteristics that all density curves have:

- The total area under the curve equals 1 (which corresponds to 100% of the distribution).
- The area under the density curve within any interval is the proportion of the distribution in that interval and, thus, represents the probability that a randomly selected value from that distribution falls in that interval.

A density curve is a graphical representation of the distribution for a continuous random variable (i.e. sample space with probabilities).

Normal distribution

It is not always easy to produce a bootstrap or randomization distribution to answer the statistical question of interest. However, in many circumstances, a theoretical model can be used in place of the randomization distribution. Most of the bootstrap and randomization distributions we have seen in this class have been symmetric and bell-shaped. This type of distribution shape occurs frequently in statistics and is also a characteristic of a family of continuous probability distributions called the *normal distributions*. The normal distribution is density curve that has been well-studied and has a number of very nice properties.



All normal distributions are centered at their mean, μ , and the standard deviation, σ , tells us how spread out the distribution is. We will use the notation $X \sim N(\mu, \sigma)$ to indicate that the random variable X follows a normal distribution with mean, μ , and standard deviation, σ .

The letter Z is used to represent the *standard normal distribution*, denoted $Z \sim N(0, 1)$, which means that Z comes from a normal distribution with mean, $\mu = 0$, and standard deviation, $\sigma = 1$.

The *standard normal distribution* is denoted $N(0, 1)$

Standardizing and Unstandardizing

A *z-score* is a measure of relative standing. Often times, we want to compare two different observations relative to their populations. To do so, it is helpful to first standardize them and then compare the *standard score* or *z-score*. The *z-score* tells us how many standard deviations the observation is from the mean. We can standardize X by using the following formula:

The *standard score* or *z-score* indicates how many standard deviations a particular value is from the mean.

$$Z = \frac{X - \mu}{\sigma}$$

We can unstandardize Z by using the following formula:

$$X = \mu + Z\sigma$$

Class Example 3.1.1 *Standardizing to compare*

For each of the following, compare the results using their standardized score

- Benny and Aaron are on the same track team. They have recently competed at a track and field day, and have each won their respective events. Benny won the high jump with a jump of 75 inches, and Aaron won the 100 m dash with a time of 12 seconds. Let $J \sim N(65, 2)$ be the height of the jumps and $D \sim N(15, 1)$ be the times of the 100 m dash. Who performed better Benny or Aaron?
- A car manufacturer has a small car that gets 36 mpg and a truck that gets 22 mpg. Let $C \sim N(35, 4)$ be the distribution of mpgs for small cars and $T \sim N(19, 3)$ be the mpgs for trucks. Which vehicle performs better relative to the other vehicles in its class?

Class Example 3.1.2 *SAT scores*

Let X be the SAT math scores for science and engineering majors. $X \sim N(609, 80)$

- a) What proportion of SAT scores should be under 569?
- b) What is the probability that a randomly selected student with a science and engineering major has an SAT score of at least 684?
- c) What proportion of SAT scores should be between 500 and 630?

Class Example 3.1.3 *Rainfall in Ithaca*

Let R be the yearly rainfall (in inches) in Ithaca, NY. Data from Cornell's Northeast Regional Climate Center indicates that $R \sim N(35.4, 4.2)$.

- a) How much rain should Ithaca expect to get in the year that is the 99th percentile of rainfall?
- b) How much rain should Ithaca expect to get in the year that marks the top 20 percent of rainfall?
- c) What should the IQR be for yearly rainfall in Ithaca? (*i.e.* the middle 50%)

Class Example 3.1.4 *Furniture assembly*

A company that markets build-it-yourself furniture sells a computer desk that is advertised with the claim "less than an hour to assemble." Let T be the time (in hours) it takes to assemble the furniture. Through

post-purchase surveys, it appears that $T \sim N(1.29, 0.43)$.

- a) What percentage of people should be able to assemble the desk in less than an hour?
-
- b) What is the probability that a randomly selected person can assemble the desk in between 1 and 1.5 hours?
-
- c) The company would like to change the advertising claim to read “less than _____ to assemble” so that 90% of people can assemble the desk in the time listed. What time should they use (as the cutoff for the bottom 90%?
-

4.1 Inference for Single Proportions

Key Concepts

- Find the mean and standard error for a distribution of sample proportions
- Identify when a normal distribution is an appropriate model for a distribution of sample proportions
- Use a normal distribution, when appropriate, to compute a confidence interval for a population proportion
- Use a normal distribution, when appropriate, to test a hypothesis about a population proportion.

Sample statistics

One of the best known theorems in statistics is the *Central Limit Theorem* (CLT) which states that for random samples with a sufficiently large sample size, the distribution of possible sample statistics for a proportion or mean follows a normal distribution that is centered at the value of the population parameter. This theorem is very powerful because it basically says that provided we have a large enough sample size, we can use the normal distribution to find confidence intervals and conduct hypothesis tests without needing to put together a bootstrap or randomization distribution. In chapter 6, we will cover some of the more basic applications of the CLT as it relates to single proportions, single averages, differences in averages, and averages of differences.

Distribution of a single proportion

For categorical data, the goal is often to learn about a population proportion from a sample proportion. We can use the CLT to conduct inference about sample proportions if we have a large enough sample size. If a sample size is large enough, the CLT says that the distribution of all sample proportions ($\hat{p}$) will follow a normal distribution that is centered at the population proportion (p). In chapter 3, we discussed estimating the standard error of a statistic using a bootstrap distribution. When we use the CLT, we can estimate the standard error for a sample proportion by

$$SE(\hat{p}) = \sqrt{\frac{p(1-p)}{n}}$$

where p is the population proportion, and n is the sample size.

Class Example 4.1.1 Use the CLT to estimate the standard error
For each of the following, find $SE(\hat{p})$.

a) Samples of size $n = 100$ from a population with $p = 0.5$

b) Samples of size $n = 200$ from a population with $p = 0.5$

c) Samples of size $n = 500$ from a population with $p = 0.5$

How large do we need?

In order to be able to use the CLT, we need a large enough sample size. For proportions, we need to check that $np \geq 10$ and $n(1 - p) \geq 10$. If the sample size is not large enough, you *cannot* use the CLT to conduct inference.

To use the CLT, we need
 $np \geq 10$ and $n(1 - p) \geq 10$

Class Example 4.1.2 *Describe the sampling distribution*

For each of the following questions, identify if you can use the CLT. If so, fully describe the distribution. If not, identify the smallest sample size you need in order to use the CLT.

- a) Samples of size $n = 10$ from a population with $p = 0.4$.

$\hat{p} \sim N(\text{_____}, \text{_____})$ or Smallest n : _____

- b) Samples of size $n = 100$ from a population with $p = 0.6$.

$\hat{p} \sim N(\text{_____}, \text{_____})$ or Smallest n : _____

- c) Samples of size $n = 40$ from a population with $p = 0.9$.

$\hat{p} \sim N(\text{_____}, \text{_____})$ or Smallest n : _____

CI for single proportion

Provided that the sample size is large enough so that $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$, a confidence interval for a population proportion p can be constructed using

We can only use the CLT for confidence intervals if $n\hat{p} \geq 10$ and $n(1 - \hat{p}) \geq 10$

$$\hat{p} \pm z^* \sqrt{\frac{\hat{p}(1 - \hat{p})}{n}}$$

where $\hat{p}$ is the sample proportion and z^* is the appropriate confidence multiplier from a $N(0, 1)$ distribution.

Finding z^*

To find the appropriate z^* for the confidence interval for a population proportion, we need to use technology to find the corresponding endpoints for the confidence interval.

Class Example 4.1.3 Finding z^*

Find the z^* values for the following confidence intervals.

a) 92%:

b) 85%:

c) 80%:

Some of the more common z^* are given in the table below.

Confidence	z^*
90%	1.645
95%	1.960
99%	2.575

Commonly used z^* values. { .striped }

Class Example 4.1.4 *Left-handed desks*

A university would like to estimate the proportion of its students that are left-handed to purchase new desks for a building renovation. They take a random sample of 300 students and find that 32 people in the sample are left-handed.

- a) Using the CLT find a 95% confidence interval for the proportion of students that are left-handed.

- b) Provide an interpretation for your interval in a).

- c) The Vice Chancellor for facilities decides that 15% of the desks they order should be left-handed. Use the confidence interval in a) to argue whether or not this will be an adequate proportion of left-handed desks.

Class Example 4.1.5 *Moon Landing Conspiracy*

A survey of 500 American adults conducted in 2019 stated that 1 in 10 U.S. adults believe the moon landing was fake.

- a) Using the CLT find a 99% confidence interval for the proportion of American adults who think the moon landing was fake.

- b) Provide an interpretation for your interval in a).

- c) Use the confidence interval in a) to argue whether or not a majority of American adults believe the moon landing was fake.

Class Example 4.1.6 *Fish democracies revisited*

In a school of fish, it is believed that a strongly opinionated minority can influence the path of the majority of the fish. In a study on fish democracies, 165 out of 200 times, the strongly opinionated minority were able to convince the majority to follow their lead.

- a) Using the CLT, find a 90% confidence interval for the proportion of times the strongly opinionated minority

will be able to convince the majority to follow their lead.

b) Provide an interpretation for your interval in a).

c) Use the confidence interval in a) to argue whether or not the opinionated minority can influence the majority.

Hypothesis test for a single proportion

Recall that provided the sample size is large enough, then according to the CLT,

$$\hat{p} \sim N \left(p, \sqrt{\frac{p(1-p)}{n}} \right).$$

We can use this fact to conduct a hypothesis test. As with the hypothesis tests we discussed in chapter 4, there are a specific set of steps to conduct a hypothesis test using the CLT.

Step 1: Check conditions for the test

Check whether both $np_0 \geq 10$ and $n(1-p_0) \geq 10$ are true.

Step 2: State the hypotheses

For a single proportion, the null hypothesis always has the form

$$H_0 : p = p_0$$

where p_0 is a number that represents the proportion claimed or *status quo* value. The alternative hypothesis has one of the following three forms:

$$H_a : p < p_0$$

$$H_a : p > p_0$$

$$H_a : p \neq p_0$$

Step 3: Calculate the test statistic

To conduct the hypothesis test using the CLT, we use a standardized test statistic of

$$z = \frac{\hat{p} - p_0}{\sqrt{\frac{p_0(1-p_0)}{n}}}$$

which follows a $N(0, 1)$ distribution.

We can only use the CLT for a hypothesis test if $np_0 \geq 10$ and $n(1-p_0) \geq 10$

Step 4: Find the p -value

To find the p -value, you must use technology to find what proportion of the standard normal distribution is more extreme than the test statistic (in the direction of the alternative hypothesis). If your alternative is two-sided, then you multiply the one-sided p -value by 2.

Step 5: State your decision and interpret in the context of the hypotheses

As with the randomization hypothesis test, you make your decision based upon the p -value from step 3.

- If $p\text{-value} < \alpha$
 - Decision: *reject* H_0 ,
 - Interpretation: We have significant evidence to suggest that the alternative hypothesis is correct.
- If $p\text{-value} \geq \alpha$
 - Decision: *fail to reject* H_0
 - Interpretation: We do not have significant evidence to suggest that the alternative hypothesis is correct.

You should *always* write your interpretation in the context of the original hypothesis.

How to draw a sketch to illustrate the p -value

Class Example 4.1.7 *Root, Root, Root for the Brewers!*

The Milwaukee Brewers played 81 home games in 2023, and won 49 of those games. If we consider the games played in 2023 as a sample of all Brewers home games, conduct a hypothesis test (using $\alpha = 0.01$) to see if the Brewers win more than half of their home games.

- a) Define the relevant parameter(s) and state the null and alternative hypotheses. Be sure to check any relevant conditions.

- b) What is the test statistic.

- c) Find the p -value, and give a sketch to illustrate your p -value.

- d) State your decision and interpret your findings in the context of the study

Class Example 4.1.8 *Union membership*

The government claimed that in 2005, 12.5% of U.S. workers belonged to unions (*Wall Street Journal*, January 21, 2006). A random sample of 1,445 workers was taken in 2023 by the U.S. Department of Labor to see if membership had decreased. In the sample, 144 of the workers belonged to unions. Conduct a hypothesis test (using $\alpha = 0.01$) to see if membership increased.

- a) Define the relevant parameter(s) and state the null and alternative hypotheses.

- b) What is the test statistic.

- c) Find the p -value, and give a sketch to illustrate your p -value.

- d) State your decision and interpret your findings in the context of the study.

Class Example 4.1.9 *Plato's Republic: Syllable Patterns*

Prose rhythm is characterized as the occurrence of five-syllable sequences in long passages of text. This characterization may be used to assess the similarity among passages of text and sometimes the identity of authors. Syllables are often categorized as long or short. On analyzing Plato's *Republic*, researchers found that 26.1% of the five-syllable sequences contained two short and three long syllables (Wishart and Leach, *Computer Studies of the Humanities and Verbal Behavior*, Vol.3, pp. 90-99). Suppose that Greek archaeologists have found an ancient manuscript dating back to Plato's time (about 427-347 B.C.E.). A random sample of 317 five-syllable sequences from the newly discovered manuscript showed that 61 contained two short and three long syllables. Do the data indicate that the population proportion of this type of five-syllable sequence is different from the text of Plato's *Republic*? You may either use a 95% confidence interval or hypothesis test (use $\alpha = 0.05$) to support your conclusion.

Class Example 4.1.10 *Rabid Raccoons*

In 2019, the United States Department of Agriculture (USDA) Wildlife Services reported that among the 4,305 positive cases of rabies in wild animals, 1,545 of them were found in raccoons. Is there evidence that more than a third of rabid wild animals are raccoons? You may use either a 95% confidence interval or hypothesis test (use $\alpha = 0.05$) to support your conclusion.

4.2 Inference for Single Means

Key Concepts

- Find the mean and standard error for a distribution of sample means
- Recognize when a t -distribution is an appropriate model for a distribution of sample means
- Use a t -distribution, when appropriate, to compute a confidence interval for a population mean
- Use a t -distribution, when appropriate, to test a hypothesis about a population mean

Distribution of a single mean

For quantitative data, the goal is often to learn about a population average from a sample average. We can use the CLT to conduct inference about sample averages if we have a large enough sample size. If sample size is large enough, the CLT says that the distribution of all sample averages ($\bar{x}$) will follow a normal distribution that is centered at the population average (μ). In chapter 3, we discussed estimating the standard error of a statistic using a bootstrap distribution. When we use the CLT, we can estimate the standard error for a sample average by

$$SE(\bar{x}) = \frac{\sigma}{\sqrt{n}}$$

where σ is the population standard deviation, and n is the sample size.

For averages, we almost never know the population standard deviation (σ), so we estimate it by using the sample standard deviation (s). When we use s instead of σ , $\bar{x}$ no longer follows a normal distribution. Instead, $\bar{x}$ follows a t -distribution.

We estimate the standard error of $\bar{x}$ with $\frac{s}{\sqrt{n}}$

The t -distribution

The t -distribution is based on a new parameter called the degrees of freedom (d.f.).

What he found is that provided we have met one of two conditions, then

t_{n-1} means a t -distribution with $n - 1$ degrees of freedom

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1},$$

or t follows a t -distribution with $n - 1$ degrees of freedom. There are two possible ways you can justify using the t -distribution for averages.

1. For a large enough sample size ($n \geq 30$), you can use the t -distribution
2. If $n < 30$, then the t -distribution is only a good approximation if the population is approximately normal.

To use the t -distribution, you need either
 $n \geq 30$ or - a population that is approximately normal

Class Example 4.2.1 *Describe the sampling distribution*

For each of the following questions, identify whether or not a t -distribution would be appropriate. If so, identify the number of degrees of freedom that you would use for the t -distribution.

- a) Samples of size $n = 10$, from a population that is approximately normal.
- b) Samples of size $n = 15$, from a population that is heavily skewed to the right.
- c) Samples of size $n = 50$, from a population that is heavily skewed to the left.

As $n \rightarrow \infty$, the t -distribution becomes closer and closer to the standard normal distribution.

CI for a single mean

Provided that the sample size is large enough ($n \geq 30$) or the population is approximately normal, a confidence interval for a population average μ can be constructed using

We can only use the t -distribution for confidence intervals if
 $n \geq 30$ or - the population is approximately normal

$$\bar{x} \pm t^* \frac{s}{\sqrt{n}}$$

where $\bar{x}$ is the sample average and t^* is the appropriate confidence multiplier from a t_{n-1} distribution.

*Finding t^**

To find the appropriate t^* for the confidence interval for a population average, we need to use technology to find the corresponding endpoints for the confidence interval.

Class Example 4.2.2 *Doggo Heart Rates*

The article “Changes in Pulse Rate, Respiratory Rate and Rectal Temperature in Working Dogs before and after Three Different Field Trials” (*Animals (Basel)*, 2020) documents the physiological conditions of nine Search-and-Rescue (SAR) dogs. The pulse rate (beats/min) for each dog was recorded immediately after a simulated search. They found a sample mean and standard deviation of 167.42 and 41.40, respectively.

- a) What is the variable in this study? Is it categorical or quantitative?

- b) Assuming that the distribution of pulse rates is (approximately) normal, what is the 99% confidence interval for the mean pulse rate of SAR dogs during a search? ($t^*_{99} = 3.355$)?

- c) Provide an interpretation for your interval in b).

- d) If we had calculated a 90% confidence interval instead of a 99% confidence interval, would it have been larger or smaller than the 99% confidence interval? Explain.

Class Example 4.2.3 *Dinosaur Diets*

A recent study on the digestibility of Jurassic plants (Howell *et al.*, 2023) found that in four samples of *Equisetum hyemale* (a fern), the digestibility (measured in milliliters of gas per 200 milligrams of plant dry matter) had a sample mean and standard deviation of 58.33 and 3.45, respectively.

- a) What is the variable in this study? Is it categorical or quantitative?

- b) How many degrees of freedom are there?

- c) Assuming that the digestibility of pieces from this fern is (approximately) normal, What is the 98% confidence interval estimate for the average digestibility?

- d) Provide an interpretation for your interval in c).

- e) Is it plausible to suggest that average digestibility is the same as *Marattia attenuata* (another fern) which has a digestibility of 9.52?

Class Example 4.2.4 *Taylor Swift!*

Taylor Swift's *The Eras* tour took the country (and world) by storm in 2023 and 2024. From a survey of 596 attendees, the average expenses for travel, tickets, costumes, etc. was \$1,327.74. Suppose the standard deviation was approximately \$500.

- a) What is the variable in this study? Is it categorical or quantitative?
- b) What is the 95% confidence interval estimate for the mean expenditure per attendee?
- c) Given an interpretation of your interval in b).
- d) Using the average attendance at a Swift concert of 75,000 people, estimate the average amount spent by all attendees at a single concert.
- e) Assuming the average sales tax rate is approximately 6%, estimate the tax revenue generated from a single Swift concert.

Hypothesis test for a single mean

Provided that the sample size is large enough ($n \geq 30$) or the population is approximately normal, then according to the CLT

$$t = \frac{\bar{x} - \mu}{s/\sqrt{n}} \sim t_{n-1}.$$

Remember, t_{n-1} means a t distribution with $n - 1$ degrees of freedom

We can use this fact to conduct a hypothesis test. As with the hypothesis test we have discussed thus far, there are a specific set of steps to conduct a hypothesis test

using the CLT.

Step 1: Check conditions for the test

Check whether either $n \geq 30$ or the population is approximately normal.

Step 2: State the hypotheses

For a single average, the null hypothesis always has the form

$$H_0 : \mu = \mu_0$$

where μ_0 is a number that represents the averages claimed or *status quo* value. The alternative hypothesis has one of the following three forms:

$$H_a : \mu < \mu_0$$

$$H_a : \mu > \mu_0$$

$$H_a : \mu \neq \mu_0$$

Step 3: Calculate the test statistic

To conduct the hypothesis test using the CLT, we use a standardized test statistic of

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$$

which follows a t_{n-1} distribution.

Step 4: Find the p-value

To find the p -value, you must use technology to find what proportion of the t -distribution is more extreme than the test statistic (in the direction of the alternative hypothesis). If your alternative is two-sided, then you multiply the one-sided p -value by 2.

Step 5: State your decision and interpret in the context of the hypotheses

As with the randomization hypothesis test, you make your decision based upon the p -value from step 3.

- If $p\text{-value} < \alpha$
 - Decision: *reject* H_0 ,
 - Interpretation: We have significant evidence to suggest that the alternative hypothesis is correct.
- If $p\text{-value} \geq \alpha$
 - Decision: *fail to reject* H_0
 - Interpretation: We do not have significant evidence to suggest that the alternative hypothesis is correct.

You should *always* write your interpretation in the context of the original hypothesis.

How to draw a sketch to illustrate the p-value

We can only use the t -distribution for confidence intervals if $n \geq 30$ or the population is approximately normal.

Class Example 4.2.5 *Food Deserts*

Food Deserts are geographical areas that have limited access to affordable and nutritious foods. An analysis of accessibility of grocery stores in the City of Vancouver was published in 2022. Using a sample $n = 4,041$ city blocks, they found the average number of grocery stores within a 15-minute walk was 3.8, with a standard deviation of 3.2. Suppose the city claims that, on average, a resident has access to two grocery stores within a 15-minute walk. Conduct a hypothesis test (using $\alpha = 0.05$) to test the city's claim.

- a) Define the relevant parameter(s) and state the null and alternative hypotheses. Be sure to check any relevant conditions.

- b) What is the test statistic.

- c) Find the p -value and give a sketch to illustrate your p -value.

- d) State your decision and interpret your findings in the context of the study.

Class Example 4.2.6 *First-time home buyers*

In 2022, the National Association of Realtors reported that from a sample of 1,262 first-time home buyers, the average age was 36, and suppose the standard deviation was 4.2 years. Conduct a hypothesis test (using $\alpha = 0.05$) to test whether or not the average age of first-time home buyers was different than 2021, where the average age was 33.

- a) Define the relevant parameter(s) and state the null and alternative hypotheses. Be sure to check any relevant conditions.

- b) What is the test statistic.

- c) Find the p -value and give a sketch to illustrate your p -value.

- d) State your decision and interpret your findings in the context of the study.

Class Example 4.2.7 *Ski Patrol: Avalanches*

Snow avalanches can be a real problem for travelers in the western United States and Canada. A very common type of avalanche is called the slab avalanche. These have been studied extensively by David McClung, a professor of civil engineering at the University of British Columbia. Slab avalanches studied in Canada had an average thickness of 67 (in cm) (*Avalanche Handbook*, by David McClung and P. Schaerer.) The ski patrol at Vail, Colorado, is studying slab avalanches in their region. A random sample of avalanches in spring gave the following thicknesses (in cm).

59	51	76	38	65	54	49	62	68	55	64	67	63	74	65	79
----	----	----	----	----	----	----	----	----	----	----	----	----	----	----	----

Assuming that slab thickness has an approximately normal distribution, is there evidence that the slab thickness in the Vail region differs from that in Canada? You may either use a 95% confidence interval or a hypothesis test (use $\alpha = 0.05$) to support your conclusion.

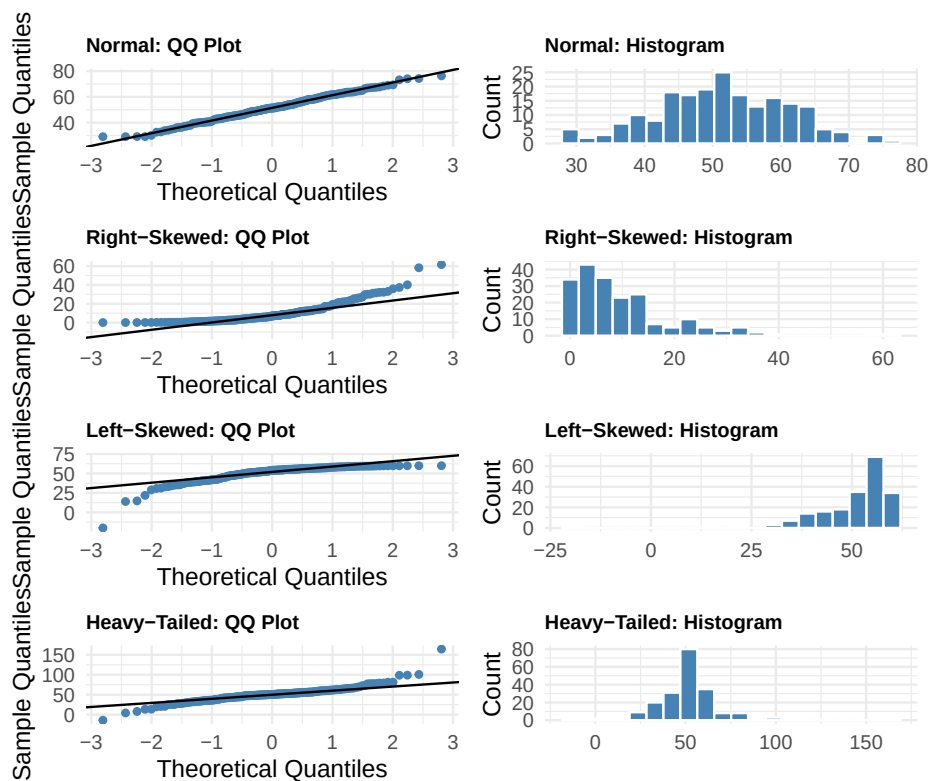
Assessing model conditions

In the examples above, we've either assumed that the population values were normally distributed or had large enough sample sizes to assume normality. These assumptions aren't always apparent (or known) in research studies. It is important to investigate whether normality is reasonable before performing a statistical test. The t -test is *robust* to lack of normality, so when we check this condition we are looking for clear evidence that the distribution is non-normal. If normality is reasonable, we can proceed with a t -test.

There are two primary ways that we can check for normality: histograms/dot plots or quantile-quantile plots. For histograms/dot plots, simply create a histogram/dot plot of the sample data and determine whether normality is reasonable (if the sample data is approximately normal, then normality is reasonable). Quantile-quantile plots, or QQ plots, are commonly seen when using statistical software (such as SPSS). They are created by first ordering the sample data from smallest to largest then comparing these ordered sample data to percentiles of the normal distribution. This information is plotted on an x-y graph (scatterplot) as points. If the resulting graph follows a straight line (that goes through the origin), we can say that the normality assumption is reasonable. The figure below provides some examples of QQ plots and their corresponding histograms.

The t -test is *robust* to lack of normality, but we still need to see if normality is a reasonable assumption.

Create a histogram/dot plot or a QQ plot to assess the normality condition for a t -test.



4.3 Inference for Difference in Proportions

Key Concepts

- Find the mean and standard error for a distribution of differences in sample proportions for two groups
- Recognize when a two-sample z -interval for proportions is the appropriate statistical inference procedure
- Use a standard normal distribution, when appropriate, to compute a confidence interval for the difference in proportions between two populations
- State the conditions for the two-sample z -test for a difference in population proportions
- Recognize when it is appropriate to use the two-sample z -test for a difference in population proportions
- Use a z -distribution, when appropriate, to conduct a hypothesis test for the difference in population proportions

Sampling distribution of a difference in sample proportions

Consider two samples of sizes n_1 and n_2 that come from populations with proportions p_1 and p_2 , respectively. It is a property of sample proportions that the distribution of all differences in sample proportions ($\hat{p}_1$ and $\hat{p}_2$) will be centered at the true population difference in proportions ($p_1 - p_2$). When the samples are *independent* of each other, the standard error of $\hat{p}_1 - \hat{p}_2$ is

$$SE(\hat{p}_1 - \hat{p}_2) = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Using statistical notation, the sampling distribution of $\hat{p}_1 - \hat{p}_2$ for “large” samples can be summarized as

$$\hat{p}_1 - \hat{p}_2 \sim N\left(p_1 - p_2, \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}\right)$$

As with single proportions, the population proportions (p_1 and p_2) are rarely ever known (which is why we are estimating them in the first place!), so they are estimated by the corresponding sample proportions $\hat{p}_1$ and $\hat{p}_2$. According to the Central Limit Theorem, when the sample sizes are large enough the sampling distribution of $\hat{p}_1 - \hat{p}_2$ follows a normal distribution.

Class Example 4.3.1 *Proportion of Elderly, Florida and Texas*

From the 2010 US Census, we learn that 17.6% of the residents of Florida are 65 years old or older while 10.5% of the residents of Texas are 65 years old or older. If independent, random samples of 300 Florida residents and 200 Texas residents are selected and the difference in the proportion of elderly residents ($p_{FL} - p_{TX}$) is calculated:

- a) Describe the shape of the sampling distribution of differences in sample proportions.

- b) What is the mean of the distribution of differences in sample proportions.

- c) Find the standard error of the distribution of differences in sample proportions.

Conditions

In order to be able to use the standard normal distribution to construct a CI for a difference in proportions, the following conditions must be satisfied.

1. The two samples must be randomly selected.
2. The two samples must be independent of each other.
3. The samples must be large enough for the sample proportions to be normally distributed; $n_1\hat{p}_1 \geq 10, n_1(1 - \hat{p}_1) \geq 10, n_2\hat{p}_2 \geq 10$, and $n_2(1 - \hat{p}_2) \geq 10$ is a good guideline to follow. This means having at least 10 successes and 10 failures in each sample.

Two-sample z-interval for a difference in population proportions

Provided that the conditions in the previous section are met, the difference between two unknown population proportions can be estimated with a confidence interval as

$$(\hat{p}_1 - \hat{p}_2) \pm z^* \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}}$$

where z^* is the appropriate critical value for the desired level of confidence from a $N(0, 1)$ distribution.

Class Example 4.3.2 *Mobile Connections to Libraries*

In a random sample of 2,252 Americans age 16 and older, 11% of the 1,059 men and 16% of the 1,193 women said they have accessed library services via a mobile device.

- a) What must be *assumed* in order to construct a confidence interval for the difference in proportions?

-
- b) Find and interpret a 95% confidence interval for the difference in proportions accessing libraries via mobile devices, between men and women.
-
- c) Does your interval in (b) suggest that there is a difference in the proportions of men and women accessing libraries via mobile devices? Which population has a higher proportion?
-

Two-sample z-test for a difference in population proportions

To conduct this test, the following conditions must be true:

1. The two samples must be randomly selected,
2. The two samples must be independent of each other,
3. The samples must be large enough for the sample proportions to be normally distributed; having at least 10 successes and 10 failures in each sample is a good guideline to follow. That is, $n_1\hat{p}_1 \geq 10$, $n_1(1 - \hat{p}_1) \geq 10$, $n_2\hat{p}_2 \geq 10$, and $n_2(1 - \hat{p}_2) \geq 10$.

Two unknown population proportions can be compared using a hypothesis test by following the steps below.

Step 1: State the hypotheses

For a difference in averages, the null hypothesis always has the form

$$H_0: p_1 - p_2 = 0$$

The alternative hypothesis has one of the following three forms:

$$H_a: p_1 - p_2 < 0$$

$$H_a: p_1 - p_2 > 0$$

$$H_a: p_1 - p_2 \neq 0$$

$H_0: p_1 - p_2 = 0$ is equivalent to writing $H_0: p_1 = p_2$.

Step 2: Calculate the test statistic

To conduct the hypothesis test, a standardized test statistic of

$$z_0 = \frac{\hat{p}_1 - \hat{p}_2}{\sqrt{\frac{\hat{p}(1-\hat{p})}{n_1} + \frac{\hat{p}(1-\hat{p})}{n_2}}}$$

where

$$\hat{p} = \frac{x_1 + x_2}{n_1 + n_2}.$$

$\hat{p}$ represents the overall proportion of successes under the presumption that H_0 is true. Since $p_1 - p_2 = 0$ implies that $p_1 = p_2$, then the two independent random samples could be seen as coming from one sample of size $n_1 + n_2$. Furthermore, x_1 and x_2 denote the observed number of successes in samples 1 and 2, respectively. The sampling distribution of this test statistic is a standard normal distribution for large samples.

Step 3: Find the P-value

Recall that the P -value is the probability of randomly observing at least as much evidence against H_0 as what is currently observed, given that H_0 is true. This will depend on which H_a is used. Using the $N(0, 1)$ distribution, the following probabilities will be the P -values for each of the three alternatives:

If $H_a: p_1 - p_2 < 0$, then $P\text{-value} = P(Z < z_0)$

If $H_a: p_1 - p_2 > 0$, then $P\text{-value} = P(Z > z_0)$

If $H_a: p_1 - p_2 \neq 0$, then $P\text{-value} = 2P(Z > |z_0|)$

These probabilities can be found using technology like StatKey, SPSS, a calculator with statistical software, or even looked up in a table of standard normal probabilities.

Step 4: State your conclusion and interpret in the context of the hypotheses

As with the randomization hypothesis test, you make your decision based on the P -value from step 3.

- If $P\text{-value} < \alpha$
 - Conclusion: reject H_0 ,

- Interpretation: There is sufficient evidence to suggest that . . . (the alternative hypothesis is true).
- If $P\text{-value} \geq \alpha$
 - Conclusion: *fail to reject* H_0
 - Interpretation: There is not significant evidence to suggest that . . . (the alternative hypothesis is true).

Always write the interpretation in the context of the original hypothesis.

Class Example 4.3.3 *Accuracy of Lie Detectors*

Participants in a study to evaluate the accuracy of lie detectors were divided into two groups with one group reading true material and the other group reading false material, while connected to a lie detector. The two way table indicates whether the participants were lying or telling the truth and also whether the lie detector indicated they were lying or not.

	Detector says lying	Detector says not	Total
Person lying	31	17	48
Person not lying	27	21	48
Total	58	38	96

- Define the two population proportions p_1 and p_2 .
- Are the conditions met for the two-sample test for proportions? Verify your answer.
- Find the three sample proportions for the proportion of times the lie detector says the person is lying (the proportion for the lying people, the proportion for the truthful people, and the pooled proportion)
- Test to see if there is a difference in the proportion of times the lie detector says the person is lying, depending on whether the person is lying or telling the truth. Use a 5% level of significance and show all details of the test.

Class Example 4.3.4 *Tagging Penguins*

A study was conducted to see if tagging penguins with metal tags harms them. In the study, 100 penguins were randomly assigned to receive a metal tag or (as a control group) an electronic tag. One of the variables studied is survival rate ten years after the penguins were tagged. The scientists observed that 20% of the 50 metal tagged penguins survived while 36% of the 50 electronically tagged penguins survived.

- Are the conditions met for using the normal distribution? Verify your answer.
- Define the two population proportions p_1 and p_2 .
- Test at $\alpha = 0.05$ to see if the survival rate is lower for all metal tagged penguins than for all electronically tagged penguins. Do metal tags appear to reduce survival rate in penguins?

4.4 Inference for Difference in Means

Key Concepts

- Find the mean and standard error for a distribution of differences in sample means between two groups
- Recognize when a t -distribution is an appropriate model for a distribution of the standardized difference in two sample means
- Use a t -distribution, when appropriate, to compute a confidence interval for the difference in means between two groups
- Use a t -distribution, when appropriate, to test a hypothesis about a difference in means between two groups

Distribution of a difference in means

For quantitative data, we often want to learn about a difference in averages from two groups. We can use the CLT to conduct inference about differences in averages if we have a large enough sample size. Consider two groups n_1 and n_2 that come from populations with means μ_1 and μ_2 and standard deviations σ_1 and σ_2 respectively. If sample size is large enough, the CLT says that the distribution of all differences in sample averages ($\bar{x}_1 - \bar{x}_2$) will follow a normal distribution that is centered at the true population difference in averages ($\mu_1 - \mu_2$). When we use the CLT, we can estimate the standard error for a sample average by

$$SE(\bar{x}_1 - \bar{x}_2) = \sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}$$

As with single averages, we almost never know the population standard deviations (σ_1 and σ_2), so we estimate them by using the sample standard deviations (s_1 and s_2).

When we use s_1 and s_2 instead of σ_1 and σ_2 , $\bar{x}_1 - \bar{x}_2$ no longer follows a normal distribution. The difference follows a t -distribution and the degrees of freedom are approximately equal to the smaller of $n_1 - 1$ and $n_2 - 1$.

We estimate the standard error of $\bar{x}_1 - \bar{x}_2$ with $\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$

Conditions

In order to be able to use this distribution for a difference in averages, you need to make sure that each group satisfies the conditions for an average. Recall from the 6.4 – 6.6 notes:

Each sample must satisfy the necessary conditions for averages.

1. For a large enough sample size ($n_1 \geq 30$ and $n_2 \geq 30$), you can use the t -distribution
2. If $n < 30$, then the t -distribution is only a good approximation if the population is approximately normal.

Class Example 4.4.1 Describe the sampling distribution

For each of the following questions, identify whether or not a t -distribution would be appropriate.

- a) Comparing averages from two samples of sizes $n_1 = 10$ and $n_2 = 15$, from populations that are approximately normal.
- b) Comparing averages from two samples of sizes $n_1 = 15$ and $n_2 = 40$, where population 1 is heavily

skewed right, and population 2 is approximately normal.

- c) Comparing averages from two samples of sizes $n_1 = 50$ and $n_2 = 50$, from two populations that are heavily skewed to the left.

CI for a difference in means

Provided that for each group the sample size is large enough ($n \geq 30$) or the population is approximately normal, a confidence interval for a difference in population averages $\mu_1 - \mu_2$ can be constructed using

Remember, each sample must satisfy the necessary conditions for averages.

$$(\bar{x}_1 - \bar{x}_2) \pm t^* \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

where $\bar{x}_1 - \bar{x}_2$ is the difference in sample averages and t^* is the appropriate confidence multiplier.

Class Example 4.4.2 Highball vs tumbler

Two common types of glassware at a bar are “highballs” (tall and thin) and “tumblers” (short and wide). Researchers randomly assigned participants either a “highball” glass or a “tumbler,” each of which held 355 ml. Participants were asked to pour a shot (1.5 oz = 44.3 ml) into their glass. The summary statistics are given in the table below.

	n	$\bar{x}$	s
Highball	99	42.2 ml	16.2 ml
Tumbler	99	60.9 ml	17.9 ml

Summary statistics for amount poured {,striped}

We want to know about the difference in the average amount of liquid poured into the two types of glasses.

- a) What are the variables of interest in this study? Are they categorical or quantitative? Which is the explanatory variable and which is the response?

- b) What is the parameter of interest?

- c) Find and interpret a 95% confidence interval estimate for the difference in the average amount poured in the two different types of glasses. Use highball for group 1 and tumbler for group 2.

- d) If you were the owner of a local bar, from a profit perspective, do you think there is any reason to prefer the highball to the tumbler glass?

Class Example 4.4.3 *Swimming heats at the Olympics*

The data in the table below are finishing times randomly selected from the preliminary heats for the 100m women's backstroke in the 2020 Tokyo Olympics for heats 5 and 6.

Heat 5	57.96	59.02	59.88	60.07
Heat 6	57.88	58.69	59.99	60.07

Randomly selected finishing times for 100m women's backstroke {,striped}

We want to know about the difference in the average finishing times for the two preliminary heats.

- a) What are the variables of interest in this study? Are they categorical or quantitative? Which is the explanatory variable and which is the response?
- b) What is the parameter of interest?
- c) Find and interpret a 95% confidence interval estimate for the difference in the average time between the two heats. You may assume that the finishing times for all participants in the heats are approximately normally distributed.
- d) Does your interval in c) suggest that there is a difference in the average finishing time between the heats?

Hypothesis test for a difference in means

Provided that for each group the sample size is large enough ($n \geq 30$) or the population is approximately normal, then according to the CLT

Each sample must satisfy the necessary conditions for averages.

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

follows a t -distribution.

We can use this fact to conduct a hypothesis test. As with the hypothesis test we have discussed thus far, there are a specific set of steps to conduct a hypothesis test using the CLT.

Step 1: Check conditions for the test

Check that for both groups $n \geq 30$ or the population is approximately normal.

Step 2: State the hypotheses

For a difference in averages, the null hypothesis always has the form

$$H_0 : \mu_1 - \mu_2 = 0$$

The alternative hypothesis has one of the following three forms:

$$H_a : \mu_1 - \mu_2 < 0$$

$$H_a : \mu_1 - \mu_2 > 0$$

$$H_a : \mu_1 - \mu_2 \neq 0$$

Step 3: Calculate the test statistic

To conduct the hypothesis test using the CLT, we use a standardized test statistic of

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

which follows a t -distribution.

Step 4: Find the p -value and give a sketch to illustrate the p -value

To find the p -value, you must use technology to find what proportion of the standard normal distribution is more extreme than the test statistic (in the direction of the alternative hypothesis). If your alternative is two-sided, then you multiply the one-sided p -value by 2. The sketch that you give should be the same as the sketch for a single mean; however, you do not need to give the degrees of freedom.

Step 5: State your decision and interpret in the context of the hypotheses

As with the randomization hypothesis test, you make your decision based on the p -value from step 3.

- If $p\text{-value} < \alpha$
 - Decision: *reject* H_0 ,
 - Interpretation: We have significant evidence to suggest that the alternative hypothesis is correct.
- If $p\text{-value} \geq \alpha$
 - Decision: *fail to reject* H_0

- Interpretation: We do not have significant evidence to suggest that the alternative hypothesis is correct.

You should *always* write your interpretation in the context of the original hypothesis.

Class Example 4.4.4 *Highball vs tumbler revisited*

Recall the following summary statistics from the highball vs tumbler example.

	n	$\bar{x}$	s
Highball	99	42.2 ml	16.2 ml
Tumbler	99	60.9 ml	17.9 ml

Summary statistics for amount poured {,striped}

Conduct a hypothesis test to see if there is a difference between the average amounts poured in the two types of glasses (use $\alpha = 0.05$).

- a) Define the relevant parameter(s) and state the null and alternative hypotheses. Be sure to check any relevant conditions.

- b) What is the test statistic.

- c) Find the p -value and give a sketch to illustrate your p -value.

- d) State your decision and interpret your findings in the context of the study.

- e) Do the findings from your hypothesis test agree with your confidence interval?

Class Example 4.4.5 *100 m backstroke times revisited*

The summary statistics from the 100m backstroke example are given in the table below.

	n	$\bar{x}$	s
Heat 5	4	59.23 s	0.96 s
Heat 6	4	59.16 s	1.06 s

Summary statistics for finishing times for 100 m women's backstroke. {striped}

Conduct a hypothesis test to see if there is a difference between the average finishing times for the two heats (use $\alpha = 0.05$).

- a) Define the relevant parameter(s) and state the null and alternative hypotheses. Be sure to check any relevant conditions.

- b) What is the test statistic.

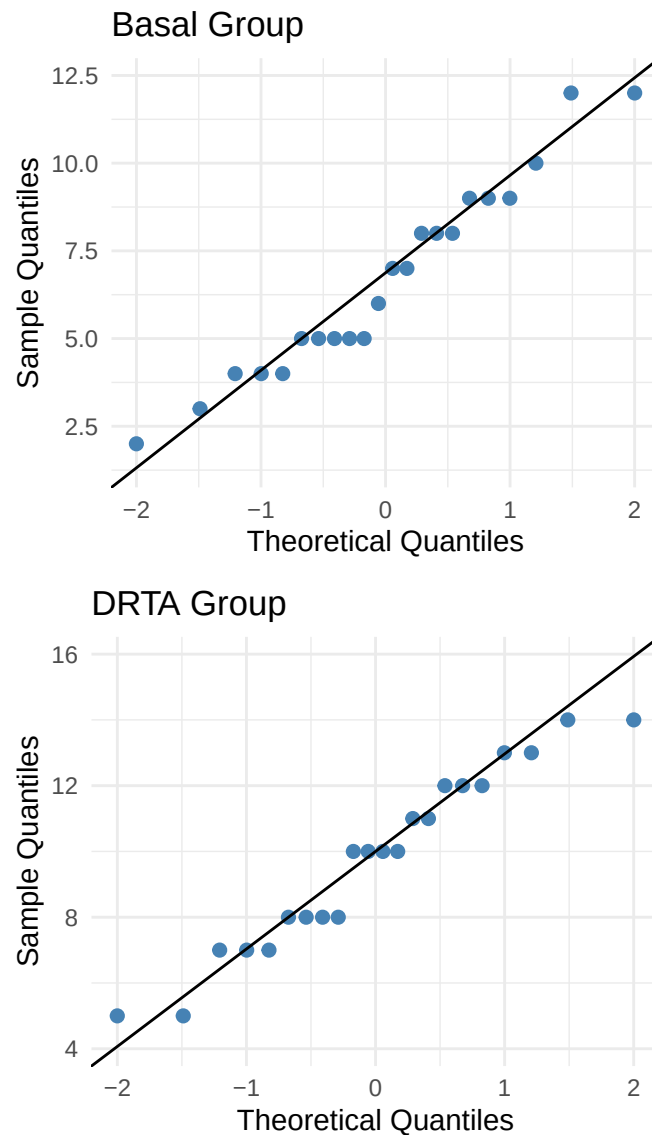
- c) Find the p -value and give a sketch to illustrate your p -value.

- d) State your decision and interpret your findings in the context of the study.

- e) Do the findings from your hypothesis test agree with your confidence interval?

Class Example 4.4.6 *Teaching methods and post-test scores*

The following SPSS output is from an analysis of data from an experimental study conducted by Baumann and Jones, as reported by Moore and McCabe (1993). Students were randomly assigned to one of two experimental groups: “Basal” represents the traditional method of teaching, and “DRTA” represents an innovative teaching methods. The results are based on post-test scores for the two groups.

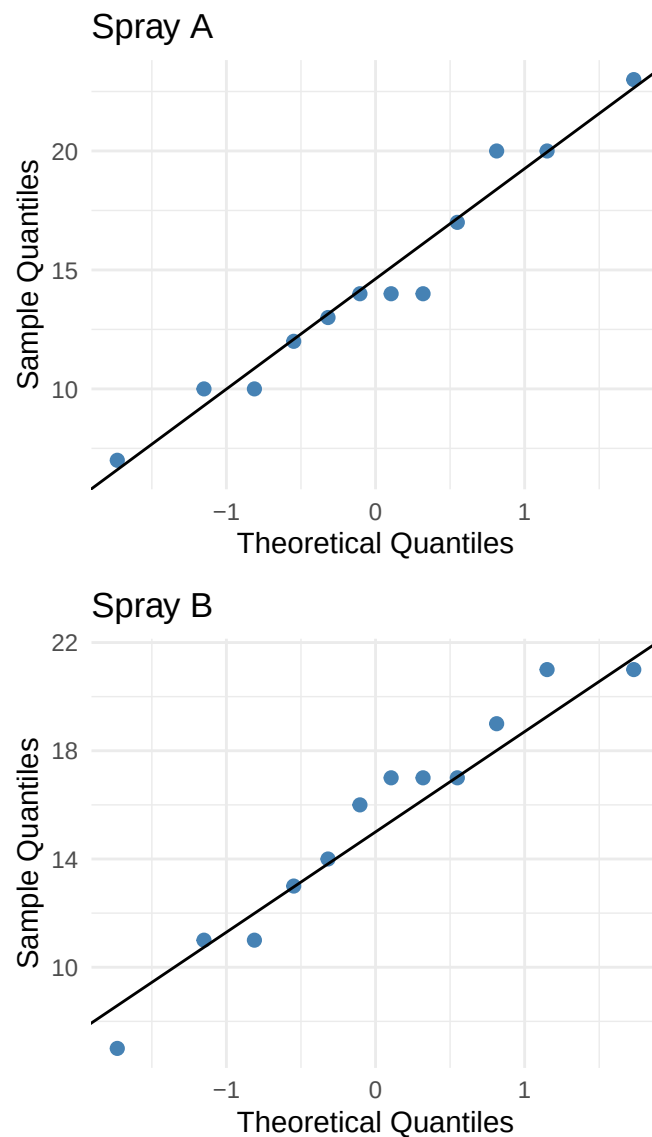


- What are the variables recorded in this study? Are they categorical or quantitative? Which of the variables is the independent variable? Dependent variable?
- Conduct a hypothesis test to determine whether there is a difference post-test scores between the two methods using the SPSS output above. Use $\alpha = 0.05$.

- c) Compare your results to the 95% confidence interval in the output. Do the test results agree with the interpretation of the confidence interval?

Class Example 4.4.7 *Insecticide effectiveness*

The counts of insects in agricultural experiments were taken between two types of insecticides (Beall, *Biometrika* 1942). The following SPSS output represents a test to determine if the two insecticides differed in effectiveness.



- a) What are the variables recorded in this study? Are they categorical or quantitative? Which of the variables is the independent variable? Dependent variable?

- b) Conduct a hypothesis test to determine whether there is a difference in insecticide effectiveness between the two insecticides using the SPSS output above. Use $\alpha = 0.05$.
- c) Compare your results to the 95% confidence interval in the output. Do the test results agree with the interpretation of the confidence interval?

4.5 Paired Difference in Means

Key Concepts

- Distinguish between separate samples and paired data when comparing two means
- Identify the advantages of using paired data when estimating a difference in means
- Use a t -distribution, when appropriate, to compute a confidence interval for a difference in means based on paired data
- Use a t -distribution, when appropriate, to test a hypothesis about a difference in means based on paired data

Inference for paired data

When examining differences in averages, typically, we use one of two methods. One option is to use the tools we discussed for the difference in averages. However, if there is a reason that the groups are not separate groups, then there is a better technique to use. If there is some kind of natural pairing between subjects in a study, then we can use techniques based on the concept of *paired differences*. Most often, paired differences occur when the subjects of the study receive two different levels of a factor (most commonly a *control* and a *treatment*). However, another way this appears is in a study in which two participants have been paired based on a number of different characteristics.

When analyzing paired data, we want to consider the *average of the differences* not the *difference of the averages*.

To decide between a two-sample t and paired difference t , consider how many times each case was measured. If each case is measured once, and we break the cases into two groups, we use two-sample t procedures. If each case is measured twice, we use paired difference t procedures.

Class Example 4.5.1 Deciding between difference in averages and paired difference

For each of the following, identify the cases for the study and decide if it would be more appropriate to use a difference in averages or paired difference.

- a) In a medical study to learn about the difference in the efficacy of generic versus brand name blood pressure medications, 50 men and 50 women were randomly assigned to receive either a generic medication or a brand name medication.
- b) Researchers want to learn about the average age differences between partners who have been together more than 10 years. They randomly select 50 sets of partners and ask them their age.

- c) In a study on the average difference in calories between servings of strawberry and vanilla yogurt, a food scientist randomly selected 12 brands of yogurt and recorded the calories in their strawberry and vanilla yogurts.
- d) The Office of Greek Life randomly selects 45 fraternity and sorority members and looks to see if there is a difference in the average GPAs for fraternities vs sororities.

Conditions

In order to be able to use the t -distribution for a difference in averages, you need to be sure that the differences satisfy the conditions for an average. Recall from the 6.4 – 6.6 notes:

1. For a large enough sample size ($n_d \geq 30$), you can use the t -distribution
2. If $n_d < 30$, then the t -distribution is only a good approximation if the population is approximately normal.

In this section, we will use a subscript d to indicate that we are working with differences.

We either need
 $n_d \geq 30$ or - the population of differences to be approximately normal

CI for an average difference

Provided that the sample size is large enough so that $n_d \geq 30$ or the population is approximately normal, a confidence interval for the average difference in the population μ_d can be constructed using

$$\bar{x}_d \pm t^* \frac{s_d}{\sqrt{n_d}}$$

where $\bar{x}_d$ is average of the differences in the sample and t^* is the appropriate confidence multiplier from a t -distribution with $n_d - 1$ degrees of freedom.

Hypothesis test for an average difference

Provided that the sample size is large enough so that $n_d \geq 30$ or the population is approximately normal, then according to the CLT

$$t = \frac{\bar{x}_d}{s_d / \sqrt{n_d}}$$

follows a t -distribution with $n_d - 1$ degrees of freedom.

We can use this fact to conduct a hypothesis test. As with the hypothesis test we have discussed thus far, there are a specific set of steps to conduct a hypothesis test using the CLT.

Step 1: Check conditions for the test

Check whether $n_d \geq 30$ or the population of differences is approximately normal.

Step 2: State the hypotheses

For a difference in averages, the null hypothesis always has the form

$$H_0 : \mu_d = 0$$

The alternative hypothesis has one of the following three forms:

$$H_a : \mu_d < 0$$

$$H_a : \mu_d > 0$$

$$H_a : \mu_d \neq 0$$

Step 3: Calculate the test statistic

To conduct the hypothesis test using the CLT, we use a standardized test statistic of

$$t = \frac{\bar{x}_d}{s_d / \sqrt{n_d}}$$

which follows a t -distribution with $n_d - 1$ degrees of freedom.

Step 4: Find the p -value and give a sketch to illustrate the p -value

To find the p -value, you must use technology to find what proportion of the t -distribution is more extreme than the test statistic (in the direction of the alternative hypothesis). If your alternative is two-sided, then you multiply the one-sided p -value by 2. The sketch is the same as for a single mean (sections 6.4 – 6.6).

Step 5: State your decision and interpret in the context of the hypotheses

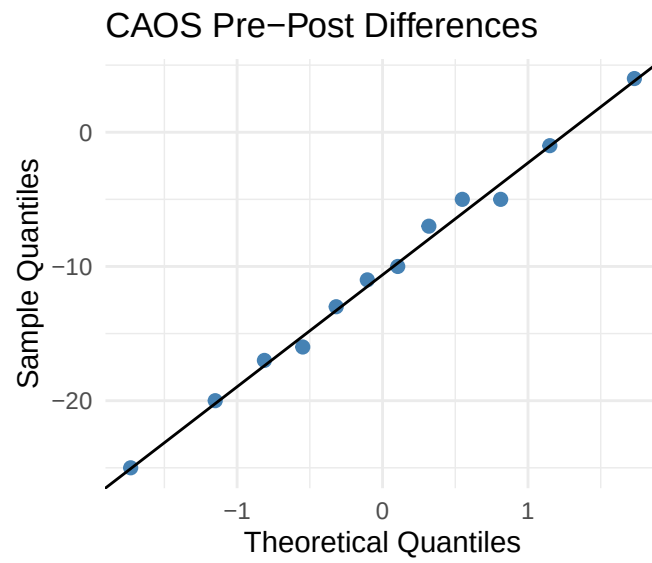
As with the randomization hypothesis test, you make your decision based upon the p -value from step 3.

- If $p\text{-value} < \alpha$
 - Decision: *reject* H_0 ,
 - Interpretation: We have significant evidence to suggest that the alternative hypothesis is correct.
- If $p\text{-value} \geq \alpha$
 - Decision: *fail to reject* H_0
 - Interpretation: We do not have significant evidence to suggest that the alternative hypothesis is correct.

You should *always* write your interpretation in the context of the original hypothesis.

Class Example 4.5.2 *CAOS Comparisons*

The CAOS (Comprehensive Assessment of Outcomes in Statistics) exam is an online multiple-choice test on concepts covered in a typical introductory statistics course. Students take one version before the start of the course and another version after the course ends. The data are summarized in the following QQ plot and SPSS output.



- a) Using the output, record the 95% confidence interval for the average difference in test scores? Interpret this interval.
-
- b) Conduct a test to determine if the test scores change, on average, between the pre- and post-test. Use $\alpha = 0.05$.
-

Class Example 4.5.3 *Body image and weight preoccupation*

A study was conducted to determine how males and females report their weight and height. 82 males and 99 females were asked their height and weight, then height and weight were measured for each participant. (Davis, C. (1990). "Body image and weight preoccupation: A comparison between exercising and non-exercising women." *Appetite*, 15, 13–21.)

The study reported that "women appear to accurately report their weight; we found no evidence of a difference between weight and reported weight ($\bar{x}_d = 1.62\text{lbs}, s_d = 10.14, t = 1.45, p\text{-value: } 0.1513$)."

The study also reported that "men, on the other hand, appear to underestimate their weight ($\bar{x}_d = -0.59\text{lbs}, s_d = 2.79, t = -2.13, p\text{-value: } 0.036$)."

Finally, the study went on to say that "there is also a significant difference between men and women when measuring the differences in height and reported height. Men overestimate their height ($\bar{x}_d = 1.76\text{in}, s_d = 2.17, t = 7.41, p\text{-value} < 0.001$) while women do not overestimate their height ($\bar{x}_d = 1.28\text{in}, s_d = 11.04, t = 1.15, p\text{-value: } 0.2517$)."

There are five statements in this study that indicate statistical findings. For each of the following, evaluate the validity of the claim based on the statistical evidence, and correct the statement if necessary. Only two of the conclusions are correct.

- a) Women appear to accurately report their weight; we found no evidence of a difference between weight and reported weight.

- b) Men, on the other hand, appear to underestimate their weight.

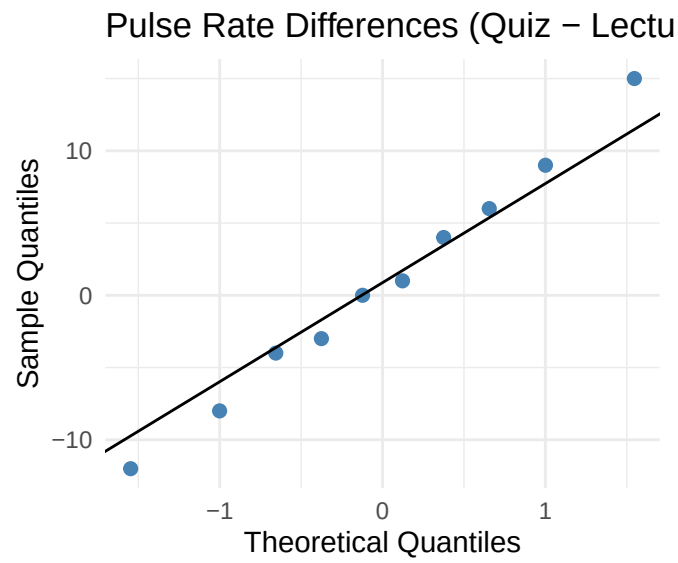
- c) Men overestimate their height.

- d) Women do not overestimate their height.

- e) There is also a significant difference between men and women when measuring the differences in height and reported height.

Class Example 4.5.4 *Pulse rates in class*

Do you think your pulse rate is higher when you are taking a quiz than when you are sitting in a lecture? The data in Table 6.27 show pulse rates collected from 10 students in a class lecture and then from the same students during a quiz. The data are summarized in the following output.



- a) Record the 95% confidence interval for the average difference in pulse rates? Interpret this interval.
-
- b) Conduct a test to determine if pulse rates are greater, on average, during a quiz than during a lecture. Use $\alpha = 0.05$.
-

5.1 Goodness of Fit for a Single Categorical Variable

Key Concepts

- Test a hypothesis about a categorical variable using a chi-square goodness-of-fit test
- Recognize when a chi-square distribution is appropriate for testing a categorical variable

Testing more than two groups

Up until now, we have discussed testing for categorical variables when there are only two groups. However, there are many statistical questions that require us to test categorical variables with three or more groups. In this chapter, we will learn about two new tests that give us additional methods to learn about categorical variables.

χ^2 test for goodness-of-fit

In this lecture, we will be discussing a new hypothesis test called the χ^2 (chi-square) goodness-of-fit test. The χ^2 goodness-of-fit test is used to test the relative frequency (or proportion) of observations for three or more groups.

The χ^2 goodness-of-fit test is used to test proportions when we have more than two groups in a single categorical variable.

Class Example 5.1.1 Relative frequencies

Javier recently took an exam in a chemistry class that had 100 multiple choice questions. There were four options per question (a, b, c, and d). Out of the 100 questions, he answered “a” 40 times, “b” 20 times, “c” 15 times, and “d” 25 times.

a) Find the relative frequencies for each of the answers on his exam.

b) If he was just guessing on each question, what proportion of the time should he answer each option.

Hypotheses

The χ^2 goodness-of-fit hypothesis test lets us see if Javier’s responses follow a different breakdown than what we expect to see if he were just guessing. The hypotheses that we want to test in this example are as follows:

$$H_0 : p_a = 0.25, p_b = 0.25, p_c = 0.25, p_d = 0.25$$

$$H_a : \text{At least one } p_i \text{ is different}$$

Note that the alternative only states that at least one is different. If we reject the null hypothesis here, we do not know which one is different, we only know that one of them is.

Expected counts

As with all hypothesis tests, we assume that the null hypothesis is true when we conduct our test. For the χ^2 goodness-of-fit test, we use this assumption to figure out the counts we *expect* to see in the sample. To find the expected count for group i :

$$\text{Expected count} = n \cdot p_i$$

Class Example 5.1.2 *Expected counts*

In the previous example, we identified the proportion of the time Javier should answer each option. Use these proportions to find the expected count of times he will answer each letter in a 100 question exam.

Test statistic

When we conduct the test, the goal is to see if the counts we *observed* are farther from the *expected* counts than we would reasonably expect to see due to random chance. We use the following test-statistic for the χ^2 test

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

We want to see if the *observed* counts are far enough away from the *expected* counts

χ^2 distribution

There is a χ^2 distribution (like a normal or t -distribution) that we can use if we have a large enough sample size.

In order for us to be able to use the χ^2 distribution, all of the expected counts need to be at least 5 or larger. If we have met this condition, then we can use a χ^2 distribution with $a - 1$ degrees of freedom where a is the number of groups in the categorical variable.

To use the χ^2 distribution, each expected count must be at least 5 or larger

Our null distribution has $a - 1$ degrees of freedom if there are a categories.

Class Example 5.1.3 *Finish the hypothesis test*

Check the conditions, identify the degrees of freedom, give the test statistic, and use the χ^2 distribution to find the p -value for this hypothesis test.

Class Example 5.1.4 *Puppies!*

Every Labrador retriever has a set of genes (called a genotype) composed of B and E genes that control their color. The three primary coat colors are black, brown, and yellow. Without genetic testing it's difficult to know the genotype of a particular dog, however, if the genotypes of both parents are known, the percentage of puppies that fit into each color is known. Suppose a breeder believes their chocolate lab is type Eebb and their yellow lab is eeBb. If so, their puppies should be 25% black, 25% chocolate, and 50% yellow. The table below gives the counts of puppies of each color from the past two litters. Perform a test to determine if there is evidence the parental genotypes differ from what the breeder believes. Use $\alpha = 0.05$.

Puppies	Black	Chocolate	Yellow	Total
Observed (counts)	6	6	8	20
Expected (counts)				

Sample of puppy colors for two litters

Class Example 5.1.5 *Age discrimination on U.S. juries*

It is often said that older people are overrepresented on juries in the United States. The *UCLA Law Review* randomly selected cases in one county in California, and recorded the number of jurors in various age groups given in the table below. Do the following data suggest that the *UCLA Law Review* should expand the scope of their study to more counties (they would do this if they observed discrimination)? Use $\alpha = 0.05$.

	21-40	41-50	51-60	61+	Total
Countywide Proportion (%)	42.0	23.0	16.0	19.0	100.0
Grand Jurors (counts)	10	18	38	66	132
Expected (counts)					

Actual ages of jurors in California county

Class Example 5.1.6 *U.S. ethnicities*

In a sample of 74 flights in 2023, the La Crosse Regional airport experienced 7 schedule modifications. According to the Bureau of Transportation Statistics, 21.53% of flights in the U.S. experienced a schedule modification.

- Is the proportion of scheduled modifications from the La Crosse Regional Airport different than the national rate? Use a χ^2 goodness-of-fit test at $\alpha = 0.05$.
- What other kind of hypothesis test might we conduct to answer this question? How do the results from that hypothesis test compare with the χ^2 test?

5.2 Independence for Two Categorical Variables

Key Concepts

- Test a hypothesis about an association between two categorical variable using a chi-square association test
- Recognize when a chi-square distribution is appropriate for testing an association between two categorical variables

Association Between Categorical Variables

In this lecture, we will be discussing a new hypothesis test called the χ^2 (chi-square) association test. The χ^2 association test is used to test about the relative frequencies (or proportions) of observations for two categorical variables.

The χ^2 association test is used to test the relationship between two categorical variables

Hypotheses

The hypotheses for the χ^2 association test tends to be described more generally than the other tests that we have seen so far. In fact, rather than using parameters in the hypotheses, we will typically express our hypotheses in words. The hypotheses that

we want to test in this test are as follows:

H_0 : *Variable 1* is not associated with *Variable 2*

H_a : *Variable 1* is associated with *Variable 2*

If we reject the null hypothesis here, we can only say that there is an association, but we cannot state exactly what that relationship is.

Class Example 5.2.1 *Traffic cameras and injuries*

Many cities have introduced traffic cameras at high-risk street intersections in an effort to reduce the severity of traffic accidents. Citizen rights groups have complained about the efficacy of these cameras. In fact, they doubt they have any impact whatsoever. A sample of 2938 reported accidents was collected including the type of accident (Death/Disabling Injury, Evident Injury, or Probable Injury) and camera status at the intersection (Before Camera, After Camera). What are the hypotheses to test the relationship between accident severity and presence of a traffic camera?

Expected counts

As with all hypothesis tests, we assume that the null hypothesis is true when we conduct our test.

For the χ^2 association test, if the null hypothesis is true, the two categorical variables are *independent*. We use this assumption to figure out the counts we *expect* to see in the sample. To find the expected count for (Variable 1 group i , Variable 2 group j):

The expected counts come from the probability rules. If A and B are independent, then $P(A \cap B) = P(A) * P(B)$

$$\text{Expected count} = \frac{\text{Row total} \times \text{Column total}}{\text{Total total}}$$

Class Example 5.2.2 *Traffic cameras and injuries, continued*

The table below summarizes the traffic camera data. Use this data to compute a table of expected counts.

	Before Camera	After Camera
Death/Disabling	61	27
Evident	210	136
Probable	1659	845

Traffic camera status and accident severity

Test statistic

When we conduct the test, the goal is to see if the counts we *observed* are farther from the *expected* counts than we would reasonably expect to see due to random chance. We use the same test-statistic for the χ^2 association test that we used for the χ^2 goodness-of-fit test:

We want to see if the *observed* counts are far enough away from the *expected* counts

$$\chi^2 = \sum \frac{(\text{Observed} - \text{Expected})^2}{\text{Expected}}$$

Class Example 5.2.3 *Traffic cameras and injuries, continued*

Using the traffic camera data, find the χ^2 test statistic using the traffic camera data to test if there is an association between traffic cameras and injury severity.

χ^2 test for association

In order for us to be able to use the χ^2 distribution, all of the expected counts need to be at least 5 or larger. If we have met this condition, then we can use a χ^2 distribution with $(r - 1) \times (c - 1)$ degrees of freedom where r is the number of groups in the first categorical variable (number of rows) and c is the number of groups in the second categorical variable (number of columns). To find the p -value using the χ^2 distribution, you can either use StatKey or a TI-83/84. The p -value calculation here works the same as for the goodness-of-fit test.

To use the χ^2 distribution, each expected count must be at least 5 or larger

Class Example 5.2.4 *Traffic cameras and injuries, continued*

Check the conditions, identify the degrees of freedom, and use the χ^2 distribution to find the p -value for the hypothesis test for the traffic camera study.

Class Example 5.2.5 *Star Trek*

In Star Trek, the colored uniforms worn by the crew identify their work area. Among Trekkies (fans of the show), they have begun to notice a pattern - they believe there is an association with uniform color and if the character suffers a fatality on the show. The table below gives a breakdown of the various crew colors and the number of fatalities - does this give evidence to suggest they are related?

	Blue	Gold	Red
Fatality	7	9	24
Survival	129	64	263

Uniform color and fatality count

Class Example 5.2.6 *Racial steering*

A subtle form of racial discrimination in housing is “racial steering.” Racial steering occurs when real estate agents show prospective buyers only homes in neighborhoods already dominated by that family’s race. This violates the Fair Housing Act of 1968. According to an article in *Chance* magazine (Vol. 14, no. 2 [2001]), tenants at a large apartment complex recently filed a lawsuit alleging racial steering. The complex is divided into two parts: Section A and Section B. The plaintiffs claimed that white potential renters were steered to Section A, while African-American renters were steered to Section B. The table below displays the data that were presented in court to show the locations of recently rented apartments. Do you think there is evidence of racial steering?

	White	Black
Section A	87	8
Section B	83	34

Apartment sections broken down by race

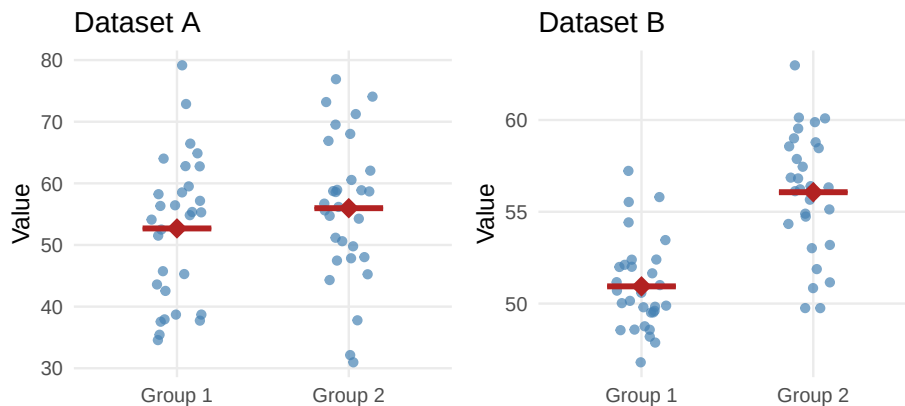
6.1 Analysis of Variance

Key Concepts

- Use ANOVA to test for a difference in means among several groups
- Explain how variation between groups and variation within groups are relevant for testing a difference in means between multiple groups.

Motivation

In a two-sample t -test (test for difference in means), we record values of one categorical variable (with two groups) and one quantitative variable. Our goal is to compare the means between the two groups.



Two sets of sample data are displayed in the figure above. In which set of sample data, A or B, is there likely to be stronger evidence of a difference between the two sample means? Why?

Analysis of Variance

In the example above, the sample mean for Group 1 was the same for both datasets, and likewise for Groups 2. However, the *spread* within each sample was different for the two datasets. We used the spread to decide which provided stronger evidence of a difference. Recall, the test statistic for a two-sample t -test is:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}}$$

In the numerator of the test statistic, we calculated the *difference* between the two sample means, or the spread between the two groups. In the denominator, we calculated the standard error, which is a measure of spread within the two groups. How would this test statistic change if we wanted to test if there was a difference in

means between three or more groups?

The Analysis of Variance (ANOVA) procedure can be regarded as an extension of the two-sample t -test to more than two samples. In fact, the test statistic is formed in much the same way. The test statistic for ANOVA is an F -statistic, and has the following form.

ANOVA is an extension of the two-sample t -test to more than two groups.

$$F = \frac{\text{Spread between group means}}{\text{Spread within groups}}$$

This is the same concept as our t -statistic from a two sample t -test; in fact, we can think of the difference in means (numerator of t) to be a measure of spread between the means.

Statement of hypotheses

The ANOVA hypothesis test is used to see if groups have significantly different averages. If we are considering k groups, then the hypotheses for the ANOVA test are

$$\begin{aligned} H_0 : \mu_1 &= \mu_2 = \dots = \mu_k \\ H_a : &\text{At least one } \mu_i \text{ is different} \end{aligned}$$

Class Example 6.1.1 *Quality control for lenses*

A company that produces glass lenses would like to know if the three presses they use are producing the same average thickness in lens. They hire an engineer to test to see if the machines should be calibrated. Write the hypothesis to test this using an ANOVA hypothesis test.

Finding the test statistic

The test statistic for an ANOVA hypothesis test is an F -statistic. To find an F -statistic, we will use an ANOVA table. An ANOVA table is used to help us compare the variability *between* the sample means and the variability *within* the samples.

SS stands for *sum of squares*

MS stands for *mean squares*

Source	df	SS	MS	F	p-value
Group	$k - 1$	SS_{Group}	$MS_{\text{Group}} = \frac{SS_{\text{G}}}{k-1}$	$F = \frac{MS_{\text{Group}}}{MS_{\text{Error}}}$	$\text{Fcdf}(\text{F-stat}, 9999, \text{df}_{\text{Group}}, \text{df}_{\text{Error}})$
Error	$n - k$	SS_{Error}	$MS_{\text{Error}} = \frac{SSE}{n-k}$		
Total	$n - 1$	SS_{Total}			

Sample ANOVA table

Here are a few relationships to note as you are working with these tables:

$$SS_{\text{Total}} = SS_{\text{Group}} + SS_{\text{Error}}$$

$$df_{\text{Total}} = df_{\text{Group}} + df_{\text{Error}}$$

$$MS_{\text{Group}} = \frac{SS_{\text{Group}}}{df}$$

$$MS_{\text{Error}} = \frac{SS_{\text{Error}}}{df}$$

$$F = \frac{MS_{\text{Group}}}{MS_{\text{Error}}}$$

Finding the p-value

To find the p -value, you need to know the degrees of freedom for the numerator and denominator. You can find the p -value using technology like Statkey, SPSS, or a calculator.

Class Example 6.1.2 *Quality control continued*

Fill in the remaining spots in the ANOVA table below.

Source	df	SS	MS	F	<i>p</i> -value
Group		10.30			
Error					
Total	14	34.40			

Quality control ANOVA table

Find the *p*-value for the *F*-statistic that tests to see whether or not average glass thickness differs for each press. What are the degrees of freedom for the numerator and denominator? State your decision and provide an interpretation in the context of the problem.

Conditions to use the *F*-test

For the *F*-test, we need to check that each average is approximately normal (i.e. $n_i \geq 30$ or sampling from a normal population). We also need to know that the standard deviations in each group are similar enough. To check if the standard deviations are similar enough, it is enough to make sure that no group has a standard that is larger than twice the standard deviation for another group.

Conditions to use the *F*-test:

- We need to know that each average is approximately normal
- No group standard deviation can be more than 2 times the others

Class Example 6.1.3 *Quality control continued*

Have the conditions to use an *F*-statistic to test if any of the presses has a different average thickness been satisfied? Explain.

Class Example 6.1.4 *Rotten Tomatoes*

Rotten Tomatoes is a website providing movie ratings and reviews. Each movie gets a “Tomatometer” score, which is “The percentage of Approved Tomatometer critics who have given this movie a positive review.” A sample of movies for seven categories was taken, and the summary statistics are provided in the table below. Does the average Tomatometer score differ by movie genre?

Genre	$\bar{x}$	s	n
Action	44.97	25.10	31
Animation	60.58	26.73	12
Comedy	49.78	26.35	27
Drama	67.90	18.06	21
Horror	41.29	27.04	17
Romance	47.20	31.57	10
Thriller	62.38	28.66	13

Summary statistics for random sample of tomatometer scores for seven categories

- a) What are the variables recorded in this study? Are they categorical or quantitative? Which of the variables is the independent variable? Dependent variable?

- b) State the appropriate hypotheses for this test.

- c) Are the conditions for using the F -distribution satisfied? Why or why not?

- d) Regardless of your previous answer, test to see whether the Tomatometer score differ significantly by movie genre (use $\alpha = 0.05$)? Make a conclusion in context.

Source	df	SS	MS	F	p -value
Group		11407.4			
Error					
Total		93268.1			

Tomatometer ANOVA table

Class Example 6.1.5 *Caffeine and finger tapping*

Does caffeine consumption have an effect on twitch movements? To investigate this question, the following experiment was conducted. 30 subjects were randomly assigned to one of three groups, with 10 subjects in each group. The subjects were then given a tablet containing 0mg, 100mg, or 200mg of caffeine. Two hours later, they were asked to tap their finger as quickly as possible for one minute. Summary statistics are provided

in the table below.

	0 mg	100 mg	200 mg
$\bar{x}$	244.8	246.4	248.3
s	2.394	2.066	2.214

Summary statistics for caffeine and finger tapping experiment

- a) What are the variables recorded in this study? Are they categorical or quantitative? Which of the variables is the independent variable? Dependent variable?

- b) Based on the summary statistics above, do you think that caffeine has an effect on finger tapping?

- c) State the appropriate hypotheses for this test.

- d) Are the conditions for using the F -distribution satisfied? Why or why not?

- e) Regardless of your previous answer, test to see whether caffeine consumption cause a difference in the average number of finger taps (use $\alpha = 0.05$)? Make a conclusion in context.

Source	df	SS	MS	F	p -value
Group		61.4			
Error		134.1			
Total					

Caffeine and finger tapping ANOVA table

- f) Is there a *practical* difference between the groups? Is this the same as a *statistical* difference?

6.2 Analysis of Variance - Multiple Comparisons after ANOVA

Key Concepts

- Create confidence intervals for single means and differences of means after doing an ANOVA for means
- Test specific pairs of means after ANOVA indicates a difference in means, recognizing the problem of multiplicity

Pairwise comparisons and inference after ANOVA

Although an ANOVA hypothesis test will tell you that there is a difference in at least one of the population means, it does not specify which averages are different. If H_0 is not rejected in an ANOVA, then there is no need to look for differences among population means. However, if H_0 is rejected, the next step is to see which means are actually different from the others. For a significant ANOVA test, there are a number of methods to compare the averages for the different groups.

When we compare these groups, we do what is called *pairwise comparisons* of the means. The phrase *pairwise comparisons* means conducting hypothesis tests between all the possible pairs of means to see which pairs are different. We can find the number of comparisons among k population means by using the formula

$$\# \text{ of comparisons} = \binom{k}{2} = \frac{k!}{2!(k-2)!}$$

where k is the number of groups. Remember from the notes on 11.1, that this is read as “ k choose 2” and you can use the `nCr` function on your calculator to do the calculation.

If H_0 is not rejected in an ANOVA, then there is no need to look for differences among population means

Pairwise comparisons is just a fancy way of saying consider all the possible pairs of means and conduct hypothesis tests to see if they are different.

Pairwise comparisons are also called *multiple comparisons* or *post-hoc tests*

Class Example 6.2.1 *How many pairwise comparisons*

Recall example 8.1.5 in which we discussed comparing the average tomatometer scores for 7 different genres of movies. How many pairwise comparisons would you need to do in order to compare all the genres with one another?

Problem of multiplicity

If you use a two-sample t -test to compare two averages at $\alpha = 0.05$, then there is a 5% chance that we will make a Type I error. However, as we conduct two-sample t -tests to compare more pairs of averages, then we don't want to consider the Type I error rate for only a single test. We want to know the *overall Type I error rate*, or the chance that we make a Type I error in at least one of the tests we conduct. To find the overall Type I error rate when comparing all pairwise comparisons for k groups, we use the following formula:

The *overall Type I error rate* is the chance that we made a Type I error in at least one of our tests.

$$\text{Overall Type I error rate} = 1 - (1 - \alpha)^{\binom{k}{2}}$$

Class Example 6.2.2 *What is the overall Type I error rate*

If we conduct pairwise comparisons of the average tomatometer scores for the 7 genres of movies at $\alpha = 0.05$, what is the overall Type I error rate?

Inference for population means after ANOVA

Population means can be compared using simultaneous hypothesis tests among pairs or simultaneous confidence intervals among pairs. Confidence intervals for individual population means can also be constructed using information from the ANOVA table.

Bonferroni correction for multiplicity

There are a number of ways to correct for the overall Type I error rate, but one of the easiest is the Bonferroni correction. When we do this correction, we make an adjustment to the significance levels for the hypothesis tests.

To use the Bonferroni correction with hypothesis tests, we take the desired overall Type I error rate, and divide by the number of pairwise comparisons.

$$\text{Bonferroni corrected } \alpha = \frac{\text{Desired overall Type I error rate}}{\text{Number of pairwise comparisons}}$$

Class Example 6.2.3 *Bonferroni correction for tomatometer scores*

If we want the overall Type I error rate to be 0.10, what is the Bonferroni corrected α we should use when comparing the average tomatometer scores for the 7 genres of movies?

Bonferroni simultaneous t-intervals

After H_0 has been rejected in an ANOVA, confidence intervals for comparing μ_i with μ_j are constructed using

$$(\bar{x}_i - \bar{x}_j) \pm t^* \sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}$$

where MSE is the mean square error from the ANOVA table. The degrees of freedom for t^* in this interval is the degrees of freedom for error in the ANOVA table ($n - k$).

After confidence intervals have been constructed for all pairs of differences, we can find population means that are different by looking to see which intervals for the differences in means do *not* contain zero. You are not be required to do this computation by hand, but you should be able to do this using SPSS.

$\sqrt{MSE}$ is also called the *pooled standard deviation*

This confidence interval uses $n - k$ degrees of freedom

Class Example 6.2.4 *Salary and Sales*

A company that employs a large number of salespeople is interested in which group of salespeople sells the most: those strictly on commission, those with a fixed salary, or those with a reduced salary plus a commission. A random sample of the previous month's records for salespeople are inspected, and the amount of sales (in dollars) is recorded for each. The ANOVA table and output for the Bonferroni simultaneous CIs at a 95% overall level of confidence is also below.

Pay Type	$\bar{x}$	s	n
Commission	473.50	29.87	6
Fixed Salary	423.20	14.43	5
Commission Plus Salary	476.25	31.64	4

Summary statistics, ANOVA output, and Bonferroni simultaneous 95% confidence intervals for salary and sales data

-
- a) Are the conditions for this test satisfied?
-
- b) Despite your answer to a), conduct a test (using $\alpha = 0.05$) to see if there any differences among different average sales for the three salary type.
-
- c) Use the 95% Bonferroni CIs to determine which pairs of treatment groups have different population means. Use a complete sentence to state how they differ (*i.e.* which is larger or smaller, which are the same) and, if they differ, state the interval for the estimated difference.
-

Bonferroni simultaneous t-tests

After we have conducted an ANOVA and found a significant result that at least one of the averages is different, we can conduct a two-sample t -test using pooled standard deviation. The test statistic is very similar to the one we discussed in chapter 6, but the standard error is different. To compare group i with group j , you should use

$$t = \frac{\bar{x}_i - \bar{x}_j}{\sqrt{MSE \left(\frac{1}{n_i} + \frac{1}{n_j} \right)}}$$

For the degrees of freedom for this test, you should use the degrees of freedom for error from your ANOVA ($n - k$), and you will need to use the Bonferroni corrected α when making your decision.

Class Example 6.2.5 *Restaurant tips*

In this example, we will be conducting many different tests from a dataset that contains information on the tipping patterns of patrons of the *First Crush* bistro in northern New York state. The data from 157 bills include the amount of the bill, size of the tip, percentage tip, number of customers in the group, whether or not a credit card was used, day of the week, and a coded identity of the server (A, B, or C). The first four variables are quantitative and the last three are categorical. For each test, you can use a hypothesis test or confidence interval (if appropriate). On each test, be sure to verify the necessary conditions.

- a) Most restaurant patrons leave a tip between 15% and 20% of the bill, and some restaurant customers determine the percent tip to leave based on the quality of the service. Summary statistics for mean tip percentage left for each of the three servers and a partial ANOVA table are given in the tables below.

Server	Sample size	Mean	Std. Dev.
A	60	17.543	5.504
B	65	16.017	3.485
C	32	16.109	3.376

Mean tip percentage by server

Source	df	SS	MS	F	p-value
Server			41.57		
Error					
Total		3001			

ANOVA table for mean tip percentage

- i. Conduct the appropriate hypothesis test to see if there a difference in mean tip percentage between the three servers (use $\alpha = 0.05$).
-
- ii. Is it appropriate to examine pairwise differences to see which of the servers differs in average tip percentage? Explain.
-
- iii. Regardless of your answer in a.ii, if we want the overall Type I error rate to be 0.05, what is the Bonferroni corrected α we should use when comparing the average tip percentage for the servers?
-

- b) Are some servers given the big spenders (or large groups) while other tend to those having only a cup of coffee or a glass of wine? Summary statistics for average bill for each of the three servers and a partial ANOVA table are given in the tables below.

Server	Sample size	Mean	Std. Dev.
A	60	22.76	12.71
B	65	21.14	10.19
C	32	25.92	14.35

Average bill by server

Source	df	SS	MS	F	p-value
Server				1.671	
Error			146.5		
Total					

ANOVA table for average bill

- i. Conduct the appropriate hypothesis test to see if there a difference in average bill between the three servers (use $\alpha = 0.05$).
-
- ii. Is it appropriate to examine pairwise differences to see which of the servers differs in average bill amount? Explain.
-
- iii. Regardless of your answer in a.ii, if we want the overall Type I error rate to be 0.10, what is the Bonferroni corrected α we should use when comparing the average bill for the servers?
-

- c) The table below summarizes the number of bills by server. Conduct the appropriate hypothesis test to see if the servers serve equal numbers of tables.

Server	A	B	C	Total
Number of bills	60	65	32	157

Number of bills by server

- d) The table below summarizes how patrons paid (cash or card) for each server. Conduct the appropriate hypothesis test to see if there is a relationship between server and how patrons pay.

Server	A	B	C	Total
Cash	39	50	17	106
Card	21	15	15	51
Total	60	65	32	157

Cash or card by server

7.1 Inference for Slope and Correlation

Key Concepts

- Use computer output to make predictions and interpret coefficients using a fitted simple linear model
- Construct a confidence interval for the slope in a regression model
- Test a hypothesis about the slope of a regression model
- Test for evidence of a linear association between two quantitative variables using a sample correlation
- Compute and interpret the value of r^2 for a regression model
- Check a scatterplot for obvious departures from a simple linear model

Modeling with two quantitative variables

In the last chapter we discussed how to use a categorical variable to model a quantitative response using ANOVA, and in this chapter we will be discussing how to use a quantitative variable to model a quantitative response. In statistics, when both the explanatory and response variables are quantitative, we use *regression* in order to explain the response variable. However, before we get into the details for regression, it's helpful to first discuss the concept of correlation.

Exploring correlation with a scatterplot

To get a sense of how one quantitative variable impacts another, it is helpful to first plot the data. To graphically display two quantitative variables, we use a *scatterplot*, or a graph that displays the relationship between two quantitative variables. When looking at a scatterplot, we should consider four different aspects of the plot:

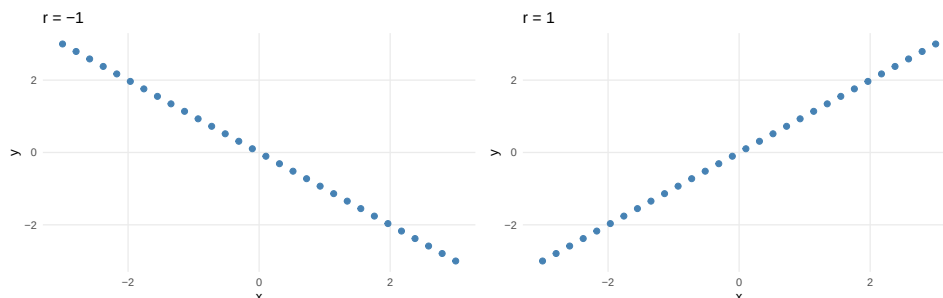
1. Do the points form a clear trend with a particular direction, are they more scattered about a general trend, or is there no obvious pattern?
2. If there is a trend, is it generally upward or generally downward as we look from left to right? A general upward trend is called a *positive* association while a general trend downward is called a *negative* association.
3. If there is a trend, does it seem to follow a straight line, which we call a *linear association*, or some other curve or pattern?
4. Are there any outlier points that are clearly distinct from a general pattern in the data?

A *scatterplot* is a graph of the relationship between two quantitative variables

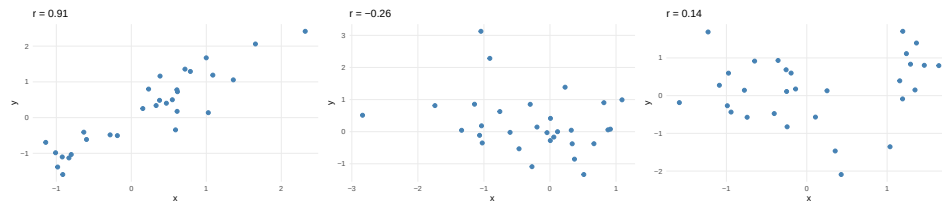
Properties of the linear correlation coefficient

There are a few properties about the correlation coefficient, r , you should know, and we will illustrate them with some pictures.

a) r is always between -1 and 1 , inclusive: $-1 \leq r \leq 1$



- b) The sign of r (positive or negative) indicates the direction of association.
- c) Values of r close to -1 or $+1$ show a strong linear relationship, while values of r close to 0 show no linear relationship.

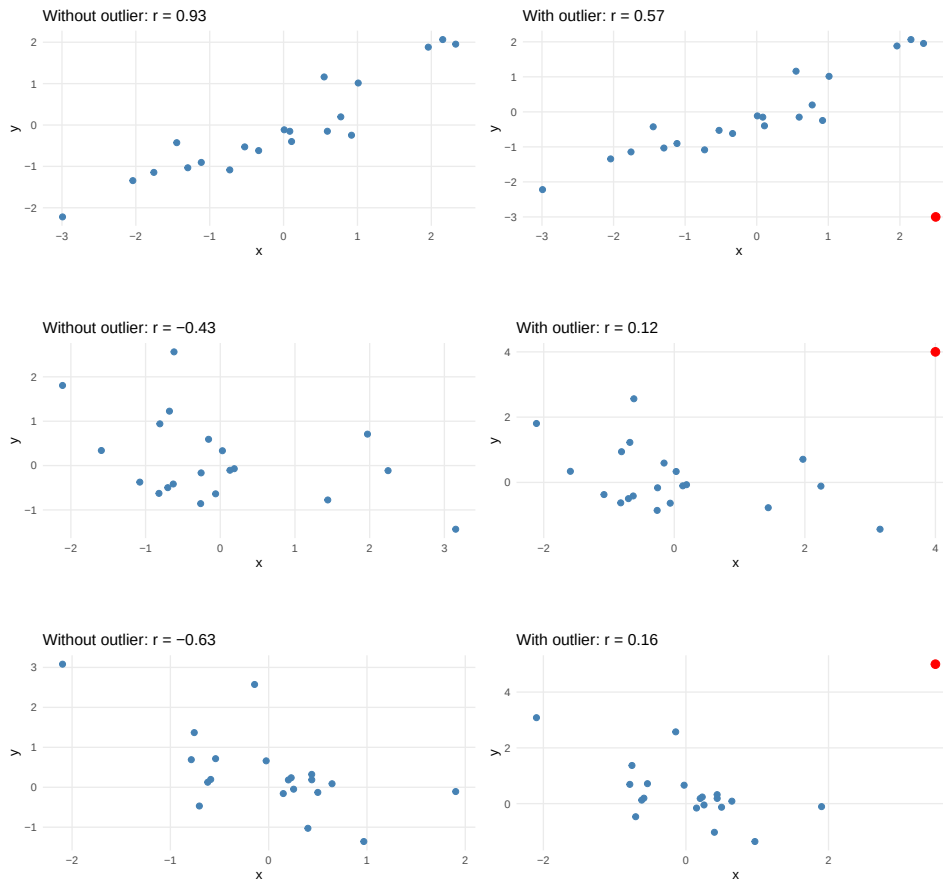


- d) r has no units and is independent of the scale of either variable.
- e) Correlation is symmetric: The correlation between variables x and y is the same as the correlation between y and x .

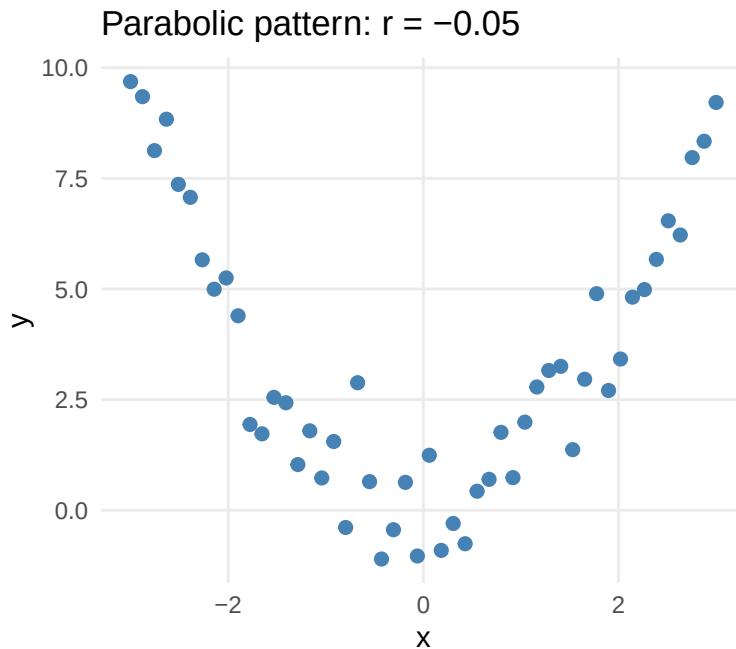
Correlation coefficient cautions

Here are a few things to keep in mind when working with the correlation coefficient:

a) r can be heavily influenced by outliers.



- b) A strong positive or negative correlation coefficient does not (necessarily) imply a cause and effect relationship between the two variables.
- c) A correlation coefficient near zero does not (necessarily) mean that the two variables are not associated, since correlation measures only the strength of a *linear* relationship (see side figure).



Linear regression

Now we are going to introduce a new statistical method that can be used to fit a line to the dataset. To begin with, it's important to first define a *residual*. To find a residual for a single observation, we use the following formula:

$$\text{Residual} = \text{Observed} - \text{Predicted} = y - \hat{y}$$

We say that the line that fits the data best is the *least squares line*, which minimizes the sum of the squared residuals.

Equation of the line

A sample regression has the following form:

$$\hat{y} = b_0 + b_1x$$

where $\hat{y}$ is the predicted value for a specific value of the explanatory variable, x , b_0 is the intercept, and b_1 is the slope. The *slope*, b_1 , represents the average predicted change in the response variable, y , given a one unit increase in the explanatory variable, x . The *intercept*, b_0 , represents the predicted value of the response variable, y , if the explanatory variable, x , is zero. The equation for the intercept and slope are given as

$$b_1 = r \frac{s_y}{s_x} \quad b_0 = \bar{y} - b_1 \bar{x}.$$

One thing to note from this formula is that the *slope* and *correlation* should always have the same sign.

A *residual* is the difference between an observed value and a predicted value of a response.

The *least squares line* is the equation that has the smallest sum of squared residuals.

Notation:

$\hat{y}$: predicted value $-b_0$: intercept estimate $-b_1$: slope estimate

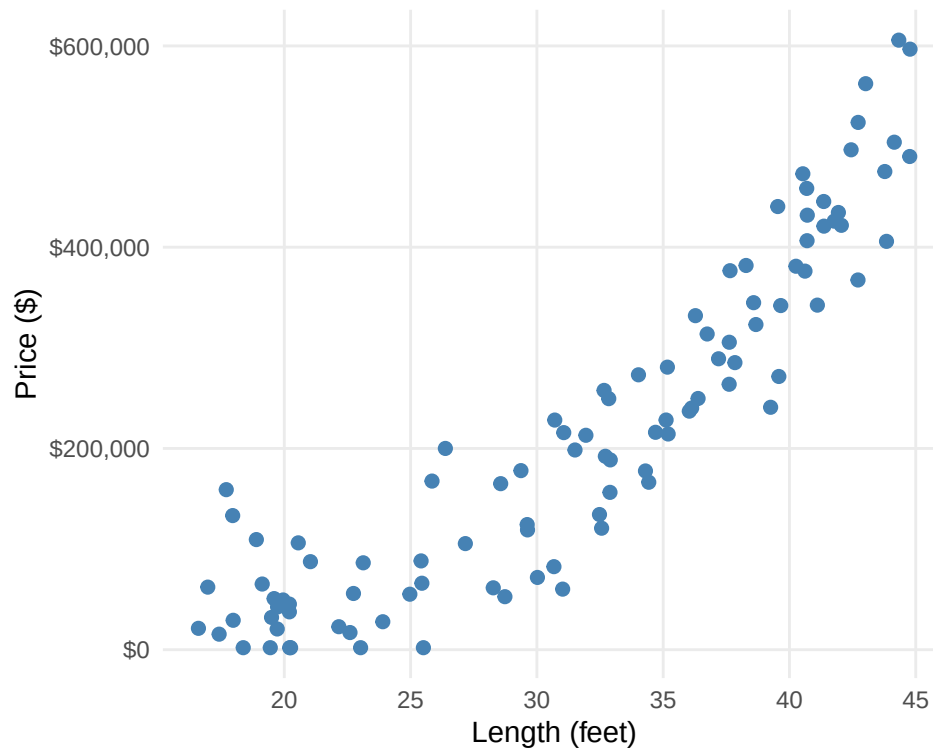
The *slope* and *correlation coefficient* should always have the same sign.

Class Example 7.1.1 *Boats*

A 2024 dataset from BoatTrader.com reveals the relationship between the listing price of a boat and its corresponding length in feet. Some summary statistics and a scatterplot of the data are given below.

Variable	Average	Std. Dev.
Price	51,498.28	66,128.95
Length	22.61	8.07

$$r = 0.364$$



- Looking at the plot, would you categorize the correlation as positive or negative? Strong, moderate, or weak?
- Which variable should be the explanatory variable, which variable should be the response?
- Use the summaries to find estimates for the slope and intercept for the regression line. Record your answer in the form $\hat{y} = b_0 + b_1x$.
- If one of the boats had a length of 20 feet and a price of \$50,000, find the predicted value and residual for this person.

e) Provide an interpretation of the slope in the context of the problem.

Inference for slopes

As with means and proportions, the regression coefficients, b_0 and b_1 , are statistics and are estimates of the population intercept, β_0 , and population slope, β_1 . Most of the time, we will not be interested in inference for β_0 , so the rest of the lecture on regression will focus on inference for the population slope β_1 . To conduct a hypothesis test for the slope, we will be testing

$$H_0 : \beta_1 = 0 \qquad H_a : \begin{cases} \beta_1 < 0 \\ \beta_1 > 0 \\ \beta_1 \neq 0 \end{cases}$$

As with all statistics, we need to report a standard error in order to determine if our findings are significant. This will be provided to you with software output, so you do not need to worry about calculating this on your own. However, you will need to know how to find the test statistic in order to find a p -value. For a hypothesis test about a slope, we use the test statistic

$$t = \frac{b_1 - 0}{SE(b_1)}$$

where $SE(b_1)$ stands for standard error of b_1 . To find a p -value, you should use a t -distribution with $n - 2$ degrees of freedom. The remaining steps of the hypothesis test are the same as all other hypothesis tests. The table below shows output that is typically given by software when conducting linear regression.

Coefficients	Estimate	Std. Error	t
(Intercept)	b_0	$SE(b_0)$	$t_{for b_0}$
Explanatory Variable	b_1	$SE(b_1)$	$t_{for b_1}$

Example output from software

Population parameters:

$-\beta_0$: population intercept $-\beta_1$:
population slope

$SE(b_1)$ is the standard error of b_1

Class Example 7.1.2 *Boats continued*

The table below gives the output from results for a simple linear regression of boat price with length for a 2025 dataset.

Coefficients	Estimate	Std. Error	t	p -value (2-sided)
(Intercept)	-78,776.98	4,431.58	-17.78	< 0.0001
Length	5,761.80	184.61	31.21	< 0.0001

Output from results for a simple linear regression of price with length

- a) Use the output to conduct a hypothesis test to see if there is a positive relationship between a boat's length and price. Use $\alpha = 0.05$.
-
- b) The hypothesis test revealed a statistically significant positive relationship between a boat's length and price. Is this practically significant?
-

Conditions for inference

In order to conduct inference for regression, we need to satisfy a few conditions.

1. The relationship should be linear
2. The residuals should have a normal distribution for all values of x
3. The variability in the response should be the same for all values of x
4. There should not be any outliers that unduly influence the results of the regression.

We can check conditions 1, 3, and 4 using a scatterplot of the data. Condition 2 can be checked using a histogram of residuals or a QQ plot.

Words of caution for regression

Although regression is a powerful tool, there are a couple of words of wisdom to keep in mind.

1. Like correlation, regression only tells us about the *linear* relationship between the explanatory and response variables.
2. Do not ever *extrapolate* or predict a response beyond the range of the explanatory variable. The regression equation only tells us about the relationship between the response and explanatory within the range of the explanatory variable.
3. As with correlation, just because there is a statistically significant relationship, it does not mean that there is a causal relationship. Remember *correlation does not mean causation*.

Coefficient of determination: r^2

There is another statistic from regression that is often used to compare two simple linear regressions models with each other. The *coefficient of determination*, notated r^2 , gives the percentage of variation in the *response* variable that we can explain by the linear regression using the *explanatory* variable. If we are picking between two different simple linear regression models, you should select the model that has a higher r^2 value. To find r^2 , square the correlation coefficient r .

r^2 gives the percentage of variation in the *response* variable that we can explain with the *explanatory* variable

Class Example 7.1.3 *Price and Horsepower*

In the previous example, we used the variable *length* to predict *price*. Now we're going to use *horsepower* to predict *price*. The table below gives the output from a simple linear regression of price with horsepower.

Coefficients	Estimate	Std. Error	<i>t</i>	<i>p</i> -value (2-sided)
(Intercept)	36,449.63	1,998.92	18.23	< 0.0001
Horsepower	162.27	8.50	19.09	< 0.0001

$r = 0.5245$ Output from results for a simple linear regression of drinks with classes missed

a) Using the output above, write the regression equation in the form $\hat{y} = b_0 + b_1x$.

b) Find the predicted and residual value for a boat with 500 horsepower and a price of \$90,000.

c) Interpret the slope of the line in the context of the problem.

d) Conduct a hypothesis test to see if there is a significant positive relationship between horsepower and price. Use $\alpha = 0.05$.

e) Does horsepower or length explain price better? Explain.

7.2 Confidence and Prediction Intervals

Key Concepts

- Compute and interpret the value of r^2 for a simple linear model using values in the ANOVA table.
- Compute and interpret a confidence interval for the mean response for a specific value of the predictor in a regression model
- Compute and interpret a prediction interval for the individual responses for a specific value of the predictor in a regression model

Understanding the definition of r^2 through ANOVA

When conducting a simple linear regression using the computer, often times, an ANOVA table is included in the output. The Sum of Squares column of the ANOVA table gives us a way to partition the variability of our response into two components, variability we can explain with our regression model, and variability that is unexplained. In the context of regression, this looks like

Variability in the response variable can be partitioned into two components:

$$-SS_{\text{Regression}} - SS_{\text{Total}}$$

$$SS_{\text{Total}} = SS_{\text{Regression}} + SS_{\text{Error}}$$

In the previous section the coefficient of determination, r^2 , could be found by squaring the correlation coefficient, r . It can also be computed as

$$r^2 = \frac{SS_{\text{Regression}}}{SS_{\text{Total}}}.$$

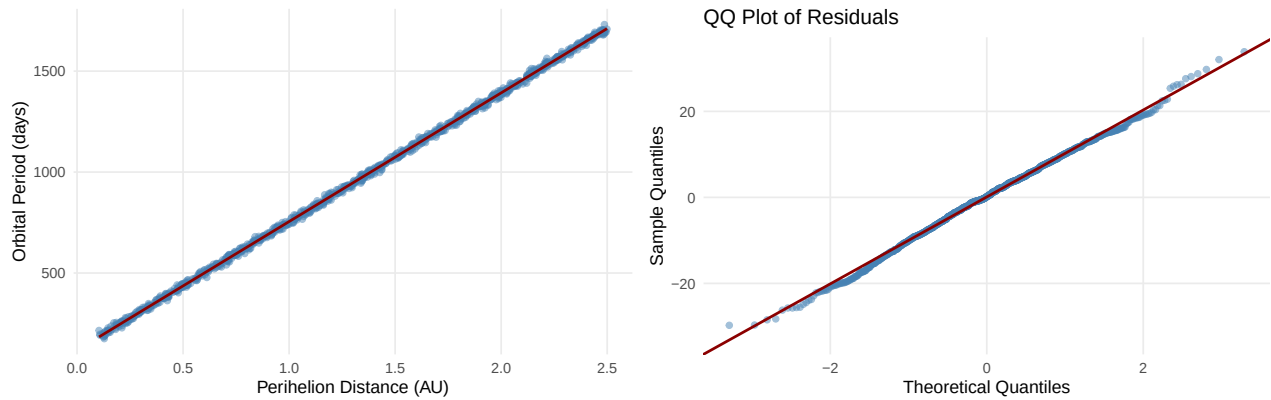
Another formula for r^2 is

$$r^2 = \frac{SS_{\text{Regression}}}{SS_{\text{Total}}}$$

This is the reason r^2 is interpreted as the percentage of variation in the *response* variable that we can explain by the linear regression using the *explanatory* variable.

Class Example 7.2.1 Asteroids!

NASA's Near Earth Object Web Service (NeoWs) tracks near earth asteroids. In a sample of 1000 near earth asteroids taken in 2024 the orbital period (number of days to orbit the earth) and perihelion distance (distance from earth when the earth is closest to the sun, measured in Astronomical Units) was recorded. A scatterplot of the data and QQ plot of the residuals are included below. Computer output is provided below.



Source	df	SS	MS	F	p-value
Per. Dist.	1	2.3×10^7			< 0.0001
Error	998	1.1×10^8			
Total	999				

Partial ANOVA results for regression of perihelion distance on orbital period

Coefficients	Estimate	Std. Error	t	p-value
(Intercept)	117.00	37.42	3.13	< 0.0001
Per. Dist.	637.90	44.29	14.40	< 0.0001

Parameter estimates for regression of perihelion distance on orbital period

a) Complete the partial ANOVA table.

b) Calculate r^2 and provide an interpretation in the context of the study.

c) Write 2 to 3 sentences describing the findings of this study. If it is inappropriate to conduct inference, simply state that and justify your decision.

d) Suppose a new asteroid was discovered with a perihelion distance of 3 AU. Assuming it was appropriate to conduct inference, would it be reasonable to use these results to predict the orbital period of this asteroid? Explain.

Confidence and prediction intervals

One of the main reasons why we conduct linear regression is to predict a response for a given value of the explanatory variable. You can easily predict the response for any value of the explanatory variable using the regression equation provided you are not extrapolating beyond the range of the data. However, when making this prediction, it is important to account for the uncertainty in the estimates for slope and intercept. We can account for this uncertainty by using an interval. There are two main types of intervals that we will use for predictions:

1. Confidence interval for the *average* response
2. Prediction interval for an *individual* response

We use *confidence* and *prediction* intervals to account for our uncertainty in making predictions using the regression equation.

Intervals for a predicted responses

Recall that we use the symbol $\hat{y}$ to represent the predicted value of y , and this is found using the least squares regression equation

$$\hat{y} = b_0 + b_1 x$$

The *confidence interval for the average response* when $x = x^*$, is

$$\hat{y} \pm t_{n-2}^* \sqrt{MSE \left(\frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2} \right)}.$$

The interpretation for the confidence interval for the average response when $x = x^*$ is:

The *prediction interval for an individual response* when $x = x^*$, is

$$\hat{y} \pm t_{n-2}^* \sqrt{MSE \left(1 + \frac{1}{n} + \frac{(x^* - \bar{x})^2}{(n-1)s_x^2} \right)}.$$

The interpretation for the prediction interval for an individual response when $x = x^*$ is:

Two facts about confidence and prediction intervals

1. As we predict further away from $\bar{x}$, the intervals get wider. For example, if $\bar{x} = 5$, a prediction interval for $x^* = 7$ will be smaller than $x^* = 10$.
2. The prediction interval for an individual response is always wider than the confidence interval for the average response.

Class Example 7.2.2 *NFL Rushing*

The tables below provide computer output for the regression line to predict number of rushing yards a player will have in a season based on their number of rushing attempts. The dataset contains all players in the NFL who attempted a rush from 2001 to 2023.

Coefficients	Estimate	Std. Error	t	p -value
(Intercept)	-4.424	0.782	-5.66	< 0.0001
Attempts	4.31	0.009	478.89	< 0.0001

Parameter estimates for regression of season rushing yards on number of rushes attempted

Source	df	SS	MS	F	p -value
Regression	1	7.14×10^8	7.14×10^8	211099	< 0.0001
Error	7514	2.54×10^7	3.38×10^3		
Total	7515	7.39×10^8			

ANOVA results for regression of season rushing yards on number of rushes attempted

- a) Is the number of rushing attempts a good predictor of the number of season rushing yards? Justify your answer using specific values in the output.
- b) Find and interpret r^2 for this regression.
- c) If a player has 300 rushing attempts in a season, how many rushing yards would you predict they would have?
- d) Suppose the 95% prediction interval for players that have 300 season rushing attempts is (1188.58, 1388.58). Provide an interpretation of this interval in the context of the question.

Class Example 7.2.3 *Predicting Mustang Prices*

The dataset MustangPrice contains information for a sample of 25 used Mustang cars offered for sale on a website. Suppose that we are interested in predicting the Price of a Mustang (in thousands of dollars) based on how old it is (Age in years). Some output for fitting this model is shown in the tables below. You may assume that any necessary conditions for inference hold.

Coefficients	Estimate	Std. Error	<i>t</i>	<i>p</i> -value
(Intercept)	30.264	3.440	8.80	< 0.0001
Age	-1.717	0.365	-4.71	< 0.0001

Parameter estimates for regression of mustang price on age in years

Source	df	SS	MS	F	<i>p</i> -value
Regression	1	1454.4	1454.4	22.15	< 0.0001
Error	23	1509.9	65.6		
Total	24	2964.3			

ANOVA results for regression of mustang price on age in years

- a) Is age of a used Mustang an effective predictor of its price? Justify your answer using the information from the output above.

- b) You are thinking about buying a 5 year-old Mustang, what would you predict the price to be?

- c) Two intervals are given below. One is the 95% confidence interval for mean price and one is the 95% prediction interval for individual prices, for Mustangs that are 5 years old. Which is which?
 (17.49, 25.86): _____ (4.40, 38.96): _____
 d) Provide an interpretation for both of the intervals given in c) above in the context of the question.

- e) The 95% prediction interval for Mustangs that are 10 years old is (9.51, 16.68). Is the average age of Mustangs in the dataset closer to 5 years or 10 years? Explain.

8.1 Probability Rules

“Probability theory is nothing but common sense reduced to calculation.” – Pierre-Simon Laplace

Key Concepts

- Compute the probability of events if outcomes are equally likely
- Identify when a probability question is asking for A and B, A or B, not A, or A if B
- Use the complement, additive, multiplicative, and conditional rules to compute probabilities of events
- Recognize when two events are disjoint

Equally Likely Outcomes

Recall that a *population* is the collection of all individuals or items under consideration. An individual could refer to a person, a playing card, or whatever object we are interested in. We typically refer to a population when referencing sampling. However, when we talk about experiments, we use the phrase *sample space*. The *sample space* is the set of all possible non-overlapping outcomes for a random experiment, and is denoted Ω . *Events* can be simple (e.g., the outcome of a single flip of a coin) or complex combinations of outcomes (e.g., the results from 10 flips of a coin). We have three important rules for probability:

P.1) Any probability must be between 0 and 1 (inclusive)

P.2) When outcomes are equally likely, the probability of the event E is given by

$$P(E) = \frac{N(E)}{N(\Omega)}$$

P.3) The sum of the probabilities for all of the experimental outcomes must equal 1

P.4) Suppose the event E is composed of several non-overlapping outcomes $E_1, \dots, E_n$, then

$$P(E) = P(E_1) + \dots + P(E_n)$$

P.5) $P(\text{not } A) = P(A^c)$ = Probability A does not occur is given by

$$P(A^c) = 1 - P(A)$$

The probability of an event is the long run frequency or proportion of times the event occurs. Probabilities are always between 0 and 1.

We use the notation $P()$ to mean *probability of*.

We use the notation $N()$ to mean *number of outcomes in* an event. When outcomes are equally likely, the probability of event is the number of outcomes in the event divided by the total number of outcomes.

Sample space, Ω , is the set of all possible outcomes.

The *complement* of a condition, written “not A” or A^c , includes events in which the condition does not happen.

Class Example 8.1.1

Suppose a fair, six-sided die is rolled twice. Determine the following.

a) $N(\Omega)$

b) $N(A)$, where A is the event that the sum of the two rolls is 5

c) $P(B)$, where B is the event that the two rolls are the same

d) $P(C)$, where C is the event that the sum of the two rolls is even

e) $P(D)$, where D is the event that the two rolls are not the same

Combinations of Events

Many of the terms we used for describing the relationship between two categorical variables is also used when describing events matching two (or more) conditions. In addition to the ideas of *joint*, *marginal*, and *conditional* proportions, we will use terms like *intersection*, and *union* of conditions.

- $P(A \text{ and } B) = P(A \cap B)$ = Probability conditions A and B both occur.
- $P(A \text{ or } B) = P(A \cup B)$ = Probability either condition A or condition B (or both) occur.

The following equations describe the probability of these complex events.

P.6) Additive Rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

P.7) Multiplication Rule:

$$\begin{aligned} P(A \cap B) &= P(B|A) * P(A) \\ &= P(A|B) * P(B) \end{aligned}$$

Venn diagrams

Venn diagrams are visual displays that are useful for investigating *intersections*, *unions*, and *complements*. These diagrams are constructed by first drawing a rectangle, representing the sample space. Within this rectangle, overlapping circles are drawn for each of the conditions of interest.

The *joint* or *intersection* of two conditions, written "A and B" or $A \cap B$, includes events that satisfy *both* conditions.

The *union* of two conditions, written "A or B" or $A \cup B$, includes events that satisfy *either* or *both* conditions.

The *conditional occurrence* of condition A given the occurrence of condition B is written "A if B" or $A | B$.

Class Example 8.1.2

Recall the following example from Section 2.1. 169 college students were asked about relationship status and gender. The results are given in the table below. For each of the following, assume that a student is chosen at random from this sample.

	Female	Male	Total
In a relationship	32	10	42
It's complicated	12	7	19
Single	63	45	108
Total	107	62	169

Contingency table for relationship status and gender.

- a) What is the probability the student is in a relationship? Is this a joint, marginal, or conditional probability?
-
- b) What is the probability the student is in a relationship given the student is female? Is this a joint, marginal, or conditional probability?
-
- c) What is the probability the student is female given the student is in a relationship? Is this a joint, marginal, or conditional probability?
-
- d) What is the probability the student is a male and single? Is this a joint, marginal, or conditional probability?
-

Class Example 8.1.3

Suppose that in a particular group of people, 10% are rich, 5% are famous, and 3% are both. Use a Venn diagram, as well as the definitions of probability, to answer the following.

- a) What is the probability that a randomly selected person in this group is not rich?
-
- b) What is the probability that a randomly selected person in this group is rich but not famous?
-
- c) What is the probability that a randomly selected person in this group is neither rich nor famous?
-

Class Example 8.1.4

Police report that when stopping a driver on suspicion of driving while under the influence that 78% of drivers are given a breath test, 36% of drivers are given a blood test, and 8% of drivers are given neither. Use a Venn diagram, as well as the definitions of probability, to answer the following.

- What is the probability that a randomly selected person in this population is not given a breath test?
- What is the probability that a randomly selected person in this population is given a breath test but not a blood test?
- What is the probability that a randomly selected person in this population is given both tests if they are given at least one test?

Class Example 8.1.5

In 1974, the University of California, Berkeley was sued for bias against women who had applied for admission to graduate schools there. The table below displays the admissions data from the Fall 1973 semester.

	Male	Female	Total
Admitted	3546	1512	5058
Rejected	4896	2809	7705
Total	8442	4321	12763

Contingency table for UC Berkeley graduate admissions in Fall 1973.

- What is the probability a randomly selected applicant was female? Is this a joint, marginal, or conditional probability?
- What is the probability a randomly selected applicant was accepted given the applicant was female? Is this a joint, marginal, or conditional probability?
- What is the probability a randomly selected applicant was accepted given the applicant was male? Is this a joint, marginal, or conditional probability?

Special cases: Independent and Disjoint

There are two special cases of rules for probability *independent* and *disjoint*. If two events, A and B , are *independent* it means that the chance of A occurring is the same regardless of whether or not B happened. If two events, A and B , are *disjoint* it means that events A and B cannot occur at the same time.

P.8) If A and B are independent, then

a. $P(A | B) = P(A)$ b. $P(B | A) = P(B)$ c. $P(A \cap B) = P(A) * P(B)$

P.9) If A and B are disjoint, then $P(A \cap B) = 0$

Class Example 8.1.6

Suppose we have a fair coin.

a) Suppose we flip the coin two times, are the flips independent of one another?

b) Suppose that the first three flips were HHT. What is the probability the 4th flip is heads?

c) Suppose that the first three flips were TTT. What is the probability the 4th flip is tails?

d) What is the probability of the event the 4 flips were HHTH?

e) What is the probability of the event the 4 flips were TTTT?

f) Is getting a head on the first flip disjoint from getting a tail on the first flip?

g) Is getting a head on the third flip disjoint from getting a tail on the first flip?

Counting Rules

It is often useful to be able to count (enumerate) the number of possible outcomes of a random experiment. Understanding a couple of simple counting rules can help to determine probabilities of events.

C.1) *Basic Counting Rule*: If event A consists of ractions that are performed in a specific order, with m_1 possibilities for the first action, m_2 for the second, etc. Then the total outcomes in A is

$$N(A) = m_1 * m_2 * \cdots * m_r.$$

C.2) *Arranging items*: If event A has n items, then the number of ways we can arrange the items in A is

$$n! = n * (n - 1) * (n - 2) * \cdots * 2 * 1.$$

C.3) *Combinations*: If event A has n items, then the number of ways we can choose k items from A is

$$\binom{n}{k} = \frac{n!}{(n - k)! * k!}$$

The *basic counting rule* provides the number of possible outcomes of a random experiment in which several actions are performed.

A *combination* is any unordered collection elements chosen from a collection of elements.

Class Example 8.1.7

Suppose that a professor gives her students a 10 question True/False quiz.

a) How many ways are there to answer the quiz?

b) How many ways are there to answer exactly 5 questions correctly?

Class Example 8.1.8

Suppose that for a particular lottery ticket, you are asked to pick 6 numbers from the numbers 1-59. No matter what order with which you choose the 6 numbers, they are rearranged to be in numeric order.

a) How many different tickets can be purchased?

b) If all 6 numbers match the official values, you win the jackpot. What is the probability of winning the jackpot?

c) If 3 or more numbers match the official values, you win. What is the probability of choosing a winning ticket?

Class Example 8.1.9

Suppose that three glasses are filled with the same cola, and then labeled C, D, and P. A randomly selected person is asked to order the three glasses by preference (this person is unaware that all three glasses contain the same cola).

a) How many ways can the cola preferences be ranked?

b) What is the probability that C is ranked first?

c) What is the probability that C is ranked first and D is ranked last?

Additional Example Questions

Class Example 8.1.10

Use the definition of independent and disjoint events to determine whether the following pairs of events are independent, disjoint, both, or neither.

- a) Rolling doubles; Rolling a sum of 8

- b) Rolling a sum of 8; Getting a 2 on the first die rolled

- c) Rolling a sum of 7; Getting a 1 on the first die rolled

- d) Rolling a sum of 5; Getting doubles

Class Example 8.1.11

We have a data from a medical study with 210 people about smoker status and cancer status. The summary counts are given in the table below.

	Cancer	No Cancer	Total
Smoker	10	50	60
Non Smoker	6	144	150
Total	16	194	210

Cancer and smoker status.

- a) What is the probability of being a smoker if you have cancer? Is this a joint, marginal, or conditional probability?

- b) What is the probability of having cancer if you are a smoker? Is this a joint, marginal, or conditional probability?

- c) Is having cancer independent of being a smoker? Explain.

Class Example 8.1.12

We have information about blood pressure and cholesterol for American males. Suppose there is a 11% chance a man has both high cholesterol and high blood pressure, a 52% chance a man has neither high cholesterol or high blood pressure, and a 27% chance that a man has high blood pressure.

- a) What is the chance a man has high cholesterol?

- b) What is the chance a man has high cholesterol or high blood pressure?

- c) If a man does not have high blood pressure, what's the chance he has high cholesterol?

- d) Is having high blood pressure independent from having high cholesterol? Explain.

Class Example 8.1.13

Fifty-six percent of all American workers have an employer-sponsored retirement plan, 68% have employer-sponsored health insurance, and 49% have both benefits. We select a worker at random.

- a) What is the probability a worker has neither employer-sponsored health insurance nor a retirement plan?

- b) What is the probability a worker has employer-sponsored health insurance or an employer-sponsored retirement plan?

- c) What is the probability a worker has employer-sponsored health insurance if they have an employee-sponsored retirement plan?

- d) Are having these two benefits disjoint? Explain.

Class Example 8.1.14

Suppose that 23% of adults smoke cigarettes. It's known that 57% of smokers and 13% of nonsmokers develop a certain lung condition by age 60.

- a) What is the probability that a randomly selected 60-year-old has this lung condition?

- b) What is the probability that a randomly selected 60-year-old does not have this lung condition?

- c) What is the probability that a randomly selected 60-year-old is a smoker if he/she has this lung condition?

Class Example 8.1.15

Lie detectors are controversial instruments, barred from use as evidence in many courts. Nonetheless, many employers use lie detector screening as a part of their hiring process in the hope that they can avoid hiring people who might be dishonest. There has been some research, but no agreement, about the reliability of polygraph tests. Based on this research, suppose that a polygraph can detect 65% of lies, but incorrectly identifies 15% of true statements as lies.

A certain company believes that 95% of its job applicants are trustworthy. The company gives everyone a polygraph test, asking, "Have you ever stolen anything from your place of work?" Naturally, all the applicants answer "No," but the polygraph identifies some of those answers as lies, making the person ineligible for a job. What's the probability that a job applicant rejected under suspicion of dishonesty was actually trustworthy?

Summary of Probability and Counting Rules

Probability rules

P.1) $0 \leq P(A) \leq 1$

P.2) $P(A) = \frac{N(A)}{N(\Omega)}$

P.3) $P(\Omega) = 1$

P.4) If the event E is composed of several non-overlapping outcomes $E_1, \dots, E_n$, then $P(E) = P(E_1) + \dots + P(E_n)$

P.5) $P(A^c) = 1 - P(A)$

P.6) Addition Rule:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

P.7) Multiplication Rule:

$$\begin{aligned} P(A \cap B) &= P(B | A) * P(A) \\ &= P(A | B) * P(B) \end{aligned}$$

P.8) If A and B are independent, then

a. $P(A | B) = P(A)$ b. $P(B | A) = P(B)$ c. $P(A \cap B) = P(A) * P(B)$

P.9) If A and B are disjoint, then $P(A \cap B) = 0$

Counting rules

C.1) Basic counting rule: If event A consists of r actions, and for each action there are $m_1, m_2, \dots, m_r$ possibilities, then the number of outcomes in A is

$$N(A) = m_1 * m_2 * \dots * m_r$$

C.2) Arranging items: If event A has n items, then the number of ways we can arrange the items in A is

$$n! = n * (n - 1) * (n - 2) * \dots * 2 * 1$$

C.3) Choosing items: If event A has n items, then the number of ways we choose k items from A is

$$\binom{n}{k} = \frac{n!}{(n - k)! * k!}$$

8.2 Binomial Probability

- $P(A)$ is the probability of event A
 - Ω is the sample space, and includes all possible outcomes - $N(A)$ is the number of outcomes in event A
 - $P(A^c) = P(\text{not } A)$
 - $P(A \cup B) = P(A \text{ or } B)$
 - $P(A \cap B) = P(A \text{ and } B)$
 - $P(A | B) = P(A \text{ if } B)$

A binomial random variable is a special kind of random variable that only has two possible outcomes for each trial, a fixed number of trials, and each trial has the same probability of success.

Key Concepts

- Identify a binomial random variable
- Compute probabilities for a binomial random variable
- Compute the mean and/or standard deviation for a binomial random variable

Binomial Random Variables

Random variables often follow simple, predictable patterns. In these cases, calculating probabilities and the mean or standard deviation are much simpler and do not require a table to be created. One type of random variable with a simple pattern is called a “Binomial Random Variable.” Suppose we call the occurrence of an event a *success*. “Success” does not necessarily indicate the outcome was positive (e.g., if we are investigating cancer in a sample of mice, success could be defined as the presence of cancer). A *binomial random variable* measures the number of successes in a fixed number of trials.

A *binomial random variable* measures the number of successes in a fixed number of trials.

In order to be considered a binomial random variable, the following properties must hold:

1. We must be measuring the number of successes in n independent trials
2. The number of trials, n , is fixed in advance
3. The probability of success, p , does not change from trial to trial
4. Each trial is independent of the others (the outcome of one trial does not affect the outcome of another trial)

Class Example 8.2.1

For each of the following, determine if the random variable is binomial. If it is, determine n and p .

- a) On average, 25% of La Crosse residents recycle. 500 residents of La Crosse were asked if they recycle. A is the number of ‘yes’ answers in the sample.
- b) A box contains 50 marbles, 35 green and 15 white. Fifteen marbles are selected without replacement. B is the number of white marbles in the sample.
- c) An American roulette wheel consists of 18 red, 18 black, and 2 green numbers. Suppose you observed red on 10 consecutive spins. J is the number of additional spins needed until you observe black.
- d) D is the number of 5s in ten rolls of a fair die.
- e) Customers arrive at Alice’s with a rate of 5 per hour. E is the number of customers that enter Alice’s between 2 A.M. and 4 A.M.

Binomial Probabilities, Mean, and Standard Deviation

When a random variable is binomially distributed, probabilities and other characteristics can be expressed as formulas. For a binomial random variable X with n trials and probability of success p ,

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k} \text{ for } k = 0, 1, 2, \dots, n. \quad \text{- Mean: } \mu = E[X] = np$$

$$\text{Standard Deviation: } \sigma = sd(X) = \sqrt{np(1 - p)}$$

Class Example 8.2.2

To test for ESP, we have 4 cards, each with different symbols. These cards are shuffled and one is chosen at random (symbol down). Each time, you guess the symbol on the card. Suppose that you repeat this test 8 times and you do not have ESP.

a) Let X be the number of times you guess correctly. Is X binomial? If so, what are n and p ?

b) What is the expected value of X ?

c) What is the standard deviation of X ?

d) To be certified as having ESP, you need to correctly identify at least 6 cards of the 8 drawn. What is the probability you get certified?

Class Example 8.2.3

Joan takes 10 phone calls per day from customers, and, on average, 30% of these callers are angry. If 7 or more of these callers in a given day are angry, Joan has a bad day.

a) Let C be the number of angry callers in a day. Is C binomial? If so, what are n and p ?

b) How many angry callers will Joan speak to on average each day? What is the standard deviation for the number of angry callers?

c) What is the probability that Friday is a good day?

d) What is the probability that 3 of 5 days in her workweek are good?

Class Example 8.2.4

In the 2023 NFL season, Jordan Love threw 579 passes. A fair-weather fan happened to only see a few games that season. The fan only saw Jordan Love throw 33 times, and of those 18 were completed (caught).

a) What would the fan estimate is Jordan Love's probability to complete a pass?

b) Based on the probability in part (a), how many passes would the fan expect Jordan Love to complete in the next 10 throws?

8.3 Data Sources

The datasets used in this course pack come from the following sources. We gratefully acknowledge the original authors and data collectors.

Lock5Data R Package (GPL-2)

The following datasets are from the **Lock5Data** R package, which accompanies *Statistics: Unlocking the Power of Data* by Lock, Lock, Lock, Lock, and Lock (Wiley). The Lock5Data package is distributed under the GPL-2 license.

- **CAOSExam.csv** — Pre-test and post-test scores from a Comprehensive Assessment of Outcomes in Statistics (CAOS) exam. 10 students, 3 variables.
- **FloridaLakes.csv** — Water quality measurements from 53 Florida lakes. Original source: Lange, Royals, and Connor (1993), *Journal of Environmental Quality*.
- **PulseRates.csv** — Resting pulse rates for 10 students before and after exercise.
- **SleepStudy.csv** — Sleep habits and academic performance for 253 college students. Original source: Onyper, Thacher, Gilbert, and Gradess (2012), *Teaching of Psychology*.
- **USStates.csv** — State-level data (50 US states) compiled from US government sources including the Census Bureau and Bureau of Labor Statistics.

R Packages (GPL-2+)

- **Baumann.csv** — Reading comprehension scores from a study comparing three teaching methods (Basal, DRTA, Strat). 66 observations. From the **carData** R package (GPL-2+). Original source: Baumann and Jones (1992).
- **InsectSprays.csv** — Insect counts for six different spray treatments. 72 observations. Built-in R dataset (GPL-2/3). Original source: Beall (1942).

Public Domain / Custom

- **HollywoodMovies2023.csv** — Opening weekend revenue and world gross for six 2023 Hollywood films. Created for this course pack from publicly available box office data (Box Office Mojo).
- **MastersWinners.csv** — Winning scores at the Masters Tournament (1934–2024). 87 observations. Compiled from public tournament records.
- **NBASalaries2022.csv** — NBA player salaries for the 2022–23 season. 467 observations. Synthetic data generated to match the published salary distribution.
- **Salaries.csv** — Monthly sales figures (in dollars) for 15 salespeople under three pay structures (Commission, Fixed Salary, Commission Plus Salary). Synthetic data created for this course pack to illustrate one-way ANOVA and post-hoc comparisons.